Production and Perception of the Epenthetic Vowel in Obstruent + Liquid Clusters in Spanish: an Analysis of the Prosodic and Phonetic Cues Used by L1 and L2 Speakers

by

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Abstract

This study hypothesizes that the Epenthetic Vowel (EV) that occurs in Spanish consonant clusters, although produced unconsciously, is part of the articulatory plan of the speaker. As part of the plan, the epenthetic vowel occurs more often in the least perceptually recoverable contexts in order to enhance them. To achieve a better understanding of the role of the epenthetic vowel, this study shows that the linguistic and phonotactic contexts condition the occurrence of these vowels. Specifically, it argues that linguistic and phonotactic contexts that are perceptually weak compel a significantly higher occurrence of EVs.

The EV was analyzed from both production and perceptual standpoints. The results show that from the production standpoint, the occurrence of the EV is affected by the type of liquid that forms the clusters: in clusters with /r/ the variables that made a statistical contribution were post-tonic position (odds ratio, 4.46), and voiceless consonants (odds ratio, 1.42). In the case of clusters with /l/ an EV has a higher probability of occurring in the context of bilabial consonants (odds ratio, 4.19), and voiceless consonants (odds ratio, 1.3). As for the effects of speech rate on the duration of EVs, the results show that speech rate accounts for 14% of the variation in an EV's length.

From the standpoint of perception, listening was divided into the tasks of perceptual identification and perceptual discrimination. The results show that the strongest predictor is the interaction voiceless x post-tonic position (odds ratio, 4.8). For the identification of the Cr clusters, the strongest predictor is the context of voiceless consonants (odds ratio, 4.42). Regarding identification of the Cl clusters, the strongest predictors are the tonic position (odds ratio, 1.54) and the labial place of articulation (odds ratio, 1.39). With regard to the discrimination of the Cr clusters, the strongest predictors for perceptual recoverability are the interaction voiceless x post-tonic position (odds ratio, 2.22), and the labial place of articulation (odds ratio, 1.37), while for the Cl cluster, the strongest predictors are the tonic position (odds ratio, 5.83) and voiceless consonants (odds ratio, 3).

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Introduction

This dissertation is concerned with the production and perception of the epenthetic vowels (EV) that occur in consonant clusters in Spanish, such as gr, gl, cr, cl, dr, pr, pl, br, bl, etc. This study examines and compares the characteristics of the cluster segments and the epenthetic vowels for production and perceptual recoverability by Spanish L2 learners and Spanish native speakers.

While, superficially, the Spanish consonant clusters seem to be composed of two consecutive consonants (CC), there is dialectological and phonetic evidence that shows that an additional vowel (an epenthetic vowel) arises in between the consonants, resulting in sequences such as $g^{v}r$, $g^{v}l$, $c^{v}r$, etc.

A number of studies have indicated that the epenthetic vowel occurs as a form to resolve a complex, marked structure (or segments with similar linguistic characteristics) during language production (Hall, 2003, and 2004, Colantoni and Steeles, 2005). On the other hand, it has also been argued that the EV is an articulatory byproduct (Blecua, 2001). In terms of perception, it is hypothesized that the EV plays a role in the perceptibility of the consonant cluster in which it occurs (Bradley, 2002; Schmeiser, 2006). However, no explanation is provided as to how it helps in the perceptual recoverability process.

I propose that the epenthetic vowel, although produced unconsciously, is part of the articulatory plan of the speaker. As part of the plan, the epenthetic vowel occurs more often in the least perceptually recoverable context in order to enhance it. To achieve a better understanding of the role of the epenthetic vowel, I show that the linguistic and phonotactic

contexts condition the occurrence of these vowels. Specifically, I argue that linguistic and phonotactic contexts that are perceptually weak compel a significantly higher occurrence of EVs.

The contribution this study makes to the literature lies in the analysis of the role that linguistic clues play in the perception of the clusters, how they evolve during the process of learning second language pronunciation, and the interactions among them during the acquisition process.

Based on the results of the present study I show that from the production standpoint, the occurrence of the EV is affected by the type of liquid that forms the clusters: in clusters with /r/ the linguistic or prosodic cues for voiceless, post-tonic position, and for dorsals, present higher probabilities of occurrence of the EV. In the case of clusters with /l/, it is the cues for voiceless, pre-tonic position, and for bilabials, that have a higher probability of occurring with the EV.

Furthermore, this dissertation presents the first study, although at a small scale, of the effects of speech rate on the duration of the EVs. It reveals that speech rate accounts for 14% of the variation in an EV's length: in Cl clusters, the EV is negatively correlated to speech rate (word length) and the nuclear vowel (NV) in the previous syllable. However, in Cr clusters there was no correlation between EVs and speech rate, nor between EVs and NVs.

I argue that the fact that speech rate affects Cl clusters but not Cr clusters can be attributed to the difference in the inherent duration and sonority of /l/ as compared to /r/. At high speech rates, the /l/ becomes shorter, which makes the cluster difficult to perceive. In such cases, more and longer EVs occur. In contrast, the /r/, which is inherently short, is not shortened substantially because it could fall under the perceptual threshold. Thus, duration is the main factor affecting EVs in Cl clusters but not in Cr clusters.

From the perception standpoint, listening is divided into the tasks of perceptual identification and perceptual discrimination. This study shows that for identification of the Cr clusters, the more salient acoustic cues are in the labial place of articulation, voiceless, pre-tonic position; and for the identification of the Cl clusters, the more salient cues are in the labial and tonic position. With regard to the discrimination of the Cr clusters, the main cues for perceptual recoverability are in the labial, voiceless, and tonic position, while for the Cl cluster, the most salient cues are in the dorsal and tonic position.

This study adopts some of the theoretical aspects brought forward by the Motor Theory. It adopts the claim that there is a speech mode of perception, and that it is better engaged by informing listeners that what they will be hearing corresponds to speech, and specifically that they are second language utterances. The results show that L2 speakers confuse EV and full vowels at a higher rate, which indicates that they perceive EVs, while L1 speakers do not appear to recognize them.

In addition, this study contributes to a new approach in the statistical analysis of EV perception and production. As far as I am aware, this is the first study that considers item and participant variation by using mixed logit models. Using this statistical analysis, both speaker/listener and item variation can be analyzed simultaneously, while avoiding the misuse of analysis of variance on proportions, or even analysis of variance on transformed proportions (see Jaeger, 2008).

Organization of the Dissertation

Chapter 1 lays out the groundwork for the present study. Specifically, I discuss relevant studies on the factors affecting first and second language perception, in addition to the perception

of the epenthetic vowel. Chapter 2 examines the phenomenon of vocalic epenthesis from the point of view of speech production. In addition, it presents the most relevant studies on the EV from both a phonetic and a phonological point of view. Chapter 3 examines the factors that affect the frequency of occurrence of EVs in Obstruent + liquid clusters as well as some of the factors affecting their duration, and Chapter 4 presents the first study on the role of speech rate on the occurrence of EVs. Chapters 5 and 6 present the results from the identification and discrimination experiments respectively. Finally, Chapter 7 examines the role of EVs on the perception of foreign accent. It presents the results of a goodness rating experiment on words containing digitally manipulated epenthetic vowels. It analyzes the factors that determine accentedness in consonant clusters and the interaction of those factors.

Chapter 1
Epenthetic Vowels

1.1 Scope of the study

This study examines the production and the perception of the epenthetic vowel in Spanish consonant clusters. In terms of perception, its goal is to determine the extent to which English native speakers perceive the epenthetic vowel that occurs in Spanish consonant clusters formed by an obstruent + liquid sequence, (i.e., [gr], [gl], [kr], [kl], [dr], [tr], [tl], [br], [bl], [pr], [pl]). The research compares the perceptual cues used by native Spanish speakers with those used by English speakers learning Spanish, and it analyses the development of the perceptibility of the epenthetic vowel from the beginner level to the advanced level of language learners. In terms of production, this study examines the factors that affect the frequency of occurrence of the epenthetic vowel.

The object of study - Spanish consonant clusters - is problematic, because while superficially, they seem to be composed of two consecutive consonants (CC), there is dialectological and phonetic evidence, as well as data from children learning Spanish, which shows that an additional vowel (an epenthetic vowel) arises in between the consonants. The examples given in (i) illustrate some cases in which Spanish speakers insert a vocalic element between the two consonants in Obstruent + *r* clusters (Malmberg, 1965).

(i)	*tigere*	[tíɣeɾe]	instead of	*tigre* [tíɣɾe]	"tiger"
	chacara	[ʧákaɾa]	instead of	*chacra* [ʧákɾa]	"small farm"

These intrusions suggest that there is a natural tendency in spoken Spanish to insert an epenthetic vocalic element between the stop and the flap (CvC) vs. (CC). Malmberg (1965) and Quilis (1993) hypothesize that this epenthetic element is an actual vowel; later in this study I will address what is meant by that claim.

The realization of this epenthetic vowel is unconscious and seems to be an intrinsic part of the syllabic structure of the Spanish language. Since native speakers are not aware of it, it is not taught in Spanish classes. The learners are therefore left to identify this structure by themselves, if they can, and to pronounce it to the best of their abilities. For those learners who fail to do so, this presumably becomes a part of their foreign accent. Later in this study I will address the issue of whether a failure to produce the epenthetic vowel really does affect perceived accentedness.

The contribution this study makes to the literature lies in the analysis of the role that linguistic cues play in the perception of the clusters, how they evolve during the process of learning second language pronunciation, and the interactions among them during the acquisition process.

1.2. Context of the study

1.2.1 Main issues in second language phonology

The task of learning a second language is especially challenging for adults. Although they may achieve a good mastery of the grammar, it is generally accepted that achieving native-like pronunciation is increasingly more difficult the later the age at which the learning starts; and it is almost impossible for adults to obtain a native-like accent. Adults, with their more developed cognitive skills, will have a head start over children, but in the long term it is the young learners

who, under normal circumstances, will finally attain native proficiency in the language, i.e. unaccented speech. (Ausubel, 1964; Snow and Hoefnagel-Hohle, 1977; Cummins, 1981; Munro, Flege, and MacKay, 1996; Piske, Flege, MacKay, and Meador, 2002).

Some of the factors that affect second language (L2) pronunciation are speech perception abilities, the age at which learning starts, the amount of exposure and experience with the language, awareness of the phonetic and phonological structure of the language as well as the influence of the first language (L1).

1.1.2.1 The role of Speech Perception in Speech Production

The effect of speech perception on phonology has been recognized as one of the factors that influence the learning of an L2. Polivanov (1931, 1964) pointed out that L2 sounds are perceived through the phonological system of the first language. Trubetzkoy (1939, 1969) argued that inexact L2 production is the result of erroneous perception, suggesting that L1 operates as a phonological filter through which the L2 sounds are processed.

An early comprehensive approach to language sound structure in terms of acoustic and auditory characteristics was proposed by Jackobson, Fant and Halle (1951), but their proposal did not resonate at the time due to the novelty of the technology and the rather small body of research on speech perception. Despite the early recognition of the role of speech perception in phonological systems, much of the research on the phonology of a second language (L2) has been done on the production of L2 speech. However, during the last 15 years there has been a revival of interest in speech perception due to two main factors: first, the technological advances that have allowed for much easier and inexpensive collection of data in the laboratory and in the field; and second, the development of theoretical frameworks that take into consideration

constraints from perception and other dynamic principles (i.e., Optimality Theory, e.g. Hume and Johnson 2001).

Although recent studies recognize the interaction of speech perception with phonological systems, it is not clear to what degree perception affects phonology and *vice versa*. This is an important consideration because of its implications for the prerequisites to accurate development of individual segments and, by extension, to the production of second language speech.

Hume and Johnson (2001) argue that there is interplay between speech perception and phonology. Speech perception affects the phonological sound system in at least three forms: a) failure to perceptually compensate for articulatory effects; b) avoidance of contrasts that are perceptually weak; and c) avoidance of noticeable alternations. In the first area, Ohala (1981) and Beddor, Krakow, and Lindemann, (2001) argue that the listeners are a source of sound change. Listeners who fail to perceptually compensate for co-articulation attribute phonetic variation to one of the segments; for instance, a vowel (V) that is nasalized in the context of a nasal consonant (N) can be perceived erroneously as a nasal vowel (~V). Thus (VN) → (~V). The second area of interaction is the avoidance of contrasts of weak perceptibility. Sound differences that are difficult to perceive tend to not be used contrastively in language. For instance, Hume and Johnson (2001) point out that in English, nearly imperceptible differences such as the apical [s] versus the laminal [s] or the voiceless inter-dental fricative [Θ] versus dental [Θ], are pronounceable, but their contrast has low salience, and therefore they are not used contrastively.

On the other hand, since contrasts are fundamental for maintaining the phonological system, weak contrasts tend to be repaired using strategies such as epenthesis, dissimilation, metathesis, deletion, and assimilation. Epenthesis consists in the insertion of a new segment

between preexisting segments; for instance, the contrastive sequence *xy* becomes → *xzy* through epenthetic repair. One example is the occurrence of an epenthetic schwa in some English dialects, as seen in words like "film" [filəm]. Dissimilation occurs when two contiguous segments have similar features; one of the elements changes its features to establish a differentiation; *xx'* → *xy*. Thus in some English dialects the word "chimney", with two consecutive nasals, is pronounced "chimley", i.e. the second nasal has become a lateral. Metathesis is a process in which elements switch positions within the string of segments to avoid having contiguous segments with similar features, *zxx'* → *x'zx*. Hume and Johnson (2001:4) point out that in some in cases of such metathesis in consonant/consonant sequences, the weaker consonant is positioned in the more robust context to repair the sequence and thus becoming more perceptible. For example, in Faroese, the sequence /sk/ metathesizes when a stop consonant follows, e.g. /baisk+ t/ becomes [baikst] 'bitter, neut.sg.' and not *[baiskt] as would be expected (Jacobsen and Matras 1961, Lockwood 1955, Rischel 1972; see Seo and Hume 2000).

Perceptibility also triggers strategies intended to avoid weak contrasts. In some cases the system opts for simplifying the contrasts through deletion, i.e. elimination of one of the segments; in other cases, by assimilation, i.e. the complete or partial copying of the features of a neighboring segment *xy* → *xx.* An example of the former case is the common pronunciation of a word like "text book" as "tex book"; an example of the latter is the pronunciation of "unbalanced" as "umbalanced", where the alveolar nasal assimilates in place of articulation to the following bilabial stop.

Of particular interest to this study is the strategy of epenthesis. The insertion of a vowel between the consonants of Spanish clusters could be motivated as a repair for the enhancement

of the perceptibility of the two consonantal segments of the cluster. Such a strategy is also used in other languages (see above re: "film"), and seems to be the main motivation for epenthesis. However, the intrusive vowel that occurs in Spanish consonant clusters is not a segment with its own phonological representation; instead it is only present at the phonetic level (Côté, 2000; Hall, 2003). Nonetheless we will refer to it as an epenthetic vowel (EV) throughout this study.

The third area in which speech perception affects the phonological system is the avoidance of noticeable alternations: perceptibility operates as a filter that affects the possible outcomes of the repair of a weak contrast. According to Hume and Johnson (2001), in this area "perception is seen as a type of filter on sound change" (6) in which changes are accepted only if they are perceived as not being too different from the intended target, and if those changes create a possible misunderstanding or a break down of communication. For instance Huang (2001) in a perceptual study of Mandarin Chinese tones by native speakers of American English and Mandarin Chinese found that low-falling-rising (214)[1] tones are perceived as simply mid-rising (35) tones. The author interprets this lack of discrimination as evidence that the phonological change is gradual, and considered, overall, as not very noticeable.

As for the influence of phonological systems on speech perception, this is inferred in cases in which speakers are better at perceiving sounds of their L1 than those of the L2 when acquired as adults (e.g. Hume and Johnson, 2001; and for the case of English vs. Catalan vowels see Cebrian, 2002).

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[1] The numbers in parentheses indicate the pitch values of the tones on a five-level scale.

1.2.1.2 Learning to perceive in a second language

Most of the research on perception has focused on the listener's ability to identify or discriminate phonetic information from an acoustic signal, and includes research on infants, children, and adults (e.g. Eimas, Siqueland, Jusczyk and Vigorito, 1971; Mattingly, Liberman, Syrdal, and Halwes, 1971; Aslin, Pisoni Jusczyk, 1983). The results have been explained from two main theoretical positions. First, the mechanism responsible for speech perception is claimed to be best described as innate and speech specific (Liberman, Cooper, Shankweiler, Studdert-Kennedy, 1967; Liberman and Mattingly, 1985; Repp, 1982; Pastore, 1981; Pisoni, 1977). Secondly, the evolution of discrimination abilities in the perception of linguistic structures in children vs. adults presents parallel patterns. Goodman, Lee, and DeGroot, 1995 argue that "the mechanisms responsible for the development of speech perception may play a role in processing through the lifespan" (4). Although there are differences between perception by infants and adults, theories of speech perception should be able to account for infants' processing mechanisms as well as adults' abilities to recognize linguistic units; in particular, theories of L2 perception should be able to explain how learners perceive the correct sound categories.

Thus models of speech perception should account for three important issues (Escudero, 2007):

(i) The initial state and the role of early linguistic experience

(ii) Perceptual learning: the developmental constraints affecting adult learners

(iii) The relationship between levels of linguistic knowledge in processing: cognitive interplay of two language systems during acquisition.

1.2.1.3 The initial state and the role of early linguistic experience

Early linguistic experience has two sorts of effects. First, there may be a time frame, a critical period for language learning (Lenneberg, 1967); and second, the knowledge of a first language affects the learning and processing of a second language (Werker and Tees 1983; Best, McRoberts and Sithole, 1988; Pisoni, Lively and Logan, 1994; Werker 1994; Best, 1995). It is generally accepted that although perceptual learning, and L2 learning in general, occurs through the lifespan, there is something special about childhood. Adults can learn nonnative contrasts, but there are limitations.

The nature of the critical period and the developmental mechanism that accounts for it can be of two types. First, the critical period may be a learning period in which some specialized neurological structures are established for perceptual processing of the language. And secondly, it may be easier to learn an aspect of linguistic structure when there is no other high-order knowledge that may impede attention at the required level of analysis. For instance Pisoni, Lively and Logan (1994), in a perceptual study of English /l/ and /r/ by Japanese listeners, found that their ability to identify the sounds differed according to word position. Goodman, Lee and DeGroot (1994) point out that these results indicate that L2 learners do not really learn a context-independent phoneme category, as is arguably the case with L1 learning. Thus, the L1 (here, Japanese) has a lifelong effect that limits perceptual learning in adults. However, Pisoni et al. point out that linguistic experience affects perception "by modifying attentional processes, which, in turn, affect the underlying perceived psychological dimensions" (156). This implies that the difficulty of learning an L2 as the individual matures may not be a sensory-based loss but rather a change in selective attention. The authors claim that in principle, all nonnative contrasts can be discriminated reliably by adults using relatively simple laboratory training

techniques. Since the underlying sensory abilities are still intact, techniques such as discrimination training only serve to modify attentional processes that are assumed to be susceptible to realignment (Aslin and Pisoni, 1980; Pisoni et al.; 1994). Terbeck (1977) provides data on selective attention as an indicator of how linguistic experience modifies different phonetically relevant dimensions based on the linguistic environment. The author used a scaling technique to measure the magnitudes of differences between pairs of vowels presented to speakers of different languages. He found that prior language experience affected vowel perception by altering "the perceived psychological distances [perceived difference] between the vowels" (156). Thus, the perceptual distance between the vowels was reported to be larger when the pair was a phonological contrast in the listener's L1 than if it was not phonologically distinctive in that language.

Pisoni et al. (1994) argue that the listener's linguistic experience "modifies the underlying psychological differences by altering the similarity relations for different perceptual dimensions" (156). Therefore, experience reshapes the psychological space favouring important distinctive contrasts in the participants' L1 and by diminishing the cues for non-contrastive distinctions. The authors also argue that, in addition to perceived psychological distance, experience results in changes in memory representation. Hence, learners have the mechanism needed "to make fine phonetic discriminations, but they cannot develop stable representations in long term memory that could be used in other tasks that require more abstract memory codes" (157). See also Cebrian (2002) for similar findings.

On the other hand, Mayberry (1994), in her study of deaf speakers' use of American Sign Language, found that if a person is not exposed to linguistic input from the early years of life, then the way language is processed will present subtle differences compared to those who have

been exposed to language. Furthermore, Cutler, Mehler, Norris and Segui (1986) and Cutler and Norris (1988) in a study of lexical segmentation by English and French speakers found that adult listeners use segmentation strategies from their L1. English is a stress-timed language formed by sequences of strong and weak syllables; strong syllables are more likely to mark content words than weak syllables. However, the prosodic structure of French is syllable based, rather than stress-based. The results of the study show that both the English and French speakers used their L1 rhythmic structures to segment their L2. English speakers used a stress-based strategy and French speakers used a syllabic-based strategy to segment units of speech. For instance the English syllabication of the cluster [bl] in the word *problem* is [prob-lem] whereas in French it is [pro-blem]. Thus, speakers use their knowledge of their L1 as strategy to segment the L2.

1.2.1.4 Differences between L1 and L2 perception

Differences between L1 and L2 perception/production have often been attributed to the influence of the L1. Interference from the L1 refers to the way the learner's existing knowledge influences the course of acquisition of L2. Trubetzkoy (1929/1965) observed that the sound system – the phonological system -- of the learner's L1 may work as a filter through which L2 sounds are analyzed. Therefore, the errors that characterize non-native speech may be the result of incorrect representation of L2 sounds (Rochet, 1995; Flege, 1995). This phenomenon has been typically described and explained through transfer or cross-linguistic influence. The concept of transfer has been used most often in the area of L2 phonology, because the L1 influence is in many cases quite obvious. Nevertheless, transfer is not exclusive to phonology at the segmental level. It also affects other levels such as phonological rules and processes, syllable structure, phonotactics, stress patterns and intonation (Sato, 1984; Tarone, 1987; Ellis, 1994; Cebrian, 2002).

Nonetheless, the nature and degree of L1 transfer is a matter for discussion: the assumption of no transfer, partial transfer or full transfer will influence the assessment of the L2 task and L2 development. Hammarberg (1997) points out that the nature of transfer is ambiguous: it could refer to a learning strategy (conscious or unconscious), a process of transferring L1 knowledge into the process of acquiring the L2 system, or to the result of such process (Escudero, 2007). Moreover, there are many documented instances of L2 phonology that are not based on transfer, but rather by the learner's current grasp of the L2 phonological structure. Frieda and Nozawa (2007), in a study of the development of English L2 by Japanese and Korean speakers, found that some perceptual categories of vowels cannot be explained in terms of categories in their L1. For example, in a discrimination task of the English contrasts /ɛ/-/I/ and /æ/-/ai/, Korean and Japanese native speakers did not assimilate to or equate them with any native categories from their L1 (Korean or Japanese respectively).

1.2.1.5 Perceptual learning: developmental constraints

Listeners must learn what acoustic input is relevant to meaningful distinctions in their L1, and during the speech perception process they attend to those while ignoring the other non-relevant information. Kuhl et al. (1992) and Best (1994) show that children as young as six months old recognize a large repertoire of contrasts in their L1, and they also recognize the vowels of the language they are exposed to. However, by ten months, children show a perceptual loss of contrasts that are not productive in their L1.

The constraints that young children face during L1 acquisition include memory, cognitive development and sensory motor control (Jusczyk, 1994). Escudero (2007) points out that the L2 learner faces other kinds of constraints: maturational constraints, input constraints and learnability constraints.

1.2.1.6. Maturational constraints

Evidence from L1 and L2 acquisition suggests that normal language learning occurs only when exposure to the language starts early in life. When exposure begins later, performance in language declines (Newport, 1990). Two main hypotheses attempt to explain maturational effects on L2 learning. On the one hand, Penfield and Roberts (1965) claimed that a child's brain is more plastic compared to the adult brain, at least before the age of about nine. The brain's plasticity allows direct learning from the input. Lenneberg (1967) argued that language acquisition is an innate process determined by biological factors that limit the *critical period* for language acquisition from two years of age until puberty. For his part, Krashen (1975) argued that Piaget's cognitive stage of formal operations, which starts around puberty, is the cause of the closing of the critical period. Lumandella (1977) introduced the term *sensitive period* -now used interchangeably with *critical period*- to emphasize that language acquisition may be more efficient during early childhood. In contrast to the generally accepted idea that there is a critical period, Johnson (1992) and Shim (1993) present evidence of late learners acquiring a high level of proficiency. The contradictory data may suggest that in addition to maturational effects, there are other factors affecting learners after puberty. Among these factors, Robertson (2002) includes personal motivation, anxiety, input and output skills, setting, and time commitment. In addition, Birdsong (2002) points out that the age variable is moderated by other variables that may decrease the number of native-like attainers. Newport (1990) argues that language learning capabilities decline because of the expansion of non-linguistic cognitive abilities. Thus, while children learn language through an implicit domain mechanism, adults often reflect on its structure and to some extent use cognitive problem-solving capabilities to learn an L2 (DeKeyser, 2000; Johnson and Newport, 1989).

1.2.1.7. Input constraints

One constraint that affects L2 proficiency levels is the type of evidence that the learners require for the learning to occur. Escudero (2007) points out that L1 learners require only positive evidence (ambient language) to develop the language, whereas L2 learners seem to need negative evidence either in terms of corrective feedback (Russell and Spada, 2006) or specific instruction in order to learn the L2 (Norris & Ortega, 2000; White, Spada, Lightbown and Ranta, 1991).

Research shows that the language learner receives modified input and modified interaction in comparison to L1 learners. These styles have been referred to as *foreigner talk*, the modified speech used when addressing a nonnative speaker, and *teacher talk*, the modified speech used by language teachers with their L2 learners (Pica, 1983; Pica and Doughty, 1985; Archibald, 1990). Another factor to be considered is the amount of exposure to the two languages (Collins, Halter, Lightbown, Spada, 1999).

1.2.1.8. Learnability Constraints

Within the framework of Generative Grammar it is generally accepted that L1 acquisition relies on innate universal principles and constraints, while L2 acquisition seems to rely more on cognitive mechanisms. Schwartz and Sprouse (1994, 1996) argue that L2 learners have access to Universal Grammar in the same way L1 learners do. That is to say, the same constraints that operate during L1 acquisition also operate during L2 acquisition. For instance, in a study of the production and perception of stress in English words by Polish and Hungarian speakers, Archibald (1990) found that adult learners seem to be able to reset their L1 parameters when learning an L2. Archibald claims that the results of this study indicate that learners have access

to the principles and parameters of Universal Grammar, and that L2 learners do not seem to violate universals of metrical phonology.

Escudero (2007) points out that in current phonological theory, the learning of L2 phonology has been proposed in terms of algorithms that "analyze and modify the rules or constraints that constitute the developing grammar" (114). One of the learning algorithms is Tesar and Smolensky's (2000) Constraint Demotion Algorithm, which operates within the description of phonological knowledge as defined by ranked sets of constraints (Optimality Theory). On the other hand, MacWhinney (1999) proposes an emergentist account according to which learning occurs in a bottom-up fashion. That is to say, the parsing of the language starts from the recognition of speech elements in the acoustic signal and from there the larger units are recognized at different processing levels until the intended word is identified. Thus, the knowledge of the L2 learners changes according to the input by means of a general cognitive learning device and not by a language specific mechanism.

1.2.1.9 The cognitive interplay of L1 and L2 systems

Knowledge of a language requires a system devoted to this function, although there is some debate as to whether there is an exclusively linguistic system or if it is part of a more general cognitive system (see MacWhinney, 1999). When a learner acquires a second language it is reasonable to assume that a second system emerges. The interaction of the two perceptual systems could affect the perceptual proficiency that the learner has in both languages (Escudero 2007).

Cook (2002) proposes three possible settings in which the L1 and L2 could interact. The three possibilities are illustrated in Figure 1.1 below. One possibility is that the two systems

operate in a separate fashion. Each system would be activated in the presence of input from the corresponding language. A second possibility is a connected system in which there is overlap between the two systems. Thus, the systems, although independent, would have perceptual commonalities. The arguments in favour of the connected system stem from the fact that most languages share at least a subset of categories (e.g. of consonants and vowels). A third possibility is a mixed system. This system may in turn have two logical possibilities: the L1 and L2 systems are completely integrated into one representational system but each one maintains its characteristics, or they completely merge into one, integrating in such a way that the initial systems are no longer identifiable. The adoption of any of those models of interaction of the perceptual systems determines the assumptions that explanatory models of L2 perception make.

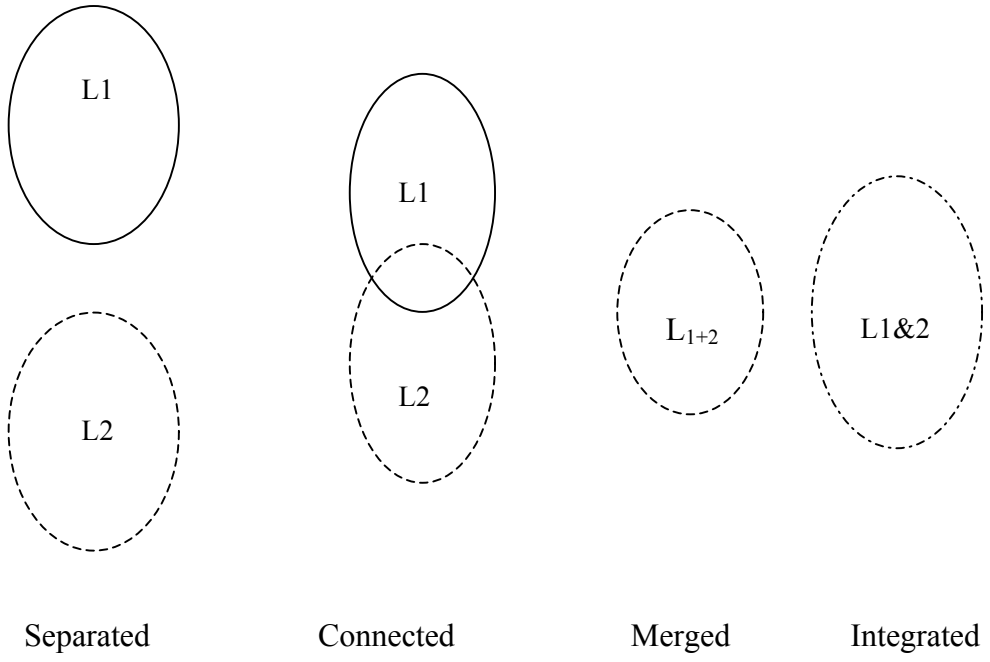

Figure 1.1. Possible cognitive status of linguistic systems in L2 learners and bilingual speakers. (Adapted from Escudero 2007:115)

The following section presents a review of the most important language perception models.

1.2.1.10 Implications of the speech perception and phonological system interplay

The primacy of speech perception over the phonological system or the phonological system over speech perception has implications for learnability. If the former is the case then is accurate perception a condition for native-like production? And if the latter is true then can an L2 learner produce native-like speech without perceiving sounds accurately? This section reviews some of the most representative experimental studies done to address these questions and discusses the implications of its findings.

A large body of research from L1 and L2 argues for the precedence of perception over production. In the area of L1 acquisition, Lehman and Sharf (1989) present a study of vowel duration cues to final consonant voicing, with implications for the perception/production relationship in children. They found that perception is consistently more advanced than production. In L2 research, Flege (1991) puts forward the "equivalence classification" hypothesis, which argues that foreign-accented speech is the result of the "development of the L1 phonetic system, which makes it increasingly unlikely that similar sounds in an L2 will evade being equated with sounds in L1" (285), thus suggesting that inaccurate perceptual representations are the cause of non-native pronunciation.

However, the experimental evidence seems to be inconclusive. For instance, studies on the perception and production of the English l/r contrast by Korean students have found that Korean learners of English perceive the distinction better than they produce it (Border, Gerber and Milsark, 1983; Kim and Park, 1995; Kim 2005). But Goto (1971), as well as Sheldon and

Strange (1982), in studies of the same l/r distinction by Japanese learners of English, found that they can produce the contrast better than they can perceive it. These and other studies challenge the primacy of perception in L2 phonology and argue that perceptual accuracy is not required for accurate production in L2 speech (Neufeld, 1988; Borrell, 1990). Nevertheless, it is important to note that the results from studies on the perception/production of consonants and vowels have to be considered separately. For instance, Bohn and Flege (1990) argue that different classes of sounds behave in different ways with respect to the interaction between production and perception. In addition, Strange (1995) argues that the articulatory nature of consonants and vowels makes them intrinsically different: consonant production often involves contact between articulators, whereas vowel articulation depends on the spatial positioning of the tongue, which makes it more difficult to obtain articulatory feedback in the latter case. She points out that it may be that "the production of vowels is more dependent on auditory feedback" (81).

The following section examines the experimental research on the production and perception of segments with special attention to vowels.

1.2.2 Experimental studies of the production and perception of L2 segments

1.2.2.1. The production and perception of L2 vowels

Experimental studies of the perception and production of vowels demonstrate a close relationship between those abilities. For example, Barry's (1989) investigation into the correlation between the production and perception of English vowels by German speakers shows that well-established perceptual categories are more likely to result in acceptable production, thereby suggesting the perceptual precedence of perception over production. In contrast, Rochet (1995), in his study of Brazilian Portuguese speakers and Canadian English speakers learning

French, found that the former tended to misidentify French /y/ tokens as /i/ while the latter tended to misidentify the same vowel as /u/. In terms of production, in a repetition task, the Portuguese speakers produced /i/ quality vowels when they heard the French /y/ tokens. The English speakers tended to produce /u/ quality vowels. Rochet claims that the difference in production can be explained by the more fronted articulation (fronting of the tongue body) of English /u/, which is closer to /y/ than the Portuguese /u/. He also claims that these results support the notion that the accented pronunciation of L2 segments by untrained speakers may be perceptually determined.

In addition to the role of perception in production, abundant research has focused on other factors that affect perception and production, such as age of L2 learning, awareness of phonetic differences, and familiarity with the language, as well as segmental features such as acoustic cues, along with supra-segmental features such as syllabic structure and phonological constraints.

Bohn (1995) proposes three learner variables that affect foreign language perception: L1 background, L2 experience, and age. Although other factors may be involved, such as cultural and motivational factors (they could have an effect on the production of sounds since learners may be reluctant to produce some sounds), Bohn finds no evidence that they can affect perception.

1.2.1.2 The role of L1

The influence of the L1 is well attested in speech production and perception. Ingram and Park (1996) studied the perception and production of Australian English vowels by Korean and Japanese learners of English. For the perception task, the subjects had to listen to recordings of

speakers reading /hVd/ words containing the Australian English vowels /i/, /I/, /e/, /æ/and /a/; and respond using a forced-choice identification task. The results of the study show that the Japanese participants could correctly identify all the vowels (92%-100% accuracy). In contrast, the Korean speakers did not show consistent judgments for /e/ and /æ/ (between 46% and 54% accuracy). The authors argue that the two groups of participants were using different strategies for making the judgments. They claim that the Japanese speakers were using duration as a perceptual cue, possibly because timing is contrastive in Japanese. Thus L1 cues helped the Japanese to identify the tokens. On the other hand, timing is not contrastive in Korean and therefore, the authors argue, the Korean group had to rely on spectral cues to identify the vowels.

In the production part of the study, participants had to read aloud the words of the perception task. Their readings were recorded and analyzed for vowel duration. The results showed a clear difference between the two groups: the Japanese participants produced vowels with internally consistent durational values, with lengths that differed across segments. The authors inferred that they were transferring moraic vowel length from Japanese to their production of the Australian English target vowels. Similarly, the Koreans produced vowels in patterns that were similar to those of their L1.

1.2.2.3. The role of Age

A great quantity of research has been devoted to the study of the relationship between the age at which a person starts learning an L2 and their ultimate attainment, i.e., their proficiency level. Carroll (1969) argued that time is a key variable in L2 acquisition, based on measured achievement. Empirical research has looked at age and the effect of experience in the L2. In a well-known study, Flege, MacKay and Meador (1999) studied the production and perception of English vowels in a group of Italian speakers who arrived into Canada at different ages. Vowel

production accuracy was assessed using an intelligibility test in which English native speakers judged the quality of the vowel tokens produced by the speakers. Vowel perception was assessed using a categorical discrimination test. The authors found that the speakers who arrived at an early age (at seven and fourteen years of age) did not differ significantly from native speakers of English either for production or perception; they achieved native-like performance. Those who arrived later (around 19 years of age and older) showed worse performance. The authors argue that their findings are consistent with the hypothesis of the Speech Learning Model (Flege, 1995), which posits that early bilinguals establish new categories for vowels found in the L2 and that this ability decreases as the age of learning increases. Similar results were found in another study of Italian speakers by Munro, Flege and MacKay (1996) in which the authors examined the English vowel production of 240 Italian speakers who arrived in Canada at different ages, between two and twenty-three. The study also included 24 native English speakers. In this study, age of arrival was considered to be their first contact with English. In an accentedness-rating test, ten English native speakers judged the degree of accent on eleven vowel tokens produced by the subjects. The authors found that perceived accentedness increased as a function of age of arrival: the late arrivers did not produce vowels that were considered consistently native-like even though they had been living in Canada on average 32 years. In the production part of the study, the results show that all the vowels, both the vowels that have an Italian counterpart and those that do not, were highly intelligible and correctly identified.

However, not all the data are consistent with the finding that early bilinguals who are highly experienced in the language will have native-like perception. Bosh and Sebastián Gallés (1997) found contradictory results. In their study, Spanish-Catalan bilinguals whose L1 is Spanish but who grew up in a bilingual setting were not able to perceive the contrast between the

Catalan vowels /e/ and /ε/ in a native-like manner. These results argue against the role of experience in perception. However, Flege *et al.* claim that these results may be due to the fact that the speakers had used their Spanish L1 more than the early Italian/English bilinguals in the studies cited above.

1.2.2.4. The role of familiarity

In addition to the role of perception and age, research suggests that familiarity with the L2 plays an important role in the production of L2 speech. For instance, Elsendoorn (1984) in his study of the production and perception of English vowel duration by Dutch speakers found that the perceptual results were consistently better as the subjects had more contact time with the L2, either by immersion in the language or through practice. Similarly, Bohn and Flege (1990) studied the production/perception of the / è / and / æ / vowels by two groups of German speakers: an experienced group, with at least five years residing in the US, and an inexperienced group with an average stay in the US of six months. The results of a production task show clear differences between the two groups. The inexperienced group was not able to produce the contrast between the target vowels. Nevertheless, in a labeling task the inexperienced group was able to differentiate the vowels. The experienced group produced the contrast between the two vowels and also obtained higher rates of accuracy in the labeling task, which confirms the effect of language familiarity in L2 perception and production.

In addition, Kashino and Craig (1994) investigated English spoken word recognition by Japanese listeners with different degrees of experience. The authors found not only that the more advanced listeners recognized words faster and more accurately than less experienced speakers, but also that they show “greater anticipatory recognition”, using the first syllable of the word to identify the remaining portion of the token. The authors point out that beginning listeners used

acoustic-phonetic information as effectively as advanced listeners; however, the advanced listeners were able to take advantage of the linguistic context to complete the recognition process faster and more accurately. This contextual information can only be used with familiar words or ones which the listener perceives or re-interprets as familiar (Fitt, 1998).

1.2.2.5 The role of linguistic cues

Another important element in L2 perception and production are the cues that listeners obtain from the signal itself. Bohn and Flege (1990) and Flege and Bohn (1989) in an identification task of synthetic vowel continua found that inexperienced German and Spanish learners of English, whose L1 does not have contrasting short/long (tense/lax) vowel sounds, relied on duration, whereas native English speakers, whose language has the long/short contrast, do rely more heavily on spectral cues. Flege and Bohn (1992) conducted a production task of the English vowels /i/ and /I/ with early and late Spanish learners. The results show that all learners produced duration contrasts but only early learners produced a significant spectral contrast. In an identification task of the same vowels (/i/ and /I/), which are not found in the Spanish system as contrastive sounds, Flege and Bohn (1989) found that Spanish learners rely on durational cues to a larger extent than spectral cues. However, the strong reliance on duration was not significant for the pair /ɛ/ and /æ/, which are identified with the Spanish vowels /e/ and /a/. Thus, the authors point out that in cases where there are not L1 counterparts, cues from duration were used. The predominance of the use of temporal cues is also reported in Spanish and Catalan learners of English for perception (Garcia-Lecumberri and Cenoz; Cebrian, 2002). In terms of the relationship between perception and production, Bohn and Flege (1990) pointed out that while spectral differences in production were related to a dominance of spectral clues in perception, strong reliance on durational cues in perception implied small durational differences in

production. Summing up, there is strong evidence that experience with the language has more influence on production than on perception (see Elsendoorn, 1984; and Llisterri, 1995).

1.2.3. Speech perception

What makes speech perception special?

Some researchers argue that the perception of speech is different from the perception of non-speech sounds and that a special mechanism has evolved for the perception of speech (Liberman, Cooper, Shankweiler, Studdert-Kennedy, 1967; Liberman and Mattingly, 1985 contra Handell, 1993). In particular it is argued that there is a "speech mode of perception" which is engaged when the listener first identifies an acoustic signal with speech-like characteristics. This section reviews the characteristics of speech that seem to require special decoding mechanisms.

The Speech Mode of Perception

Evidence of a speech mode in perception comes from a study by Remez, Rubin, Pisoni and Carrell (1981). The authors used synthetic signals, which resembled the frequency and amplitude of the first three formants of natural vowels. They played the sounds, derived from an English sentence, to two groups. The first group was not told anything regarding the sounds; they reported hearing science-fiction like sounds, electronic music, computer beeps and so on. The second group was instructed to transcribe a "strangely synthesized English sentence" (949); this group was able to hear the sounds of speech even though they sounded unnatural and they were able to transcribe the sentence. The authors point out that the instructions to the listener can help to engage the speech mode and once this mode is engaged it is difficult to reverse the process. The listeners who have heard the stimuli as speech sounds tend to continue to do so. It is

important to emphasize that the temporal patterning of the sounds is crucial. The stimuli were not identified as speech sounds if they were presented in isolation.

Nevertheless, Stevens and House (1972) argue that listeners are not set in the speech perception mode before hearing the signal, but the mode is triggered by a signal that has the appropriate acoustic properties (presumably more appropriate than the synthetic speech mentioned above). Hence the triggering of the speech mode is mainly an involuntary response to the speech-like acoustic signals.

The mode for speech perception is not unique in human perception; harmonic complex tones can be heard in two modes as well. They can be heard in an analytic mode, hearing the pitches of one or more individual partials; or they can be heard in a synthetic mode: hearing a single pitch corresponding to the fundamental frequency. However the speech perception mode is unique in that it operates for a whole class of complex and varied acoustic signals (Moore, 2003).

1.2.3.1. Problems with speech perception

One of the biggest problems that is faced when studying speech perception is the lack of invariance (Pisoni and Sawusch, 1975). There is not a one-to-one relation between the acoustic signal (the sound in the speech continuum) and the segments resulting from the linguistic analysis (phonetic segments such a *t* or a *d* for instance). That is to say that the analysis of the spectrographic signal generally does not reveal segments in the acoustic signal that correspond uniquely to segments in the message. From very early experiments in speech perception, it has been shown that a single segment in the acoustic signal (a chunk of sound) carries information about several successive phonetic segments. Very different acoustic signals were often

perceived as identical phonetic segments and identical acoustic segments were perceived as different phonetic segments according to the context in which they were embedded. The acoustic cues that determine a phonetic segment are numerous and variable depending on the phonetic context, stress, and speech rate, among others; all of those factors are natural consequences of the articulation of speech in a continuum.

Some of the theories of speech perception are based on the interaction between speech articulation (production) and perception since the speech sounds are produced by a "source which has well defined acoustic constraints for the listener [to identify]" (Pisoni & Sawusch, 1975:18). However, Stevens (1972) has found that in some areas of the vocal tract small changes in the position of the tongue resulted in large variations in the formant frequencies of the sounds; and in other areas of vocal tract, large changes in the tongue's constriction produced small variations in the frequencies of the sounds' formants. Stevens argues that the acoustic properties of a phonetic feature, such as velar or palatal for instance, could be generated by "the vocal tract without requiring a very precise positioning or manipulating of the articulatory apparatus" (18). Thus, trying to find exact correlates between acoustic data and linguistic segments needs to be reassessed.

Another major problem in speech perception is to determine the units of perceptual analysis. Different researchers have argued for the primacy of the phonetic feature (Stevens, 2005, 2002), the phoneme (Liberman and Mattingly, 1985; Pisoni and Luce, 1987; Moore, 2003), the syllable (Mehler, 1981), and the word as the basic unit in perception. However, it is now accepted that speech is processed by the linguistic system at different levels and the best perceptual unit depends on the level of perceptual analysis under consideration. Pisoni and Sawusch (1975) point out that to argue that there is a basic perceptual unit is to acknowledge that

language exists primarily in one form over the other without taking into account the fact that human language exists in many forms, some of which are not accessible to conscious inspection. For instance, we normally do not listen to phonetic details when perceiving speech since we recognize words and not sounds, although it has been shown that the first 250 msec of a word are usually sufficient for word recognition (Remez, 2002).

Consider for instance the distinction between phonemes and syllables. The fact that phonemes are better recognized in syllables than standing alone (Strange, 1995), and the abstract nature of the phoneme, have led some authors to argue that the syllable is a more basic perceptual unit (see Liberman and Mattingly, 1985 for a discussion of this topic). However, Studdert-Kennedy (1975) points out that phonemes and syllables work in a symbiotic relation: "the syllable serves as the carrier of phonetic information, and its function is to create contrast for phonetic perception so that the listener can detect sound segments and features for the subsequent operations of the perceptual system" (19). Some authors have assumed that by choosing the syllable as a basic unit the invariance problem could be solved. But there are still some problems with this approach; first, there are difficulties in defining the syllable acoustically, and second, syllables are not indivisible units; "they have a complex internal structure" that at some point in the perceptual process has to be recovered. In addition, the acoustic cues for syllables are affected by context in the same fashion as the cues for phonemes (Pisoni and Sawusch, 1975, 19).

In this study, we assume, following Fodor, Bever, and Garret (1974) and Pisoni and Sawusch (1975), among others, that perceptual information is distributed across at least the whole syllable, and that this is necessary to derive information for an accurate phonetic interpretation. This approach does not necessarily adopt a minimal unit of perception but it

acknowledges the type of information that is required in order to perceive accurately phonetic information.

1.2.3.2. The nature of perceptual cues

In this study, the term "cue" is used to mean information in the acoustic signal that allows the listener to obtain phonetic or phonological information. For instance, Liberman, Dellatre & Cooper (1952) found that a short burst of energy can be sufficient to cue a stop consonant, and that the frequency of this burst relative to the frequency of the formants of the following vowel allows the listener to differentiate between various stop places of articulation, e.g. between /p/, /t/, and /k/. However Mayo (2000) points out that cues are not static or binary. That is, they are of a more gradient nature, since articulators have also different configurations besides just open closed (in the case of mouth opening for instance). Thus Mayo (2000: 7) argues that rather than being a discrete aspect of the acoustic signal, a cue is an acoustic variable with a potential function. Thus, a short burst of energy is a potential cue to the presence of a stop consonant.

The relation between cues and the physical speech signal is not one-to-one. It has been demonstrated that for instance a rapid closing and opening of the vocal tract (specially at the beginning or the end of the articulation) has acoustic consequences, such as various rising and falling formant transitions, a period of significantly reduced sound intensity, and then a second, acoustically different set of formant transitions among others, which show a many-to-one relation. Furthermore, research has shown that an acoustic signal can cue more than one type of information. For instance Whalen and Levitt (1995) show that all else being equal, Fundamental Frequency (F0) cues vowel height and obstruent voicing, among other things (such as pitch), which suggests a one-to-many relationship. That is to say that one acoustic signal, the F0, is related to at least two phonetic cues: vowel height and obstruent voicing.

Mayo (2000: 9) also points out that even very small changes in the rate of speech of a speaker, the register, or the loudness, among other things, will have an effect on the speech stream. Therefore, cues do not only vary between speakers, "but also between two productions of the same utterance by the same speaker." (8). The author concludes that while a "speech percept must be the same each time it is heard, the aspects of the signal that cue that percept will never be invariable in a straightforward way" (8).

In addition, it has been shown that there are trading relations and cue equivalences. Fitch, Halwes, Erickson & Liberman (1980), in a study of the relative influence of two cues to the presence of post-fricative stop consonants, found cues including i) duration of silence between the fricative noise and the onset of the stop, and ii) presence or absence of vocalic onset transitions appropriate for bilabial stop closure. The authors found that the two cues interacted in what has been called a *trading relation*. This means that when there are two cues, they interact and both `carry some of the load' of creating a percept, therefore less of only one of the cues is needed. Mayo points out that in *trading relations* "the two cues are able to `trade' in the amount they are needed perceptually" (14). In the same study, Fitch et al. (1980) also showed the *perceptual equivalence* of cues to a contrast. They found that different cues could map to the same information. Also, Best, Morrongiello and Robson (1981), and Haywood (2000) point out that speech is highly redundant. That is to say that one cue rarely signals only one phonetic contrast. This redundancy is beneficial for listeners since multiple cues serve to enhance the contrast between different segments.

Dorman, Studdert-Kennedy & Raphael (1977), in a study to determine the role of stop bursts and formant transitions in the perception of place of articulation of voiced stops, found that the perceptual weight listeners gave to the burst and transition information depended upon

the place of articulation of the consonant, the quality of the following vowel, and the speaker. Similarly, Whalen (1991) found that the perceptual weight given to cues to fricative place of articulation depends on the identity of both the fricative and the following vowel. Furthermore, research has shown that cues develop as children's linguistic systems mature (see Greenlee 1980, Krause 1982, Morrongiello, Robson, Best & Clifton 1984, Nittrouer & Studdert-Kennedy 1987, Nittrouer 1992, Ohde & Haley 1997).

In sum, Mayo (2000) points out that the relative weighting of acoustic cues is "dictated both by the demands of the perceptual system to make use of the most informative cue in any given context, and by demands external to the perceptual system itself". (17). Moreover, the cue weighting changes with time as the children develop their linguistic system.

1.2.4. Models of Speech Perception

Models of speech perception have been guided by two main principles: a taxonomic principle and a dynamic principle. The models based on the taxonomic principle consider that linguistic structures, especially phonological structure, could be derived directly from segmenting the signal. Thus, according to this approach, the phonological structure could be described independently from other grammatical structures such as the syntax or semantics. Under the dynamic principle, the perceptual process is considered as an active process in which the acoustic information is used to test hypotheses about the structure of sentences. Hence, during the perceptual process, a listener is guided by phonological principles to determine the phonetic properties of the acoustic signal. What the listener 'hears' is in fact internally generated by his/her internal rules (at either the phonemic, lexical, or syntactic levels) for the hypothesized structure of the sentences. This approach considers that the interpretation that the listener makes of the speech signal is not directly determined by the physical properties of the signal since our

conscious experience of language is, almost always, driven by the semantic intention and not the form (signal). Therefore, it is difficult to be aware of small variations in the signal. In this way, what is perceived in the end depends on the listener's knowledge of his/her language as well as extralinguistic factors that determine what the listener expects to hear (Chomsky and Halle, 1968).

The following section briefly reviews several of the current theories of speech perception that are relevant to this work. In addition, it presents the different theoretical assumptions that compose the theoretical framework adopted in this study.

Motor Theory of Speech Perception

The Motor Theory was proposed by Liberman, Cooper, Shankweiler, Studdert-Kennedy (1967) and Liberman and Mattingly (1985). One of the basic assumptions of this theory is that "speech is perceived by processes that are also involved in its production [motor patterns]" (Liberman et al., 432). This view is based on the fact that the perceiver is also a speaker and it would be uneconomical to have two separate systems to process language production and perception.

The main support for the Motor theory comes from experiments in which listeners had to discriminate categorically the sound they heard -categorical perception tests- from synthetic reproductions of stop consonants and steady-state vowels. In those experiments the listeners had to identify individual phonemes or syllables. The authors found that as they continuously varied the formant transitions in the signal, listeners categorically heard the sounds *b*, *d*, or *g*. They concluded that the formant transitions are the cues for the perception of /d/ as opposed to /b/ or /g/. They also argue that the acoustic patterns occur quite generally for consonant sounds.

However, research on natural speech has shown that it is rarely possible to find invariant acoustic cues corresponding to a consonant. For example, for any given stop consonant, the slope of the F2 transition into the F2 of the vowel naturally depends on the location of the F2, and thus can take on as many different slopes as there are vowels, while being consistently heard as the same stop consonant.

For steady-state vowels, the frequency of the formants presents less variability, but vowels are rarely found in steady state in normal speech. Generally, vowels are articulated between consonants. Liberman et al. (1967), observed that in connected speech, vowels, like any speech sound, show restructuring, that is to say that "the acoustic signal at no point corresponds to a vowel, but rather shows, at any instant, the merged influences of the preceding and the following consonant" (p. 433). Thus, a section of the acoustic signal carries information about several neighboring phonemes, which makes the relation between phoneme and acoustic signal very complex. The authors have argued that the complexity of the relationship between a phoneme and the acoustic signal implies that there exists a special decoder to handle the necessary complex computations, which map multiple cues onto one percept.

Liberman et al. (1967) refer to those phonemes whose acoustic patterns present considerable context-dependent restructuring, such as stops, as "encoded", and those phonemes for which there is less restructuring, such as vowels, as "unencoded". They propose that the perception of encoded phonemes is different from the perception of unencoded ones. However, the lack of consistent supporting data is one of the major arguments against the Motor Theory: most of the data present a "lack of invariance" between the acoustic signal and phonetic perception.

Some researchers argue that categorical perception results are due to coding processes in short term memory and not to differences between various classes of speech sounds (Pisoni, 1973; Pisoni and Sawusch, 1975). A way to overcome the problem of the lack of invariance was to propose that the same perceptual response to different acoustic patterns results because the pattern would be produced by the same articulation or, in cases where the signal is not present, the underlying motor commands to the articulators (acoustic gestures). Analogously, different perceptual responses to similar acoustic patterns result from different articulations or different underlying motor commands. Liberman and Mattingly (1985) argue that " the objects of speech perception are the intended phonetic gestures of the speaker, represented in the brain as invariant motor commands that call for movements of the articulators through certain linguistically significant configurations " (3). Thus, what the listener perceives are the articulatory gestures that the speaker is intending to make when producing an utterance. However, the gestures are not reflected in the acoustic signal or in the observable articulatory units. According to the authors, the perception of the articulatory gestures is done in a specialized speech mode whose main function is to translate the signal into articulatory gestures automatically.

The Motor Theory accounts for some speech perception phenomena such as the variable relationship between the acoustic signal and perceived speech patterns, categorical perception, and a perceptual mode specific to speech. Nevertheless, the model is incomplete in that it fails to specify how the conversion from the acoustic signal into perceptual features occurs. In addition, it does not determine the level of perceptual analysis at which articulatory knowledge is employed in recognition. Those gaps in the model have led some authors to claim that the motivation for this approach is merely intuitive, since the message has to be "encoded' in the signal (Pisoni and Sawusch, 1975), or that the model is more like a philosophy than a theory (Klatt, 1989).

This study adopts some of the theoretical aspects brought forward by the Motor Theory. It adopts the claim that there is a speech mode of perception, and that it is better engaged by informing listeners that what they will be hearing corresponds to speech, and specifically that they are second language utterances. This will allow them to centre their attention on phonetic features and may prevent them from missing some features while activating their speech mode (see Morrison, 2006, for further discussion).

1.2.5. Speech perception in a second language

The perception of speech in a second language has, in addition to the factors that affect first language perception, some additional considerations as noted on section 1.2.2. above.

1.2.5.1 Analysis-by-Synthesis

The Analysis-by-Synthesis model is based in the assumption that there is a close connection between speech production and speech perception processes, and there are components and processes shared by both. Marslen-Wilson (1985, 2001) points out that in this model, the perceptual process starts with a preliminary analysis that consists in the peripheral processing of the speech signal to identify patterns. In cases where phonetic features are not strongly context-dependent, the auditory pattern will provide a relatively direct mapping of those features during the preliminary analysis. The outcome of the preliminary analysis is a rough matrix of phonetic segments and features, which is then passed on to the control system. Hence, the recognition of some features is assumed to be done by relatively direct operations on the acoustic information output from peripheral analysis.

Other approaches to the study of speech perception have adopted the aims and methods of information processing models used in the study of visual and auditory perception. Those

models are based on the assumption that perception is a sequence of events that is hierarchically organized, and it involves structures for storage and process for the transformation of information over time.

1.2.5.2 Perceptual Assimilation Model

Best (1994) and Best, McRoberts, and Sithole, (1988) propose the Perceptual Assimilation Model. According to the authors, the difficulty that a listener has with the discrimination of nonnative contrasts can be predicted from the relationship between the L1 and L2 phonologies. For example, if each element of the nonnative contrast is similar to two distinct elements of the L1, then discrimination of the forms should be very good. But if the two elements of the non-native contrast are similar to one element (phoneme) of the L1, then the discrimination abilities depend on the degree of similarity of each member of the nonnative contrast and the native phoneme. In addition, if the characteristics of the nonnative contrast are quite different from any native contrast, the nonnative items may be easily discriminated.

1.2.6. Non-linguistic factors in speech perception: Information Theory

It is generally accepted that no two spoken instances of a word are identical, even in cases when they are pronounced by the same speaker in the same conversation. Some of the factors that influence the variability are not acoustic in nature: psycholinguistic models of the speaker's speech production include what the listener knows and perceives (Bolinger, 1963, 1981; Chafe, 1974; Lindblom, 1990). Lindblom, (1990) points out that the speaker estimates the contribution of signal-complementary processes during the course of an utterance and "adjusts the articulatory accuracy to accommodate listener's residual dependence on the speech signal" (89). The author argues that when other sources of information are available, speakers "hypoarticulate", but when

there are no sources other than the sound, the speaker "hyperarticulates". This variability can be produced by conversational factors such as speech rate, stylistic register, and emphasis (new information vs. old information) among others. Furthermore, variability can be produced on first and subsequent articulations of the same word by the same speaker based on what listeners know about old or relevant information (see Bard, Anderson, Sotillo, Aylett, Doherty-Sneddon, and Newlands 2000, for discussion).

1.3. Research questions and hypotheses

This study examines second language perception, comparing the linguistic cues used by English speakers learning Spanish and by native Spanish speakers. It seeks to determine the interactions among cues for both native and non-native speakers.

The specific research questions to be addressed in this study are as follows:

1) To what extent can the role of individual acoustic cues in the perceptual process be determined?
2) To what extent do native speakers and non-native speakers use the same cues in the perception of Spanish consonant clusters, including the epenthetic vowel?
3) What interactions occur among the linguistic cues?
4) How does cue interaction develop as learners progress through the language acquisition process?

The motivation behind research question (1) is the theory that native speakers develop perceptual phonemic categories for vowels and consonants in their L1, and that they can use their intuitions to decide whether a token falls into the category of a given vowel or consonant

(Liberman, 1970; Hoopingarner, 2004). This study explores whether the perceptual cues of listeners are determined exclusively in terms of their L1 or if those cues undergo a rearrangement during the L2 learning process, in this case with respect to their perceptual interpretation of epenthetic vowels.

The basis of research question (2) is that very early in life children's perceptual systems favour the contrasts present in the language they are exposed to, and demote those that are not present. However, research has shown that L2 learners can discriminate L2 contrasts after undergoing some training, often using laboratory training techniques such as discrimination tasks, which are assumed to modify the attentional processes that are susceptible to realignment (Aslin and Pisoni, 1980; Pisoni et al.; 1994). Research question (2) addresses the issue of the perception of nonnative features through the L2 learning process with no explicit laboratory training. In this case we focus on the learner's perception of syllables containing EVs, in the absence of knowledge of their existence.

Research question (3) examines the very nature of the perceptual process. While several studies have looked at the correspondence between features in the speech signal and the category perceived, little research has been done on the interaction among linguistic cues in L2 perception and the weight that each one exerts during the interaction.

Although it is accepted that speech perception affects the phonological system and *vice versa*, it is not clear how the phonological system is re-shaped during the language acquisition process, or in an interlanguage. Major (1987) claims that interlanguage phonology behaves in a similar fashion to adult language phonology in terms of universal hierarchical relationships involving markedness. However, transfer from the first language (L1) interacts with these factors.

Research question (4) builds on the assumption that interlanguages are continuously evolving grammars that change as the learner moves through the learning process (Selinker, 1972; Tarone, 1988). Thus the perceptual process, instead of being based on a set mechanism, should be a changing mechanism that evolves along with the development of language proficiency. Thus, we foresee a gradual, unconscious sensitivity to the presence/absence of EVs.

The comparison of native and non-native speaker groups will indicate to what extent L2 learners at any of the proficiency levels (beginners, intermediate or advanced) use the same cues that native speakers employ when perceiving consonant clusters.

Finally, a comparison of production and perception will shed light on differences between articulatorily and perceptually motivated epenthetic vowels.

Each experimental study will propose particular assumptions and predictions, but the general hypotheses of the dissertation as a whole are as follows.

1. In perceptual identification and discrimination tasks the Spanish native speakers will show more accuracy than L2 learners because of their greater sensitivity to the information carried by EVs.
2. In both kinds of tasks, as well as in goodness ratings, L2 learners will present more variability due to different levels of language proficiency because sensitivity to EVs, which can improve perception, develops over time and with greater exposure to the L2.
3. In the perceptual tasks, advanced L2 learners will have native-like accuracy based on longer experience with the L2.

Chapter 2
Speech Production: Vowel Epenthesis

This chapter examines the phenomenon of vocalic epenthesis from the point of view of speech production. The sections are organized as follows. Section 1 presents a review of relevant studies on the phonetic and phonological aspects of the epenthesis in Spanish. Sections 2 and 3 present the results of a speech production experiment on the pronunciation of Spanish consonant clusters. Section 4 discusses the findings of this experiment in light of previous studies, and outlines the principles of a new type of statistical analysis method, *random effects modeling*, that is currently coming into use for this type of data.

2.1. Epenthesis

Vowel epenthesis is a widespread phenomenon. It has been documented in many languages: Dutch, Scots Gaelic and Hocank (Hall, 2004), Navajo (McDonough, 1996), Turkish, (Asci, 1996), and Spanish (Malmberg, 1965; Navarro, 1963; Quilis, 1981), among others[2]. Furthermore, cases of epenthesis have also been reported in the process of L1 and L2 acquisition, which seems to suggest that it is employed as a mechanism to resolve complex clusters and thus to facilitate learning.

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[2] For a comprehensive list of the languages that exhibit epenthesis and vowel harmony see Finley (2007).

2.1.1 Epenthesis in L1 acquisition

Research has shown that epenthesis occurs as a process in first language acquisition in languages such Dutch (Jongstra, 2003a, 2003b; Taelman, 2005), English, (Stemberger, 1993; Fee, 1996), Polish (Lukazewicz, 2006), and Portuguese (Freitas, 2003), among others. For instance, in the case of L1 acquisition of European Portuguese, in a study of the acquisition of consonant clusters of the type Consonant + Consonant (CC) by 7 monolingual children, Freitas (2003) found that in an initial stage of acquisition children used epenthesis as a strategy to "repair" the clusters such as those presented in (2.1).

(2.1) Vowel epenthesis (EV) in the acquisition of European Portuguese branching Onsets. After Freitas (2003:35).

Non-EV pronunciation		EV-pronunciation	
cobra	['kɔbɾɐ]	['kɔbiɾɐ]	'snake'
livro	['livɾu]	['liviɾu]	'book'
pedra	['pɛdɾɐ]	['pɛdiɾɐ]	'rock'

Freitas argues that children in the first stage of acquisition do not recognize or represent the cluster as two segments in the syllable onset, but rather as a complex onset such as in geminate (long) consonants – but, crucially, not as branching syllabic onsets. Thus, the epenthesis results from the child's undeveloped syllabic structure, which does not allow for an onset category that branches into two different consonants. Instead, the child breaks up the CCV structure into CVCV, which does conform to the pattern for the early stage of syllable structure. At a later stage, when the phonology of the children develops, the correct branching onsets are acquired and CCV, without an epenthetic vowel, becomes a possible representation and a possible output.

It is generally accepted that the EV does not have a phonological representation in either children's or adults' grammar. That is to say that it does not have mental representation but only a phonetic/acoustic realization created at the production stage. Freitas suggests that the choice of epenthesis is based on the Portuguese preference for Consonant +Vowel (CV) syllabic structure, which is the predominant pattern in the language. Similar results were found by Ribas (2003) in her study of 134 Brazilian Portuguese speakers.

It has been argued that EV is the result of the higher frequency of the CV structure in the input. It has been hypothesized that typical structures, such as CV syllables, occur most frequently in languages around the world because they are unmarked in language typology (Jakobson 1941/1968; Stampe, 1969). Nevertheless, in languages such as English and Dutch, where consonant clusters are highly common, children also use epenthesis at some point during the process of language acquisition. Thus it is generally accepted that the CV structure is a universal stage during the acquisition of clusters and not just a stage in languages where the CV syllables are more common (Jongstra, 2003a, 2003b; Gnanadesikan, 2004).

2.1.2. Epenthesis in L2 acquisition

In the acquisition of an L2, it has been noted that learners repair the syllabic sequences that are not allowed in their L1 either by deleting a consonant or by epenthesizing a vowel (Archibald, 2003; Weinberger, 1988). Epenthesis is a recurrent phenomenon in L2 interlanguage, and it seems to be affected by particular constraints that differ from L1 and L2 phonotactics. For instance, in Korean-English interlanguage and loanwords, Korean speakers learning English produce epenthesis in words with a sequence of vowel + consonant Coda (voiced or voiceless) [VC#]. For instance, words like 'mud' [mʌd] and 'peak' [pik] are pronounced like [mʌdɨ] and

[pikɨ]. Tak's (1996) study of Korean-English interlanguage and loanwords shows that Korean speakers learning English produce three types of epenthesis represented by the surfacing of three vowels: [u]-epenthesis appears between a bilabial and another consonant ($C_{bilabial}$VC#), [I]-epenthesis between an alveolar or a velar and another consonant ($C_{alveolar/velar}$VC#), and [i]-epenthesis between a palatal and another consonant ($C_{palatal}$VC#). The author argues that the quality of the epenthesized vowel depends on the place of articulation of the preceding consonant, which in turn spreads the place feature to the vowel. As in other cases of epenthesis, in Korean-English interlanguage the epenthetic vowel does not have a phonological representation in the speakers' L2. It is only present in the surface (phonetic) representation.

Cases of epenthesis have been reported in a variety of other interlanguages: in English L2 by Japanese speakers (Dupoux, Kahehi, Hirose, Pallier, and Mehler, 1999), as well as Bantu and Nilotic language-speaking learners of Swahili as an L2 (Musau, 1999). Cases of epenthesis in an L2 cannot be attributed exclusively to L1 transfer. For instance, in a study of the L2 English productions of Iraqi Arabic and Egyptian Arabic speakers, Broselow (1992) reports cases of epenthesis in the speakers' interlanguage, as illustrated in (2.2). She claims that the cases of epenthesis are not just production errors due to L1 transfer, or error patterns motivated by the grammar of either L1 or L2; instead, the author claims that the cases of epenthesis are consistent with principles of universal markedness constraints that are common to speakers of many different L1s.

Across-the-board epenthesis is illustrated by the pronunciation of English words by Iraqi Arabic speakers. As shown in (2.2), an epenthetic vowel appears before all initial biconsonantal clusters:

(2.2) Iraqi Arabic (after Broselow 1987, 1993)

		EV-pronunciation	
a.	study	[istadi]	'study'
b.	snow	[isnoo]	‘snow’
	plane	[ibleen]	'plane'
c.	street	[sitrit]	'street'
	splash	[siblas]	'splash'

Such an epenthesis pattern may reflect a general preference in Iraqi Arabic for ensuring that segments adjacent in the input (L1) are adjacent at output. That is to say that in cases such as 2.2a and 2.2b the clusters [st], [sn] and [pl] are not broken by the epenthetic vowel to maintain the original structure from L1. However, when there are three contiguous consonants such as in 2.2c ([str] and [spl]), an epenthetic vowel occurs after the /s/, breaking up the triconsonantal clusters in a way that is not consistent with phonetic strategies found in the L1.

2.2. The analysis of vowel epenthesis

A vowel-like segment has long been reported in consonant clusters produced by adult native speakers of Spanish. This vocalic segment is not recognized as present in the phonological representation of the consonant cluster, and it does not appear in the orthography or in broad transcriptions of Spanish speech. Moreover, it is not consistently present in all tokens of a given cluster, even in the same word within the same speaker. It has been referred to in the literature by various names: Svarabhakti vowel or ‘vocal esvarabática’ (Bradley and Schmeiser, 2003; Bradley, 2004), vocalic element or ‘elemento vocálico’ (Lenz, 1892; Navarro Tomás, 1918; Gili Gaya, 1921; Massone, 1988; Blecua, 2001), parasitic vocalic element or ‘elemento vocálico parasítico’ (Malmberg, 1965; Quilis, 1970), intrusive vowel (Hall, 2003), epenthetic or excrescent schwa (Davidson, 2003), schwa-like vocalic element (Gafos, 2002), and epenthetic vowel (Ramirez, 2002, 2006, Colantoni and Steele 2005a, b). The term adopted through this study is that of epenthetic vowel (EV).

Vowel epenthesis has been accounted for by two main approaches. On the one hand, it is hypothesized that the EV plays a role in the perceptibility of the consonant cluster in which it occurs (Bradley, 2002; Schmeiser, 2006). This suggests, but does not necessarily imply, that the EV may be produced by the speaker in order to achieve greater clarity of pronunciation as an aid to the listener. On the other hand, epenthesis may also be seen as a way to resolve a complex, marked structure during language production (Hall, 2003, and 2004).

Nonetheless, some studies argue against epenthesis both as a phonemic and a phonetic process. For instance, Sato (1984), in an investigation of syllable structure in the interlanguage of two Vietnamese learners of English, found negligible use of vowel epenthesis as a syllable modification strategy. The author interprets these results as evidence against the hypothesized universal preference for the open CV syllable and the hypothesized prevalence of epenthesis as a syllable modification strategy in interlanguage speech. Of course, one might well ask whether data from two speakers of a single language under one set of testing conditions should be considered conclusive in choosing among theoretical approaches to the EV.

The epenthetic vowel is generally reported in cases of co-occurrence of the stop consonant and the alveolar liquid flap /r/ (either in the C + /r/ order or /r/ + C) (Malmberg, 1965; Quilis, 1981). This observation stems from dialect data that show the presence of an epenthetic vowel across different Spanish dialects, as in (2.3).

(2.3)	Spelling	Gloss	Non-EV pronunciation	EV pronunciation
	tigre	‘tiger’	[tíɣɾe]	[tíɣᵊɾe]
	chacara	‘small farm’	[ʧákɾa]	[ʧákᵊɾa]
	grupa	‘croup (of horse)’	[gɾúpa]	[gᵊɾúpa]
	tarabilla	‘small clasp’	[tɾaβíʎa]	[tᵊɾaβíʎa]
	crónica	‘chronic’	[kɾónika]	[kᵊɾónika]

(2.4)	puerta	'door'	[pwerta]	[pwerəta]
	resbalar	'to slide'	[rɛsβalar]	[rɛsəβalar]

Malmberg presents some examples showing the presence of a vocalic element in clusters formed by a stop consonant followed by the lateral alveolar /l/, as in (2.5), although he does not include these examples in his discussion, and Quilis does not consider them in his study.

(2.5)	Spelling	Gloss	Non-EV pronunciation	EV pronunciation
	Inglaterra	'England'	[iŋglatera]	[iŋgəlatera]
	Indulgencia	'indulgence'	[indulxensia]	[induləxensia]

Malmberg (1965) also reports the EV in clusters formed across syllabic boundaries (heterosyllabic clusters) such as in the case of (2.4). In this study, the focus is on the EV that occurs in clusters containing a liquid consonant in syllable internal (tautosyllabic) position. It does not consider heterosyllabic clusters in order to control for the effect of syllable boundaries.

2.2.1. The segments

Tautosyllabic clusters in Spanish are formed by the concurrence of an obstruent consonant and a liquid. The obstruent consonant can be a voiceless labiodental fricative /f/, a voiced or voiceless stop (/p/, /t/, /k/, /b/, /d/, /g/), or a voiced approximant (/β/, /δ/, /γ/)[3]. The liquids are the voiced, apico-alveolar simple vibrant (flap) [ɾ] or the voiced apico-alveolar lateral [l].

[3] The voiced approximants genreally occur in intervocalic contexts, but in some dialects (i.e., Rioplatense) they also appear after pause (word initial position).

Acoustically, Blecua (2001) points out that the flap has three variants (allophones), which depend on different degrees of relaxation in the articulators. The least relaxed variant is an occlusive allophone, in which the tip of the tongue touches the alveolar ridge [ɾ], whereas a more relaxed variant is the approximant allophone, in which the tongue comes close to, but does not make contact with the alveolar ridge, and the air can flow between the two articulators [ɹ]. The greatest degree of relaxation of the articulators would produce the elision of the flap [Ø], which occurs at a higher rate in casual/fast speech. As for the lateral, acoustically, there is only one place of articulation: it is apico-alveolar, where the airstream is allowed to escape along either or both sides of the tongue. It is similar to the English [l]; however in the Spanish [l] the tongue body is significantly elevated in the oral cavity, close to the hard palate, thus making it more palatalized. The Spanish lateral does not present 'dark' or 'light' variants of /l/ the way English does in cases such as *peel* and *listen* respectively.

As for the consonants and their interaction, it has been shown that Spanish voiced stops (/b/, /d/, /g/) generally undergo a process of fricativization or spirantization surfacing as /β/, /δ/, and /γ/, respectively, when they occur in intervocalic position.

2.6.

/d/ → /ð/

a.	[káða]	*cada* 'each'.
b.	[bérðe]	*verde* 'green'
c.	[káldo]	caldo 'broth'

In (2.6a) the stop /d/ becomes the spirantized /ð/ in intervocalic position. The same process occurs in (2.6b) after the rhotic (/r/). However, in (2.6c) the spirantization is blocked by the lateral. Such a discrepancy in the behavior of the liquids is attributed to the Obligatory Contour Principle (OCP), which disallows contiguous segments in the same syllable from sharing the same place feature (in this case the coronal feature). Thus, (c) would be evidence that /r/ and /l/ differ in the place of constriction. (Harris, 1983; Mascaró, 1984; Hualde, 1989; Lipski, 1993). This observation holds true for the dialects considered in this study.

The difference between the liquids is not only articulatory in nature. Rice (2005) points out that they differ in sonority (the rhotic /r/ is more sonorant than the lateral /l/), and it is considered a marked relationship. Rice (2005) points out that there is not a fixed (universal) relationship; instead it is determined in a language-specific manner. (For further discussion on the dissimilarity of /l/ and /r/ see Frigeni, 2009). Although epenthesis is not a good test for determining the markedness relationship of liquids in Spanish, it provides information that has to be taken into account when determining their markedness ranking. It can provide information related to perceptual salience in consonant clusters as we will discuss in the next section.

The distribution of possible consonant clusters in tautosyllabic position is illustrated in Table 2.1.

Table 2.1.

Distribution of elements in tautosyllabic consonant clusters

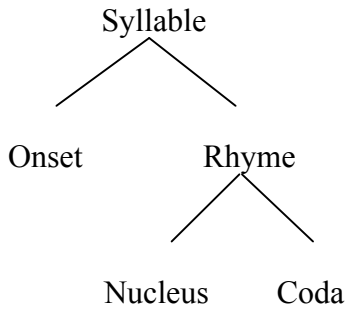
C1		**C2**	
Manner of Articulation		Alveolar flap [ɾ]	Lateral [l]
Voiceless fricative	[f]		
Voiceless occlusive	[p], [t], [k]		
Voiced occlusive	[b], [d], [g]		*
Voiced approximant (spirantized)	[β], [δ], [γ]		*

* the /dl/ and /δl/ clusters are not present in Spanish

2.2.2. Syllable structure

The syllable patterns in Spanish include V, VC, CV, CVC, CCV, and CCVC. In representing the syllabic structure, we adopt the basic structure shown in 2.7.

2.7. Syllabic structure skeleton

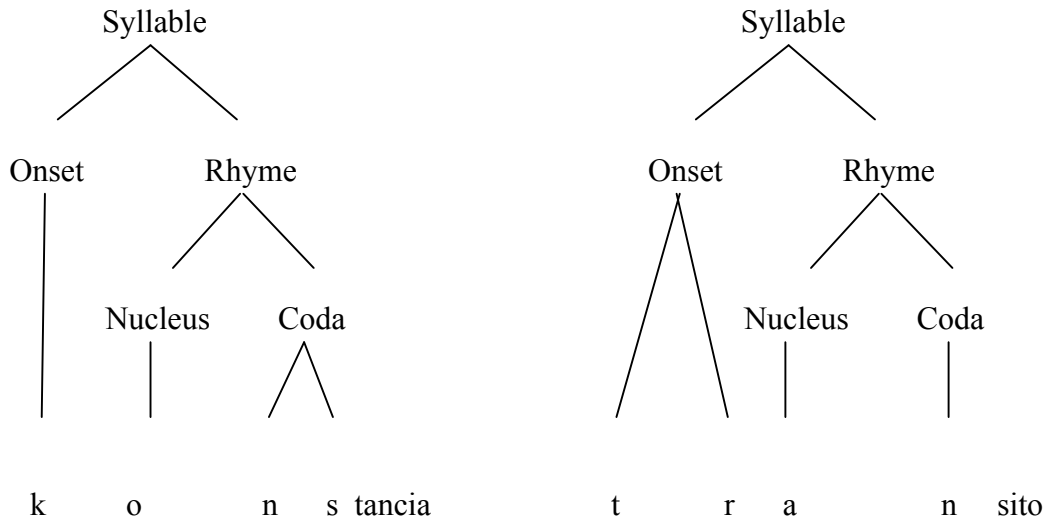


As shown in (2.8a) and (2.8b), the structure of the Spanish syllable allows the syllabic nucleus to be occupied exclusively by a vowel. As for the Codas, the constraints on the structure allow very few cases of complex codas, and it allows only /s/ in the second position of the coda, which is considered to be an extrametrical slot (see Harris 1983). A representation of a complex coda is

presented in (2.8a) for the word *constancia* 'constancy' which is syllabified as cons.tan.cia. On the other hand, complex onsets are more varied in Spanish, as shown in (2.8b). *transito* 'transit' which is syllabifies as tran.si.to. The syllabic structure is presented in 2.8b.

(2.8a) complex codas: representation of the syllable 'cons' in the word *constancia* 'constancy'.

(2.8b) complex onsets: representation of the syllable 'tran' in the word *transito* 'transit

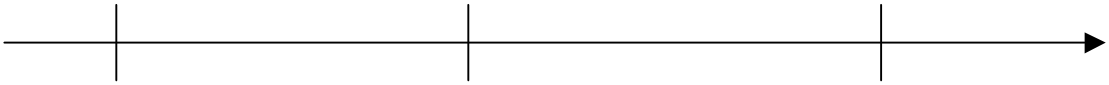


The sequences of segments that can occur in a syllable are constrained by universal principles. Sievers (1881) put forward the Sonority Sequencing Principle (SSP)[4] that observes that the sonority peak of a syllable is at its centre, while sonority decreases towards the periphery. Since the most sonorous elements are the vowels, they usually occupy the syllabic nucleus. However, there is cross-linguistic variation as to which segment combinations are

[4] Also Jespersen (1904), Sausure (1914) and Grammont (1933) noted the SSP. More recently Hooper (1976), Kiparsky (1979), Steriade (1982), Selkirk (1982), and Clements (1990) have attempted to provide formal characterizations of the SSP (Morelli, 1999).

allowed in onset or coda positions[5]. Steriade (1982) proposed the Minimum Sonority Distance Principle (see also Harris 1983), according to which the possibilities for combining segments are based on how close they are in sonority relative to one another. Steriade points out that the permissible distance in sonority is language specific, with some languages allowing segments closer in sonority to cluster while others require a greater sonority difference. Based on Steriade (1982), Archibald (2003: 150) represents the markedness relation in a continuum scale in (5).

(5) Markedness:



Obstruent + V Obstruent + Sonorant + V Obstruent + Obstruent + V

Viewing markedness as an implicational relationship, Archibald (2003) points out that the obstruent clusters are the most marked since their occurrence in a language implies the occurrence of both sonorant clusters and singleton onsets. In the continuum, the singleton onsets are the least marked because their presence does not imply the occurrence of either type of cluster. From a perceptual point of view, Wright (1996) argues that less marked sequences (e.g., [pa]) are easier for a listener to recover than more marked ones such as [sta]. That is to say that

[5] Some languages, such as English, allow consonantal segments to occupy the nucleic position. For instance /l/ can be a syllable nucleus in cases such as [litl] 'little' in which the second /l/ is syllabified.

perceptually, the more contrast between the adjacent segments, the more perceptually salient they are.[6]

In L2 acquisition, learners repair the syllabic sequences that are not allowed in their L1 either by deleting a consonant or by epenthesizing a vowel (Archibald, 2003, Weinberger, 1988). For instance, Abrahamson (1999), in a study of Spanish-speaking learners of English and Swedish, found that speakers repair the more marked sCC onsets more often than the sC onsets, and had a larger proportion of errors in clusters that violated the SSP. In a similar study, Carlisle (1991) examined Spanish learners of English and found that marked onsets such as [st] presented higher rates of error than less marked ones such as [sl]. The author points out that vowel epenthesis before the English [sC] cluster is conditioned by the environment in such a way that "epenthesis occurs more significantly after consonants than after vowels" (86).

To sum up, epenthesis is present in both L1 and L2 acquisition with similar trends: namely, it is a universal developmental stage, but it does not maintain phonological status in either one, but rather maintains only a phonetic manifestation (sound). As presented above, the case of epenthesis in Spanish is not clear-cut. It has been hypothesized that the motivation for the epenthesis could be a repair strategy to dissolve marked syllabic sequences, or merely the articulatory byproduct of the articulation of two concurrent (usually heterorganic) consonants.

Another type, and probably the most well known, of epenthesis in Spanish is that before sC clusters. Diachronically, a vowel appeared in Vulgar Latin where there had been none in Classical Latin. An *i* later to become an *e* appeared in cases such as 2.9.

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[6] Morelli (1999) argues that the sC consonant clusters (obstruent clusters) are not constrained by principles of sonority but by constraints on the Place of articulation of the segments.

(2.9) Epenthesis in the sC clusters in the evolution from Latin to Spanish.

Classical Latin	Vulgar Latin	Spanish	
schola	iscola	escuela	'school'
statuam	istatua	estatua	'statue'
scriptam	iscrita	escrita	'written'

Lathrop (1996) argues that the function of the epenthetic vowel was to "regularize the sound system" (21). He argues that in all the cases in which there was a sC cluster, the *s* was syllable final such as in *pás-tor* 'sheppard', *aus-cul-ta-re* 'to listen'. Thus, the addition of the epenthetic vowel would allow the *s* in cases such as *schola* to become syllable-final position *is-co-la.*

In a synchronic analysis within the Generative framework, Harris (1969) argues that the epenthetic vowel in sC clusters is extrametrical. That is to say that it has phonetic realization, but it does not have a spot in the metric tier -- no phonological representation -- and therefore it is invisible to changes or processes that occur at the deeper phonological level.

The epenthetic vowel in sC clusters is similar to that occurring in C+ liquid clusters in which they do not have a phonological representation. They only have phonetic realization. However, the latter is not consistent across speakers and words and it is not represented in the orthography.

The following section presents the most relevant studies on the epenthesis in C + liquid clusters from both a phonetic and a phonological point of view, with special attention to cases of epenthesis within syllabic boundaries.

2.3. Previous studies on the epenthetic vowel

Previous studies have analyzed the phenomenon of epenthesis from a phonetic or phonological point of view. The following section presents the more relevant studies from the two approaches and analyzes their contributions.

2.3.1. Phonetic approaches to the epenthetic vowel

Lenz (1892) was the first linguist to report the epenthetic vowel, in his studies of Chilean Spanish. He described the epenthetic as a perfect glottal sound (a vocalic sound with little attenuation in the vocal tract) that is observed in the context of the flap and its neighbouring consonants either in consonant-liquid or liquid-consonant sequences. For example, he found it in cases such as *arte* [aɾᵊ te] 'art', and *trabajar* [tᵊɾaβahar].

Navarro Tomás (1918) points out that the epenthetic occurs when the flap [ɾ] is in contact with either a preceding or following consonant, and that the epenthetic's pitch is analogous to that of the nucleic vowel within the same syllable. He points out that its duration is similar to that of the flap (25-30 ms). In addition, the author notes that the articulation of such an element is spontaneous and unconscious by native speakers of Spanish.

Gili Gaya (1921) also reports cases of the epenthetic vowel between the flap and the contiguous consonant. However, the author points out that the occurrence of the epenthetic is not systematic, claiming that it is a feature more characteristic of the formal style. However, Blecua and Machuca (2000), in their study of the consonant clusters in two sociolectal styles, found that the epenthetic does not occur consistently. The authors argue that the epenthetic is not

fundamental for articulatory purposes and, like Gili Gaya, consider it a stylistic feature of formal speech.

The fact that the EV is described as characteristic of formal speech is consistent with the argument that it appears most often when people are trying their best to make the listener hear the elements of the cluster.

Malmberg (1965) also observes the 'parasitic vowel' (EV) in all consonant clusters formed by /ɾ/ and any other consonant. He identifies the EV as a mid central vowel similar to schwa [ə]. The author claims that this vowel is dialectal in nature and that its motivation is diachronic – a feature maintained in the evolution from Latin to Spanish. He notes the occurrence of a schwa-like vowel in Argentinian, Colombian, and Iberian Spanish, but considers it a widespread phenomenon in all the dialectal varieties of Spanish. He adds that the EV acts to dissolve a consonant cluster that is difficult to pronounce, but he does not provide a ranking of degree of difficulty among clusters. Later studies that have looked at different dialects found it in all dialects studied (Argentinian, Colombian, Iberian Spanish; Schemeiser, 2006).

Quilis (1970) characterizes the epenthetic acoustically as a glottal sound with a mean duration of 32ms and values that range between 8ms and 56ms. Quilis, along with other researchers (Gili Gaya, 1921; Blecua 2001), attributes the variability in duration to the speed of enunciation and to the unconscious nature of this segment. However, the author does not elaborate on why something unconscious should or should not necessarily be variable; neither does he mention the rate of occurrence or non occurrence of the EV or the problems that arise when it is very short (around the 10ms mark) and thus unlikely to be perceptible.

Quilis claims that the variability in EV occurrence is not affected by the type of the first consonant in the cluster or the stress on the syllable. In terms of its acoustic structure, the epenthetic is very similar to a nucleic vowel, but with less intensity. The epenthetic vowel generally presents similar formants to the nucleic vowel: the first two formants are generally present but the third and higher formants may be tenuous, possibly due to low intensity. Although the author does not make it explicit, the assumption is that there seems to be a type of vowel harmony or anticipatory (forward) assimilation.

Nonetheless, the existence of the epenthetic vowel as such is controversial. Massone (1988), in her analysis of the liquids in Spanish, argues that the frequencies of the formants of the vocalic element (epenthetic) are similar to those of the tap, and different from those of the syllable's nucleic vowel. Thus, she claims that the flap in consonant clusters is a vibrant sound that has only one vibration and is formed by a vocalic component plus an occlusion. In other words, she posits that the vocalic element is part of the flap and not an independent segment.

In this study, we assume (along with Quilis, 1970) that the vocalic element between consonants in clusters is an epenthetic vowel. A supporting argument for this interpretation comes from the fact that the EV occurs in clusters with /l/, which is not a tap.

Quilis (1993) measured the formant frequencies of the epenthetic vowel compared to those of the nucleic vowel in Spanish. The author argues that the epenthetic vowels have spectral frequencies but that they have less intensity, as illustrated in Figure 1 below. The graph shows that, overall, the vowel space of the EVs is the same as for full vowels, but more centralized. In terms of height, the EV in the context of high vowels lowers, before the low vowels it rises, and before the central vowels it remains the same. In terms of advancement, the front vowels cause the EV to retract, the back vowels cause it to advance, and the central mid vowel does not alter

its position. However, the author does not mention whether the nucleic vowels were taken from the same tokens as the EVs, or if the EVs were charted against average values for nucleic vowels.

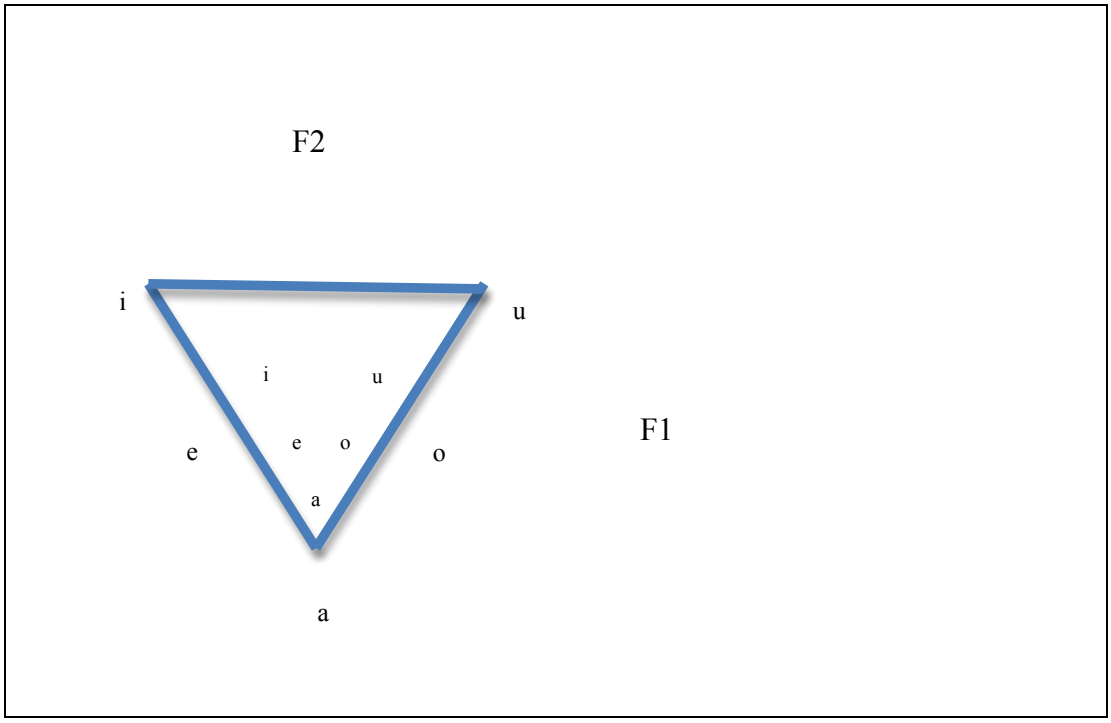

Figure 2.1. Distribution of formant frequencies (formant 1 and 2 (F1) and (F2) of nucleic and epenthetic vowels. (After Quilis, 1993).

In a comprehensive study of the Spanish rhotics, Blecua (2001) argues that the /r/ is a segment composed of an occlusion and a vocalic component (the EV), which occurs between the occlusion part of the /r/ and either the preceding consonant, such as the /t/ in *trabajo* 'work' resulting in [t^{a}rabajo], or in cases such as *puerta* 'door', where the EV occurs after the occlusion of the /r/ and before the following consonant (/t/) producing [puer^{a}ta]. The author claims that when a vowel precedes or follows the /r/, the vocalic element is either non-existent or very difficult to identify. Blecua argues that the vocalic element marks the limits of the occlusion and that it is necessary to ensure the perceptibility of the occlusion (Chap. 5, p 5). That is, the

epenthetic vowel is not really an independent segment but a part of the flap (along the lines of Cerdá, 1968; Massone, 1988; and Celdrán & Rallo, 1995). The /r/, according to Blecua, is an occlusion flanked by vocalic elements (EVs), in such a way that the flap can be represented as 'vocalic element+occlusion+vocalic element'. Furthermore, when the /r/ occurs in an intervocalic context, the vocalic elements (EVs) are indistinguishable from the vowels. They would be distinguishable only in the context of a consonant, such as with the consonant clusters.

One of the main objections to this approach is that it does not explain the high variability in the occurrence of EVs in consonant clusters. According to Blecua's characterization of /r/, we would expect a more consistent occurrence of the EV whenever /r/ occurs in a cluster; however, this is not the case. Furthermore, Blecua's approach assumes that the EV does not have its own independent formants but rather assimilates to context. If this were the case, we would expect that in consonantal contexts EVs should have different formants depending on where the tongue is coming from or going to, but this has not been attested. The main type of variability of the EV is in terms of its length; however, little research has been done in terms of its acoustic characteristics. Such studies would shed light on whether there is significant variability in its vocalic quality or not. In addition, EVs occur in consonant clusters with the lateral /l/ (i.e., /pl/, /bl/, etc), which supports the argument for the independent nature of the EV with respect to the /r/.

Blecua's assumption that EVs are part of the /r/ contrasts with the hypothesis that the EV is an independent element, that is to say, that it is not a part of the rhotic (as adopted by Gili Gaya, 1921; Malmberg, 1965; Quilis, 1970; Ramirez, 2002, 2006; Colantoni and Steele, 2005; Schemeiser, 2006). This approach also differs from that of Bradley and Schemeiser (2002) and

Bradley (2004), who assume that the occlusion of the /r/ is superimposed on the nucleic vowel, which divides the vowel into two parts: the EV and the nucleic vowel.

In her experimental study of the Spanish rhotics, Blecua (2001) analyzes the productions of two males from Madrid. As indicated previously, she identifies three acoustic variants of the rhotic based on the degree of relaxation of the articulators: the least relaxed occlusive [ɾ] or tap; a more relaxed approximant variant [ɹ]; and, with the greatest degree of relaxation of the articulators, the elision of the flap [Ø]. The distribution of these variants is presented in Table 2.2

Table 2.2.

Distribution of variants of /r/ in Cr clusters. After Blecua (2001)

	Speaker 1		Speaker 2		Total	
	# of cases	%	# of cases	%	# of cases	%
EV+occlusion/approx	166	69.2%	212	88.3%	378	78.8%
Occlusion or approx	40	16.7	18	7.5%	58	11.8%
elision	34	14.2%	10	4.2%	44	9.2%

The author mentions that the results for the two speakers are statistically different (χ^2 (1), $N = 480$) = 27.03, $p < .0001$)[7]. From the results in Table 2.2 we note that the most common form of /r/ is the variant realized as an EV plus an occlusion or an approximant (EV + occlusion/approximant) at 78.8%, followed by the occlusion or approximant with no EV (11.8%), and lastly the elided form (9.2%).

[7] Although the author does not report the results for the Chi Square, she reports the raw data from which this calculation was done.

Blecua finds that rhotics containing EVs occur at a significantly higher rate after voiced segments than after voiceless ones ($p = 0.0001$). For voiced stops, she reports an 86.7% rate of occurrence of EVs and an 85% rate of occurrence for voiced approximants. The voiceless counterparts exhibit fewer EVs (61%) and higher rates of elision of components of the rhotic (93%). However, it is important to note that the results are not consistent between the two speakers: the results for /r/ after a voiceless consonant are only statistically significant for speaker number two (Blecua 2001, 21).

Blecua also claims that the variability in the duration of the epenthetic vowel is related to the manner of articulation of the contiguous consonant (i.e., the [p] in *prosa* 'prose' or the [t] in *puerta* 'door'). The author reports that the duration of the rhotic is affected by context: it is longer after a voiceless consonant than after a voiced consonant. However the difference is not statistically significant, and thus should not be interpreted as evidence for an effect of voicing, at least in the data she gathered. Blecua argued that voiced consonants are intrinsically shorter than their voiceless counterparts, and therefore the Cr sequence, in which both the consonant and the tap are voiced, will result in longer EVs due to compensatory lengthening.

Due to the very limited number of speakers used in this experiment, and since the results for the two speakers were statistically different on some measurements, the results have to be considered with caution. The power statistics were not reported and therefore it is not possible to distinguish between more or less reliable results. These results do, however, show the variable nature of speech production, and the EV phenomenon in particular.

Ramirez (2002) presented an experimental characterization of the epenthetic vowel based on an analysis of close minimal pairs such as *prosa-porosa* 'prose - porous' produced by six native speakers of Spanish. The origin of the speakers is the following: Colombia (2), El

Salvador (1), Peru (1), Argentina (1), Spain (1). The results of this study show that, on average, the duration of the epenthetic vowel is 26 ms. Ramirez points out that the length of the EV depends on the nuclear vowel; thus, in the vicinity of [o] the EV is 1/6 of its length (19ms/120ms), but in the vicinity of [u], the EV can be up to one half of the length (35 ms/70 ms). The average duration of EVs is 26 ms and the overall nuclear vowel average is 102 ms. Thus, EVs have an average duration of 1/4 of that of the nucleic vowel. Although the analysis of the EV in relation to the nucleic vowel was not conducted, this relationship is addressed in the production study presented in chapter 3.

Bradley and Schmeiser's 2003 study reanalyzed Gili Gaya's 1921 data on epenthetic vowels. The results from Gili Gaya's 1921 study are summarized in Table 2.3.

Table 2.3.

Prosodic and segmental influences on the duration of EVs in /Cr/ clusters (based on measurements from Gili Gaya 1921:277-8) (from Bradley and Schmeiser, 2003:2)

Variable	Mean duration of EV (cs) by cluster type	
Position within the word	Word-initial 5.3	Word-internal 3.7
Stress	Stressed syllable 6.5	Unstressed syllable 5.2
Order of constriction location	Back-to-front 6.3	Front-to-back 5.5

Although the results are not statistically significant, the authors interpret them as tendencies. Thus, they suggest that the epenthetic vowel is longer in word-initial clusters than in word-internal ones, and that it also tends to be longer in stressed syllables as compared to

unstressed ones. In addition, they claim that it tends to be longer in clusters with back-to-front order of constriction (i.e. dorsal + /ɾ/ such as /kɾ/, /gɾ/) as opposed to clusters with front-to-back order of constriction (i.e. labial + /ɾ/ such as /pɾ/, /bɾ/ and /fɾ/). However, since the results are not significant they have to be taken with caution, especially since the original investigation does not report the specifics for data collection or the power statistics.

In another study on EV, Colantoni and Steele (2005) analyzed the rate of occurrence and the length of EVs in obstruent + liquid clusters produced by speakers of Argentinean Spanish and Quebec French. In their study they observed the production from ten speakers of Spanish (five male, five female) and eleven speakers of French (six females, five males) in a word-reading task. The authors argue that the EV is inserted between an obstruent and a liquid as a process of dissimilation. They argue that the "rate of epenthesis as well as the length of the epenthetic vowel are a function of the similarity between the two members of the cluster, that is, the more similar the members of the cluster the higher the rate of epenthesis" (80). As an example, they point out that Cr clusters have a higher proportion of EVs compared to Cl clusters. The authors argue that this is due to the fact that the [r] in Spanish is a non-continuant[8] and the [l] is a sonorant continuant. Therefore, according to the authors, the [r] is acoustically more similar to the obstruents (p, t, k, b, d, g) than the [l], which produces a higher occurrence of EVs in the former. However, the authors do not report the variants of /r/ found in Argentinean Spanish. One of the more common variants found in Spanish is the approximant variant [ɹ] (Blecua, 2001). If this is also the case in Argentinean Spanish, then the [r] would have more

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[8] There is no general agreement as to the classification of the liquids in Spanish; some authors classify the /l/ as [-continuant] and the /r/ as [+continuant] (Cresey, 1978; Nuñez Cedeño and Morales Front, 1998).

characteristics of [+continuant]. If this is the case then [r] would have more similarities to /l/, and therefore the rate of occurrence of EVs would be comparable. Due to the variability in the production of /r/, one way to test the authors' hypotheses would have been to break down the data into the more and less continuant versions of /r/ and then to compare the frequency of EV in those conditions. This approach would raise some more complex questions about whether the consonant has to be underlyingly or only phonetically similar to the r/l in order for the effect of continuance to emerge.

Colantoni and Steele examined the effect of the first consonant of the cluster on the EV. They analyzed its manner of articulation ([±continuant]), place of articulation (labial, coronal, dorsal) and voicing (([±voice]). They hypothesized that, in terms of manner of articulation, more cases of simplification (epenthesis or reduction) would occur in Cr clusters than in Cl clusters, since both segments in the former (i.e., [pr], [gr], etc.) share the feature [-continuant]. In terms of place of articulation, they hypothesized that more cases of simplification would occur in clusters in which the first consonant was a coronal ([dr], [tr]). With regard to voicing, they expected to find more cases of simplification in clusters that agreed in voicing ([gr], [dr], [br]). In their results, regarding manner of articulation, they found more EVs in clusters with rhotics (94%) than laterals (1.9%) and claimed that this has to do with the fact that the lateral /l/ is [+continuant], and thus more dissimilar than the first consonant that was always [-continuant].

The results reported by Colantoni and Steele (2005) constitute a large deviation from other data on the occurrence of EVs in Cl clusters (Ramírez 2002, 2006). It is difficult to explain the disparity in the data because although in the Ramirez data the Cl clusters presented fewer cases of the EV (25%) compared to Cr clusters (68%), their occurrence (based on speakers of different countries) was not negligible -- as reported in the study by Colantoni and Steele. A

possible explanation involves the variety of Spanish analyzed by the authors (Argentinean Spanish). If the dialectal differences were in terms of the realization of surrounding consonants, then the disparity could be dismissed. But if it is simply a dialect difference, with all else being equal, then this suggests that the EV is linguistic in nature and not just the consequence of vocal tract movements.

As for place of articulation, Colantoni and Steele found that Spanish homorganic clusters ([dr] and [tr]) do not present higher rates of EVs compared to other places of articulation as they had hypothesized. There was "virtually no difference" (86) in the rate of occurrence of EVs among clusters with coronal initial consonants (96.6%), labials (95.4%), and dorsals (94.9%). However, place of articulation played a role in the length of the EV. They found that the EV is significantly longer in clusters with an initial dorsal consonant ($F(2,37) = 9.3$, $p = .0001$) than in other places of articulation. The authors argue that in the clusters where the first consonant is a dorsal plus the coronal [r] (the dorsal-coronal sequence), there is "tongue displacement" (p. 89) and the longer EVs are products of consequent gestural adjustment. Thus, the results from the experiment seem to go against the proposed hypothesis, which argues for more and longer EVs in consonants with similar places of articulation. The results actually suggest that longer EVs occur when the places of articulation are farther apart.

Overall, the analysis of the statistical results reported seems contradictory; for instance, the results from the ANOVA for place report 37 degrees of freedom when the authors reported ten subjects. In the t-test results, the researchers reported 935 degrees of freedom, although there were only 40 participants. It suggests that the statistical analysis was done by analyzing all the data points (from individual tokens) as separate (independent) measurements, and not by

obtaining average values per subject. Such an approach is statistically biased and can contribute to Type 1 error.

The authors also found that position within the word had a significant effect on the length of EVs: they found longer EVs in word internal clusters than in word initial ones ($t(936) = 3.55$, $p < .0004$). Also, EVs in stressed syllables were longer than in unstressed ones ($t(936) = 3.52$, $p < .0004$). Regarding voicing, the authors found that EVs are significantly longer after voiced consonants ($t(936) = 14.99$, $p < .0000$) than after voiceless consonants. The analysis of the statistical results show exaggerated degrees of freedom, suggesting that each token was treated as an independent observation, when in fact each speaker provided many tokens. A more appropriate analysis would be to focus on means per speaker rather than on the entire raw data set and many of these seemingly significant differences would likely disappear.

Colantoni and Steele point out, along with Blecua (2001), that voiced segments are intrinsically shorter than voiceless ones. Thus, the authors propose that the EVs are longer when the initial consonant of the cluster is intrinsically short (voiced). They claim that "The presence of the […] longer epenthetic vowel in Spanish would be a compensatory effect to preserve isochrony." (90). However, the relationship is left unexplained. The authors do not analyze the correlation between the consonants' VOT and EVs, which would provide insight as to the real relationship between the voicing of the first consonant of the cluster and the EV.

Regarding the results from Colantoni and Steele's study, in the case of prosodic factors (such as longer EVs in word-medial clusters as compared to word-initial clusters), it is not clear whether this is really a matter of different positions having different durations. Perhaps it is not really a prosodic effect, but rather a consequence of the effect of prosody on segmental duration. However, the study does not report what percentage of the whole syllable the EV is. This

information is important because there could be a confound between duration and position. One needs to be able to distinguish between factors affecting EV alone and factors that also affect the nuclear vowel. If the entire syllable is longer or shorter in a given position, it would not be surprising to find that the EV is also longer or shorter because of the same factors. Not looking at the whole syllable leads to confusing an effect on the length of all the elements in the syllable with an independent effect on the length of the EV.

Ramirez (2006) carried out a descriptive analysis of the acoustic production of five native-Spanish speakers (two females, three males) from different countries in Latin America. The effect of dialect was not analyzed due to the small size of the data set. The results show an average EV duration of 27 ms, which is very similar to the results presented by previous studies (Ramirez, 2002; Blecua, 2001, Bradley and Schmeiser, 2004; Schmeiser, 2006). Regarding the effect of linguistic factors, the results show a higher rate of occurrence for EVs after dentals (78%) than after velars (43%), labiodentals (25%) and bilabials (39%). As for duration, no statistically significant difference was found by place of articulation (after dentals (26 ms), after velars (34ms), labiodentals (20ms) and bilabials (26 ms)). Regarding voicing, there was no significant difference; however, there was a slightly higher rate of EV occurrence after voiceless segments (53%) than after voiced ones (42%). In terms of duration, EVs were significantly longer after voiced consonants (35 ms) than after voiceless ones (23 ms). As for stress, the results show that EVs in post-tonic syllables are significantly shorter than those in the tonic position, but not significantly different from those in pre-tonic position.

Regarding manner of articulation, EVs occurred in clusters with stops 54% of the time, and in clusters after fricatives only 25% of the time, a difference that was statistically significant ($P < .001$, Fisher's exact test).[9]

In terms of duration, a paired samples t-test showed only a marginally significant difference in EV duration whether it was after stops (M = 27msec, SD = 4.5) or after approximants (spirantized stops, M = 19msec, SD = 6.3), $t(4) = 2.46$, $p = .081$.

In a comprehensive study of the epenthetic vowel, Schmeiser (2006) proposes "to assess what factors condition the [SV] epenthetic vowel's durational variability" (112). In his study, he examines the clusters formed by /r/ + Consonant [rC] and consonant +/r/ [Cr] as produced in a passage-reading protocol by 29 participants from six different countries (Argentina, Colombia, Ecuador, Guatemala, Mexico, and Spain). The approximately 420-word passage contained 33 /Cr/ clusters and 23 /rC/ clusters, with a total number of 1017 tokens. The results were analyzed by means of single-factor, within-subjects ANOVAs using the durational means within speakers. The author notes that "Given the exploratory nature of the study, coupled with the notion that SVs do not necessarily surface in every consonant cluster, an imbalance of speakers and/or tokens is to be expected for any given hypothesis" (46).

Based on an analysis of previous studies, the author proposes seven hypotheses about the influence of linguistic and non-linguistic factors on the duration of EVs. Then he examines the results in light of a theoretical framework based on Articulatory Phonology, Byrd's (1994, 1996b) temporal Phase Windows, and Bradley and Schmeiser's (2003) analysis.

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[9] A posteriori analysis using Fisher's exact test was done rather than a chi-square test of independence due to the small number of cases

In this study we will examine only those hypotheses regarding the /Cr/ clusters; the /rC/ clusters will be excluded from this review since they are beyond the scope of this study.

2.3.1.1 The effect of position within the word

Bradley and Schmeiser's 2003 reanalysis of Gili Gaya (1921) suggests that, although not statistically significant, the EVs tend to be longer in word-initial /Cr/ clusters than in word-medial ones (see values in Table 1). They argue that the differences in the length of the EVs according to word position are due to coarticulation effects, and in turn, coarticulation is determined by speech style. They point out that in casual speech there is a higher degree of coarticulation of adjacent segments (Cr in this case), while in formal speech there is more isolation of the segments and thus a lesser degree of coarticualtion. The authors argue, as did Chitoran et al. (2002), that the degree of coarticulation of clusters that appear word-initially (vs. word-internally) is limited by conditions imposed by perceptual recoverability. They argue that these patterns are explained by the fact that "word onsets are potential utterance onsets, in which case no preceding vowel is available to provide formant transitions into the first consonant" (9). The authors also argue that word onsets have been shown to be important for lexical access; and therefore, "it is plausible that minimal overlap is favored word initially so as to preserve more acoustic information about each consonant of the cluster". (9). However, they do not present data as to what percentage of clusters in word-initial position are also in utterance-initial position, which leaves this argument unsupported. Furthermore, it is not clear from their analysis whether the coarticulation of the consonants is due to the position within the word or if it is due to the rate of speech; if the medial syllables are shorter than initial syllables, and the increased rate affects the EVs in the same way as the nuclear vowel, a completely different picture of their findings results. Although speech rate is mentioned as a factor, it is not analyzed. Thus, the results seem

not to be conclusive. The analysis depends on whether position in the word is confounded with stress.

2.3.1.2 Prosodic stress

Schemeiser (2006), based on Bradley and Schmeiser's 2003 reanalysis of Gili Gaya's 1921 findings, suggests that, although not statistically significant, longer EVs occur more often in stressed demisyllables[10] than in unstressed ones. The researchers point out that due to the preservation of acoustic information in stressed syllables, less overlapping of the segments is expected. However as indicated above, a well known effect of stress is the lengthening of the segments in the stressed syllable. Thus longer EVs are expected in stressed syllables –where all the segments may all be longer. This assumption confounds stress with duration.

I argue that a more appropriate test would examine whether the EV is longer as a percentage of the whole syllable. The results of such analysis are presented in Chapter 3.

2.3.1.3 Order of constriction

Bradley and Schmeiser's 2003 reanalysis of Gili Gaya's data found that EV duration tends to be longer in clusters with a back-to-front order of constriction ([kr] and [gr]), although, again, the difference is not statistically significant. Bradley and Schmeiser (2003) argue that in clusters with a back-to-front order of constriction such as [gr], the acoustic release of the first consonant [g] will be "perceptually obscured because the second constriction lies ahead [more

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[10] Schmeiser (2006) adopts the term 'demisyllable' instead of 'cluster' when discussing prosodic stress because the sequence Cr does not receive the prosodic stress, but rather the nuclear vowel of that syllable.

fronted in the oral cavity] of the first constriction in the vocal tract." (9). In contrast, they argue that in clusters with a front-to-back order such as [br], "it does not obscure the acoustic release of the first consonant because the second constriction lies behind the first." (9). That is to say that in the articulation of [gr], the gesture of [r] (the tongue tip) obscures the acoustic release of [g] (articulated by the tongue body), but in clusters such as [br] the gesture from [b] (the lips) does not obscure the acoustic release of [r] (the tongue tip), which is pronounced in a further back position within the oral cavity. However, the term "perceptual obscurity" is not clearly defined. It is not clear whether clusters such as [gr], which are articulated with one active organ (the tongue), are more perceptually obscure than clusters with two active articulators (tongue and lips), such as in [br]. This seems to suggest that 'perceptual obscurity' is defined in articulatory terms. That is to say, it is defined by the articulation of the clusters, which is contradictory. It is an empirical question whether a velar release is "obscured" – i.e. less intense – than a bilabial one, and whether this has anything to do with the occurence of EVs. Even though speech gestures are rapid, as are acoustic events, there is no acoustic evidence that the /r/ obscures the /g/.

Based on these arguments from Bradley and Schmeiser (2003), Schmeiser (2006) put forward hypothesis #3. "/Cr/ clusters with a back-to-front order of constriction location (i.e., /kr/ and /gr/) will evidence longer EVs than ones with a front-to-back order (i.e., /pr/, and /br/)." (56).

2.3.1.4 Heterorganic vs homorganic clusters

Schmeiser (2004) examined the effect of place agreement (homorganic vs. heterorganic) of the first consonant in the cluster on the duration of EVs (among other factors such as position within the word, stress, order of constriction location, and voicing). He studied the production of

five speakers of Peninsular Spanish in a passage-reading task. The results of this study are presented in Table 2.4.

Table 2.4.

Mean EV (SV) duration and ANOVA p values (Schmeiser, 2004)

VARIABLE	MEAN EV DURATION (ms)		PROBABILITY (ANOVA)
1. Position within the word	Word-initial 22.59	Word-internal 22.4	p=0.93
2. Stress	Stressed syllable 22.99	Unstressed syllable 22.06	p=0.64
3. Order of Constriction location	Back-to-front 27.77	Front-to-back 20.76	p<0.05
4. Place Agreement	Heterorganic /CR/ 23.65	Homorganic /CR/ 19.41	p=0.06
5. Voicing	Voiced C1 27.33	Voiceless C1 20.07	p<0.001

Based on the marginally significant results for Place Agreement (p = 0.06), Schemeiser (2006) put forward Hypothesis #4. "Heterorganic /Cr/ clusters (i.e., /pr/, /br/, /kr/, and /gr/) will evidence longer SVs than homorganic ones (i.e., /tr/ and /dr/)." (56).

2.3.1.5 Voicing

Previous studies have found that EVs are significantly longer in clusters after the (inherently shorter) voiced consonants as opposed after voiceless consonants (Blecua, 2001; Colantoni and Steele, 2005; Ramírez, 2006; and Schmeiser, 2004). Based on these findings Schemeiser hypothesizes that "/Cr/ clusters in which C_1 is voiced will evidence longer EVs than ones in which C_1 is voiceless." (57). Nevertheless, it is not clear if the effect of longer EVs after voiced consonants is specific to EVs or if it also affects nuclear vowels. So far no studies have looked into effects that are particular to EVs and not to all vowels.

2.3.1.6 Manner of articulation

Blecua (2001), in her study of the Spanish rhotics (/ɾ/ and /r/), found that the /r/ tends to be the shortest after the voiceless fricatives (35ms) and longest after approximants (53.5ms), with stops in the middle (Table 5). Based on Blecua's 2001 findings, Schemeiser (2006) put forward hypothesis #6: Fricatives will evidence the shortest EV duration, followed by Stops and finally Approximants.

However, note that Blecua (2001) presented data for the whole /r/ since she considers the EV to be a constituent part of the [r]. She does not consider it to be an independent epenthetic element as it is assumed to be in this study and also in Schemeiser's (2006). Thus, the results reported by Blecua are for the whole [r] and not for EVs in particular. A possible explanation for the differences in duration is the change in airflow in the articulation of fricatives with respect to stops. Because that is a consistent difference in articulation, it would be assumed that if there is potential for an EV, but if it comes after a segment with airflow, it will not be possible to measure the EV accurately, and it will be 'obscured' by the airflow. That is to say that it will not have measurable formants.

Table 2.5.

Average duration of /r/ for manner of articulation and voicing (Schmeiser, 2006: 23 after Blecua 2001).

Manner of articulation	Average duration of rhotic (ms)
Fricative	35
Voiceless stop	39.5
Voiced stop	51.5
Approximant	53.5

2.3.1.7 Place of articulation

Based on the findings of Colantoni and Steele (2005a) that EV duration is statistically longer after dorsals (see discussion on Colantoni and Steele, 2005), Schemeiser proposed Hypothesis #7: "Labials will evidence the shortest EV duration, followed by coronals and finally, dorsals." (67). This hypothesis overlaps with hypothesis 3 (order of constriction) since clusters with dorsal C1 are the same clusters with a back-to-front order of constriction whereas the labials are the same as the clusters with front-to-back order of constriction.

Schemeiser (2006) analyzed the results for each hypothesis by country of origin of the speakers. Although country of origin was not considered a variable, it was used as a grouping criterion to determine the validity of each hypothesis. The author defined a supported hypothesis as "one in which the findings in the majority (i.e. four or more) of the six countries supports the hypothesis with statistically significant results," (81). Thus, for a hypothesis to be supported it was required that the results were significant for at least four countries out of six possible. A disadvantage of this criterion is that it eliminates the possibility of identifying the effects of dialectal variety. Furthermore, as noted, country was not considered as a variable in the study. Thus, when the results appeared statistically significant for a country there were no follow-up tests. The results by country by hypothesis are summarized in Table 6 (Schmeiser, 2006:81). It should also noted that "country" is too rough a measure. There are many dialects within each country, and bordering regions of different countries sometimes share dialect features (like the Rioplatense dialect that is characteristic of Argentinian and Uruguayan speakers).

Table 2.6.

P values by country[11] *for /Cr/clusters.*

Country	H1	H2	H3	H4	H5	H6	H7
Spain	0.711	0.718	**< 0.001****	0.205	**< 0.001****	**< 0.001****	**< 0.001****
Argentina	0.694	0.328	***0.060****	0.974	**< 0.001****	**< 0.001****	0.170
Mexico	0.655	0.145	**0.040****	0.978	**0.012****	***0.065****	0.281
Guatemala	0.326	0.324	**< 0.001****	0.865	**< 0.001****	**0.008****	**0.001****
Colombia	***0.052****	0.405	**0.012****	0.330	**< 0.001****	**0.003****	**0.009****
Ecuador	0.642	0.119	**<0.001****	**<0.014****	**< 0.001****	**< 0.001****	**< 0.001****

**=statistically significant with alpha set 0.05
* =marginally statistically significant with alpha set between 0.051 and 0.065

Schmeiser's results show that the rate of occurrence of EVs is 64.16% in /Cr/ clusters (compared with 84.2% in /rC/ clusters). The results confirm the previous finding in terms of variability in the occurrence of EVs (Blecua 2001), and they also show great variability across countries, ranging from 48% to 85% with the highest rate for speakers of Ecuadorian Spanish (84.9%). The results for the first two hypotheses, regarding prosodic factors, reveal that (H1) position within the word is a factor: longer EVs occur in word medial clusters, with results for one country being marginally significant (Colombia, p = .052). The results for prosodic stress (H2) were not significant. Thus the author points out that "prosodic factors do not influence SV durational variability" (109).

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[11] Despite the Despite the fact that data are grouped by country, country is considered as a control variable, with no specific hypothesis about dialect differences and how they would affect EVs.

A propos of segmental factors, Schemeiser finds that order of constriction (H3) is a factor that affects the duration of the EV. The front-to-back order presents longer EVs across countries (marginally significant for Argentina, $p = .06$) and across all participants. With regard to the homorganic vs. heterorganic factor (H4), the author finds a significant effect by subjects (overall, including all of the countries) but not by country (group), i.e. it did not apply to the required 4 countries, and this leads him to claim that the hypothesis is not supported. The analysis of voicing (H5) shows that EVs are longer after voiced consonants. In the case of manner of articulation (H6), the author finds that the EV is longer in stops than in fricatives. In terms of place of articulation (H7), the EVs are longer after dorsals than after coronals and the EVs after labials are the shortest[12].

Regarding position within the word and stress, Schemeiser, in contrast to other studies (Bradley and Schmeiser, 2003; Colantoni and Steele, 2005), found no effect on the cluster of either the position that it occupies within the word, or the stress of the syllable. Results were only marginally significant for position for Colombian speakers. These results argue for a multidimensional analysis of the EV. Considering one variable at a time in the computation – either position within the word or stress -- gives a partial analysis of the phenomenon. A multidimensional approach should look at the interaction of factors such as position within the word, speech rate and stress simultaneously. Regarding order of constriction, Schmeiser's 2006 results support Bradley and Schmeiser's 2003 claims. The results show fairly consistently that clusters with back-to-front places of articulation (i.e., /gr/, /kr/) have longer EVs. The author

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[12] The results regarding Place of articulation seem to be in contradiction with those reported for Order of constriction. Longer EVs are reported after dorsal consonants ([kr], [gr]) and shorter ones after labial consonants ([pr], [br]). However, it is also reported that longer EVs occur in front-to-back clusters ([pr] and [br]) than in back-to-front clusters ([kr], [gr]).

argues that this is due to both articulatory and perceptibility factors. In the case of a front-to-back order of constriction (i.e., /br/ and /pr/), shorter EVs are favoured because "the perceptibility of both consonants is not really in jeopardy." (127), and therefore, the articulatory gestures "accommodate the smallest amount of EV duration, but yet avoid coarticulation". Instead, the author claims that shorter EVs occur in the front-to-back clusters because the gestures require the use of two major articulators: the lips and the tongue. They allow a fast change from the consonantal gesture to the gesture of /r/. In addition, in the front-to-back constriction, the acoustic release is not obscured (it is not behind the alveolar tap). The fact that the movement from a lip gesture is faster than from a tongue body gesture also contributes to shorter EVs.

There are two confounded explanations in this argument. It seems that order of constriction can also be analyzed in articulatory terms based on the number of active articulators used in the articulation of the cluster. Back-to-front clusters are articulated using only one active articulator (the tongue), and front-to-back clusters are articulated using two active articulators (lips and tongue). When using one articulator there is an articulatory need to differentiate the two segments; therefore, the EV is used as an articulatory resource (Ramirez, 2006).

Regarding voicing, Schemeiser's results support the findings in other studies (Blecua, 2001; Schmeiser, 2004; Colantoni and Steele, 2005; Ramirez, 2006). They show that shorter EVs occur after intrinsically longer voiceless consonants while longer EVs occur after voiced consonants, which are intrinsically shorter. Schmeiser, like Blecua (2001), views longer EVs after voiced consonants in terms of intrinsic duration of the consonant. Therefore, the voicing of the consonant and the tap in the Cr sequence results in longer EVs. Schmeiser pointed out that the EV's duration is not strictly a matter of longer EVs occurring after shorter consonants. He argues that "perceptual recoverability is the determining force in EV durational variability for the

case of C1 voicing". He also argues that the Cr cluster becomes a sequence of two voiced consonants in cases where C1 is voiced. Since they are intrinsically short in duration, their perceptibility is in greater jeopardy as opposed to a cluster where the first consonant is voiceless. He further argues that shorter EVs occur after voiceless consonants because there a sequence where a tap, which is very short and voiced, is followed by a voiceless consonant, which is longer. This combination creates a sequence in which perceptibility is already optimal and therefore a short EV is employed.

Regarding place of articulation, Schmeiser argues that after a stop consonant (p, t, k, b, d, g), the EVs are "neither long nor short" (136). This 'average' duration is due to the fact that stops are both voiced, which trigger shorter EVs, and voiceless, which trigger longer EVs. In addition, as a group they have front-to-back order of constriction location, which triggers shorter EVs, and back-to-front order, which triggers longer EVs. Thus, Schmeiser argues that the combination of these factors results in an EV, which, in terms of duration, is in the middle range. Surprisingly, the results for place of articulation of the stops do not resemble those for order of constriction. As discussed above, place of articulation is a variable that subsumes order of constriction (dorsals comprise back-to-front order of constriction and labials comprise front-to-back order). However, Schmeiser's analysis of the clusters with the approximants (β, δ, and γ), showed that they present the longest EV duration. The author argues that these results are due to the fact that the approximants, which are voiced, represent the shortest intrinsic duration within manner of articulation, with an average duration of less than half that of a stop (Schmeiser, 2006; Manrique and Signorini, 1983:121). Thus, Schmeiser (2006) argues, the results seem to confirm Blecua's implicational hypothesis that intrinsically longer consonants result in shorter SV duration and vice-versa. (137).

The results from Schmeiser show that in clusters with approximant C1 the EVs are significantly longer than for stops, but for those with a stop C1 there is a significant effect on the EV's length. This difference seems to indicate that manner of articulation rather than order of constriction is a factor in the length of EVs. If order of constriction were a factor, it would have been expected that clusters with a dorsal C1 ([gr], [ɣr], and [kr]) would present longer EVs, independently of whether it were a stop ([g], [k]) or an approximant ([ɣ]).

Furthermore, no analysis is presented of whether factors such as the manner of articulation or order of constriction affect all the vowels in their vicinity or if they exclusively affect the EVs. This issue is important to address because it will reveal the factors that influence the occurrence and length of the EV. If the factors found to affect EVs also affect nuclear vowels, then the EV is no different from other vowels and may simply have the articulatory function of separating articulations where the places are similar.

Schmeiser concludes that "[EV] duration in Spanish must be viewed along a continuum and that such a variable element will not exhibit clear-cut findings among all speakers", (142). Furthermore, the author points out that the "[EV's] duration can be viewed as a trend in terms of both manner and place of articulation." (142). In other words, the duration of the EV depends on the type of the articulator and the speed at which it moves.

As for the role that the EV plays, Schmeiser argues that the EV's duration is "first a tool used by the speaker to ensure the perceptibility of dissimilating like consonants and is then subject to restrictions within the vocal tract." (142). Schmeiser, along with Quilis (1993), argues that longer EV duration is inversely proportional to higher articulatory energy. Within the elements that form consonant clusters, voiceless consonants have higher articulatory energy

"because their energy is more concentrated in the supraglottic organs", (Quilis 1993:67). That is to say that after /p/, /t/, /k/ it is expected that we would find a shorter EV duration than after their voiced counterparts. So, if a cluster has a voiceless stop it is more readily perceived than if it has a voiced stop (e.g. [pr] is more perceptible than [br]). Furthermore, the author argues that if a cluster has a voiced stop, it is more likely to need the enhancement of an EV in order to be perceived.

In turn, the voiceless fricatives have the highest articulatory energy, which makes them more perceptible. Thus, they should have the fewest EVs and the shortest EV durations. As for manner of articulation, Schemeiser, along with Manrique and Signiori (1983), points out that fricatives are the class with the longest average duration. The class with the next longest average duration is the stops, which require "greater occlusion than the voiced approximants, and require the lowest articulatory energy", (Schemiser, 143).

Schemeiser proposes that, based on the short duration and the voicing quality of the tap, longer EVs are "needed as articulatory energy decreases to ensure perceptibility of both consonants" (143). Thus, based on the inherent articulatory energy of the different segments, the author proposes a continuum of the duration of the epenthetic vowel as illustrated in (2.5).

(2.5) Proposed continuum of EV's duration based on articulatory energy (Schemeiser, 143)

Manner:	Fricatives		Stops		Approximants
	/f/ →	/p/→ /t/→ /k/→	/b/→ /d/→ /g/ →		[β] → [δ] → [γ]
	shorter EV				longer EV

A summary of the studies reviewed is presented in Table 2.7. The methods used most often to collect data are sentence reading, where the target word is inserted in a carrier sentence (i.e., say ____ again), and passage reading. These methods of passage and sentence reading are preferred over interviews or collections of spontaneous speech (i. e., spontaneous conversations) because in the latter it is difficult to control for prosodic and linguistic contexts. Other methods such as word lists, in which the target item is presented as a single word and then presented as a list, create the word-list effect. This effect results when the subjects are more conscious of their pronunciation and they overemphasize it; in addition, the list of words tends to be read with a pause after each word, creating an irregular intonation pattern which may cause variation in intensity and duration. Blecua (2001), who uses a passage-reading task, reports avoiding placing the target item in a word that is before a pause to avoid variations in intensity and duration. Other studies (Colantoni and Steele, 2005; Schmeiser, 2006) that use the same protocol do not report using any controls to avoid prosodic effects.

The number of subjects in the experiments varies between two and 26. Although results in phonetic experiments are generally consistent, the data from a number of subjects as small as two has to be taken with caution. Furthermore, the statistical analyses used in some studies present problems which may distort the results. (i.e., each response from a subject is considered independently without averaging the results for each condition for each subject).

Although the effect of speech rate is widely recognized as having an important role in segment co-articulation (Bradley and Schmeiser, 2003), there are no studies that have controlled for it or analyzed it in as an individual factor, or analyzed its effect on other linguistic or prosodic variables. I argue that the analysis of EVs in relation to speech rate has to look at the duration of EV as compared to the duration of the entire syllable and also the duration of the nucleic vowel.

Table 2.7.

Summary of the methodologies used by some relevant studies

Study	**Data collection protocol**	**Corpus and participants**	**Control for prosodic variations**	**Control for speech rate**
Blecua 2001	Passage-reading task	Paragraphs containing 370 target segments. They were read by two participant males (740 tokens)	Target items did not appear in words before pause to avoid variation in intensity and duration	None
Schmeiser (2001) Reanalysis of Gili Gaya (1921)	Word list reading	Analysis of EV occurrence in Peninsular Spanish (unspecified number of speakers)	None	None
Ramirez (2001)	Sentence reading task. In this task, the target word was inserted in a carrier sentence	Six participants	The target word is inserted in the middle of the sentence to avoid prosodic effects from the sentence's	None

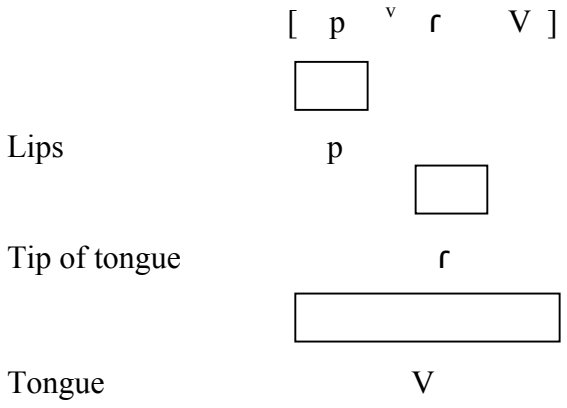
			edges that may cause variation in intensity or duration.	
Colantoni and Steele (2005)	Passage-reading task	Eleven participants	The target word is inserted approximately in the middle of the sentence	None
Colantoni and Steele (2006)	Sentence-reading task.	40 participants	The target word is inserted approximately in the middle of the sentence to avoid prosodic effects on intensity or duration.	None
Schmeiser (2006)	Passage-reading task	420 word long paragraph containing 33 Cr cluster; 29 participants	None	None

2.3.2. Phonological approaches to the epenthetic vowels

Within the framework of articulatory phonology[13], Bradley (2002) argues that epenthesis is a phenomenon of variation in the timing of articulatory gestures associated with the elements of the consonant cluster (consonant (C) + liquid (l) + vowel (V)). He argues that the consonantal gestures overlap with the vocalic gestures. Thus, the epenthetic vowel has the same pitch as the nucleic vowel: both vowels are derived from the same articulatory gesture, and the consonantal gesture creates an interruption, breaking them apart as illustrated in (2.6).

[13] For a detailed description of the principles of articulatory phonology see Browman and Goldstein (1989, 1990, 1991, 1992)

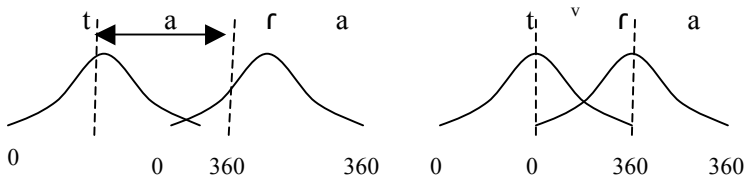
(2.6) Partial overlap in [pʷɾ ʌ]



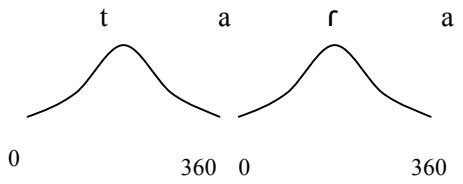
Hall (2003) considers the epenthetic vowel to be an "intrusive vowel" which is non-syllabic and non-segmental. It occurs in consonant clusters that contain a sonorant or a guttural (pharyngeal or uvular) sound, where the consonants in the cluster do not share place of articulation (they are heterorganic), and they tend to disappear in fast speech. In addition, speakers are unaware of their existence. The author argues that in cases where it is perceived, it is due to the overlapping of a vowel gesture and neighboring consonants. Thus, the epenthetic vowel is not the result of a synchronic process of vowel epenthesis.

For instance, the possible realization of /tɾa/ and its diachronic evolution into /taɾa/ are illustrated in (2.7).

2.7 (a) Production of epenthetic vowels with variable durations



(b) Listener reinterprets longer fragment as lexical vowel



Hall claims that epenthetic vowels differ from real vowels in their phonological behavior.

Since real vowels can form the nuclei of syllables, they can influence a large range of syllable-conditioned phonological processes. However, epenthetic vowels cannot form syllables regardless of their duration, and they are not syllabic by phonological diagnostics such as language games (Kekchi), syncope (Scotts Gaelic), allomorph conditioning (Finnish), reduplication (Hocank), and stress (Spanish). In the case of Spanish, the epenthetic vowel and the nucleic vowel are counted as one for stress purposes. In (2.8) the stress is assigned to the antepenultimate –proparoxyton- syllable regardless of the duration of the epenthetic.

(2.8) Metrical cohesion of the Spanish epenthetic vowel.

(a)	hidrómetro	[i.ðɾó.me.tɾo]
(b)		[i.ðɾó.me.toɾo]
(c)		*[i.ðɾó.me.to.ɾo]

Hall claims that the motivation for the occurrence of the epenthetic is language specific. For example, in Scottish Gaelic the epenthetic vowel avoids the overlap between certain kinds of consonants and in this way it enhances the perception of the cluster. In Hocank, on the other hand, the occurrence of the epenthetic avoids complex onsets. That is to say, it is motivated by syllable structure.

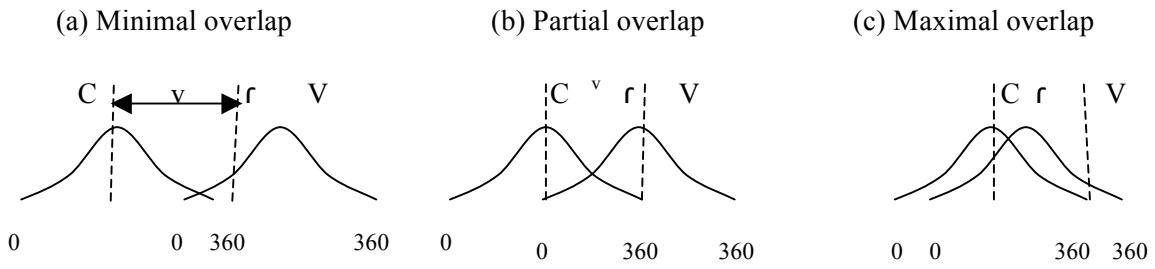
Hall's cross-linguistic survey shows that vowel intrusion occurs more often with liquids than with other sonorants, and more often with rhotics than with laterals, except the alveolar trill. The proposed implicational hierarchy is illustrated in (2.9).

(2.9) Vowel intrusion triggers (Hall 2003: 28).

(obstruents, if ever) > other approximants, nasals > [r] > [l] > [ɾ], [ʁ] > gutturals
Among nasals: [m] > [n]

Bradley and Schmeiser (2003) use the Articulatory Phonology (AP) framework to analyze coarticulation of the speech continuum, in particular the consonant-consonant sequences where epenthesis occurs. Schmeiser (2006) notes that within the AP, the gestures are articulatory movements that produce a constriction in the vocal tract and are defined in terms of the dynamics of the articulators; each articulatory gesture is defined as a phonological primitive as well as a unit of articulation, and they have internal duration – a property represented in a graph in terms of a 360º cycle (Figure 10). Adjacent gestures are temporally coordinated with respect to each other and may exhibit varying degrees of overlap; the consonant articulations are superimposed on vocalic gestures, which are themselves articulatorily adjacent (Gafos, 1999). The different degrees of overlap, which result in different epenthetic vowels durations, are illustrated in (2.10) (Gafos, 1999; Bradley, 2002; Schmeiser, 2006).

(2.10). Patterns of gestural overlap for /Cɾ/ clusters



In (2.10a) and (2.10b), the gesture for the tap must start within the Phase Window (represented by the vertical dotted line). In cases of minimal overlap (2.10a), the vocalic gesture is clear and the epenthetic vowel is noticeable. In cases of partial overlap (2.10b), the vowel is reduced, and in maximal overlap (2.10c), the vocalic gesture is completely covered by the consonantal gestures. Bradley (2002) points out that a model that incorporates functional phonetic factors such as intergestural timing accounts for the variation in duration of the EV. In addition, the fact that the consonantal gestures overlap the vocalic ones offers an explanation as to why the pitch of the epenthetic and the nucleic vowel are similar: they are derived from the same articulatory gesture which is interrupted by a consonantal gesture (see also Steriade, 1990).

Furthermore, the authors analyze the presence and duration of the EV within the gestural timing of the consonant clusters in the framework of the Optimality Theory. They adopt two faithfulness constraints from Cho (1998a, b):

(11) a. IDENT(timing)

The relative timing of gestures in the output must fall within the lexically specified Phase Window, which determines a permissible range of gestural overlap

b. OVERLAP

Adjacent consonantal gestures must be maximally overlapped.

The authors formalize the results: the tendency of epenthetic vowels to be longer in word-initial and stressed /Cɾ/ clusters than in word-internal and unstressed clusters as illustrated in (4).

(4) Tableau for constraints IDENT and OVERLAP

	IDENT (timing)	OVERLAP
☞ a. /CɾV/ → CvɾV		*
☞ b. Cᵊɾ̃V		*
c. CɾV	*!	
d. /#CɾV/ → #CvɾV		*
e. #Cᵊɾ̃V	*!	*
f. #CɾV	*!	
☞ g. /CɾV̀/ → CvɾV̀		*
☞ h. Cᵊɾ̃V̀		*
i. CɾV̀	*!	
☞ j. /CɾV́/ → CvɾV́		*
k. Cᵊɾ̃V́	*!	*
l. CɾV́	*!	

V̀ = short epenthetic, V́ = long epenthetic

Schmeiser (2006) points out that in the canonical realization of the /Cɾ/ cluster, the ranking of IDEN (timing) over (>>) OVERLAP allows the candidates with epenthetic vowels to win (a, b). For the outputs d-f, a longer epenthetic vowel in word-initial /Cɾ/ demisyllables[14] is optimal. Outputs e. and f. violate IDEN (timing); the same violations occur for k. and l. due to longer epenthetic vowels in stressed syllables. The candidates e. and k. violate IDEN (timing) because of the lexically-specified constraints.

On the other hand, Bradley (2007), within the framework of Articulatory Phonology, analyses cross-dialectal phonetic variation in Spanish clusters. The author proposes a

[14] I adopt the term demisyllables (along Bradley and Schmeiser, 2003; and Schmeiser, 2006) when discussing prosodic stress since the stress does not fall on the cluster /Cɾ/ but on the nucleic vowel of the syllable. The term demisyllable represents the parts of the syllable: the initial and final parts, in this case the onset and the nucleus.

phonetically-motivated explanation in terms of the coordination of consonantal gestures of phenomena such as the obstruent + rhotic cluster realization, vowel intrusion and rhotic assibilation induced by coarticulation. Bradley, along the lines of Hall (2003), argues that gestural timing is not exclusively a "low-level phonetic implementation component", (15); instead, he proposes a unified model in which "gestural and non-gestural constraints are present in the same level of the phonology" (15).

Bradley points out that there are conflicting results on the phonetic detail in the realization of the Spanish /Cr/ clusters: multiple studies have found that different factors affect the duration of the epenthetic vowel to a significant degree. The author attributes these results to the variable nature of the epenthesis and differences among speakers and dialects. However, there have not been comparative studies on dialect differences, and the statistical analyses employed do not account for variability within individual speakers.

Bradley (2007) extracts generalizations from the previous studies. These generalizations are presented in (12) below.

Generalizations in phonetic variation in onset clusters across Spanish dialects. (Bradley 2007:22-23)

(2.12) "a. An intrusive vowel of variable duration typically occurs in /CrV/ but not in /ClV/.

b. The formant structure of the intrusive vowel in /CrV/ demisyllables is similar, but not identical, to that of the tautosyllabic nuclear vowel.

c. Anaptyctic (EV) vowels that arise historically from /CrV/ copy the quality of the tautosyllabic nuclear vowel.

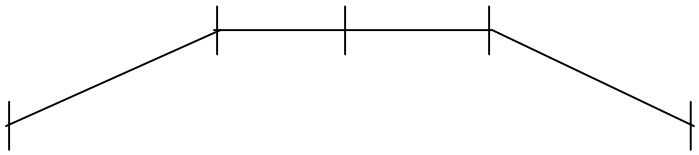
d. Coarticulation in casual speech of heterorganic /Cr/ in a given dialect entails coarticulation of homorganic /Cr/ (where C1 is noncontinuant) but not vice versa."

Generalization (a) does not take into consideration that Cl clusters have EVs, although at a lower rate (Ramirez, 2002, 2006). A model of EVs should explain its occurrence in all the possible contexts: Cr as well as in Cl clusters.

In his account, Bradley follows Gafos' 2002 Articulatory Phonology within a constraint-based grammar, which proposes that gestural coordination is determined by alignment constraints with reference to temporal landmarks of the articulatory gesture as illustrated in (2.13).

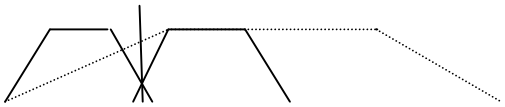
(2.13) a. ALIGN(G_1, LANDMARK$_1$, G_1, LANDMARK$_1$) (Bradley, 2007)
Align landmark$_1$ of gesture$_1$ with landmark$_2$ of gesture$_2$.

b.



Bradley (2007), following Davidson (2003), argues that variation in the duration of EVs can be accounted for by "specifying a range of landmarks in the initial consonant gesture with which /r/ may be aligned with any point between the release and the offset of the preceding consonant", (25). That is to say that the beginning (ONSET) of the /r/ gesture can be "aligned with any point between the RELEASE and OFFSET of the preceding consonant", (25) as illustrated in (2.14).

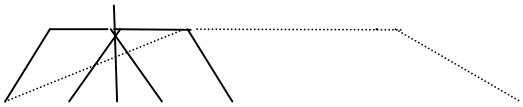
(2.14) a. ALIGN(C, OFFSET, /r/, ONSET) (Bradley, p 23)
In /Cn/, align the offset of C with the onset of /r/.
b. *Coordination*: C OFFSET = /r/ ONSET



Percept: [C ə r V]

In (2.14) the consonantal gestures (represented by the solid lines) are superimposed on the tongue body gesture of the vowel (represented by the dotted line) using the models of Browman and Goldstein (1990) and Steriade (1990). The open transition between the consonants (/C/ and /r/) allows the vowel to be perceived. In contrast, the closed transition of the adjacent consonants does not allow the emergence of the vowel and, therefore, the perception of the EV, as illustrated in (2.15).

(2.15) a. ALIGN(C_1, RELEASE, C_2, TARGET)
In /C_1C_2/, align the release of C_1 with the target of C_2.
b. *Coordination*: C_1 release = C_2 target



Percept: [C ə C V]

Bradley (2007) argues that from a functional point of view, perceptibility motivates the gestural alignment in (2.14) while effort minimization motivates (2.15). Bradley, following Hall (2003), points out that open transitions and vowel intrusion, such as in (2.14), ensure clearer perceptual clues for the neighbouring consonants while greater overlap "yields a relatively faster, more efficient overall articulation of the cluster" (24).

Bradley points out that the structure represented in (2.15) is characteristic of Cl clusters. However, as will be discussed in Chapter 3., the rate of occurrence of the epenthetic vowel in this type of cluster is not negligible.

The alignment constraints provide a formalization of the gestural coordination, and this approach explains the duration of EVs as the result of two opposing forces: minimal effort and maximal perceptibility. However, it does not provide a model that responds to phonetic questions such as when an EV should be shorter or longer.

The studies presented above address the phenomenon of the EV in relation to its linguistic context. However, they do not examine the effect of the context on the nuclear vowel. I argue that a better approach is to analyze the effect of the linguistic context on both the EV and the nuclear vowel. This analysis can shed light on whether the EV has a special status or if its behavior is predictable from speech production models.

Furthermore, there is no analysis or control for the effects of word duration or speech rate on the occurrence and duration of EVs. The analysis of these factors will provide information as to their effect on segmental coarticulation and the duration of segments and particularly EVs.

In addition, there has been little research on dialect differences in the realization of the consonants and their effect on the mechanisms that enhance perceptibility. The dialect differences could in turn be used to predict different rates of occurrence and durations of the EVs.

Regarding methodology, the analysis of the EV in contrast to the full vowel would be better done by studying minimal pairs where the linguistic context is kept constant for the two types of vowels. Furthermore, this analysis of EV has to examine the roles of the speaker and

hearer since perceptibility is a two-way street. The analysis has to try to answer the question of whether the speaker is trying to be clear by means of an enhanced EV.

Chapter 3

Speech Production: The Occurrence of the Epenthetic Vowels

This chapter examines the factors that affect the frequency of occurrence of EVs in Obstruent + liquid clusters as well as some of the factors affecting their duration. In this chapter I argue that the degree of lenition or fortition of the individual consonants contributes to the overall degree of lenis or fortis of the consonant clusters. In addition, I argue that the degree of fortition or lenition of the nucleic vowel also contributes to the strength of the following liquid and therefore of the whole cluster. Regarding EVs, I argue that more and longer EVs occur in weak contexts, whereas fewer and shorter EVs occur in strong contexts.

This chapter is organized as follows. The first section presents the working assumptions upon which the study is based. The second part reports the results of a production experiment investigating the effects of linguistic factors on the frequency of occurrence of EVs.

3.1. Working assumptions

Based on the results of previous studies on the occurrence and duration of EVs (Ramírez, 2002, 2006; Bradley, 2003; Colantoni and Steele, 2006; Schmeiser, 2006), we will examine the effects of linguistic factors on the occurrence of EV. The factors to be analyzed are the following:

1. Type of liquid: presence of the rhotic /r/ versus presence of the lateral /l/ as the second in the cluster
2. Place of articulation of the Obstruent: dorsal (/k/, /g/), coronal (/t/, /d/) or bilabial (/p/, /b/)

3. Voicing of the onset element of the cluster (voiceless, voiced)
4. Stress: position of the cluster with respect to the stressed syllable in the word (pre-tonic, tonic or post-tonic)
5. Quality of the nucleic vowel (/a/, /e/, //i/, /o/, /u/)

Some variables analyzed in previous experimental studies are not considered in the present study because they present statistical redundancy[15]. For instance, Bradley and Schmeiser (2003) and Schmeiser (2007) examine the variables 'Order of constriction' and 'Homorganicity'. The variable 'Order of constriction' divides the data into clusters with back-to-front movement of the articulators (/g/, /k/ + liquid) vs. front-to-back order (/b/, /p/ + liquid). The variable Homorganicity examines the clusters based on the division of homorganic (/d/, /t/ + liquid) and non homorganic clusters ((/k/, /g/, /p/, /b/ + liquid). However, the factor 'Place of articulation' covers all of these variants, since the back-to-front clusters are the dorsal variant, clusters with front-to-back order are labial, and the homorganic clusters are coronal. Since these are not independent factors, a single variable, Place of Articulation, will suffice, and conclusions regarding order and homorganicity can be addressed as part of the interpretation of the results.

The experimental analysis presented in this chapter is based on the same data used for the analysis of the effect of speech rate on the EV (Chapter 4).

3.2. Hypotheses and Assumptions

We begin with the assumption that the main factor determining the occurrence of EVs is the perceptual recoverability of the cluster (Côté, 2001; Hall, 2003; Schmeiser, 2006; Bradley, 2007). Since EVs by hypothesis aid in cluster identification, we examine the additional

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[15] In a pilot study the redundancy of variables was confirmed statistically by the results of high colinearity among Homorganicity and Order of constriction with Place of articulation.

possibility that the prosodic context of the cluster can also either enhance or reduce discriminability. The basic idea is that if discrimination ability for consonants is enhanced by prosodic context, EVs will be either less likely to appear, or will appear with shorter durations.

We assume, along with previous researchers, that the perceptibility of a cluster is determined by the phonetic features of the segments forming the cluster. However, in contrast to previous studies, I propose that EV enhances the perceptibility of clusters with poor recoverability (weak contexts).

The basic assumptions stem from considering the lenition (weakening) of a segment as a change that usually entails a decrease in occlusion or an increase in sonority (Trask, 1996). More commonly it has been accepted as a scalar change. For instance, Lass (1984) considers that lenition is defined by "a scale of sonority (increased output of periodic acoustic energy) and a scale of openness (less resistance to airflow). Although lenition has usually been seen as a diachronic phenomenon, it has also synchronic effects, Lavoie (2001) points out that lenition has been mainly seen as a step on the way to deletion, as an increase in sonority, as a decrease in effort, and as a decrease in the duration and magnitude of the articulatory gestures. As a working definition, I adopt Lavoie's (2001) notion of lenition as "any alternation which yields a consonant that is articulated with a more sonorous manner of articulation or with less marked structure" (6).

From a production point of view it seems counterintuitive that the more sonorous segments are considered more lenis than less sonorous segments, which are considered fortis. Nonetheless the following section presents a scale of lenition (weakening) based on the degree of voicing of the segment.

3.2.1. Voicing

The hierarchy of consonant strength based on voicing is determined by the consonant's sonority levels: consonants with a high degree of voicing (sonority) are considered more lenis (weaker) than those with a low degree of voicing. Consonants with a high degree of voicing resemble vowels, which are more lenis than voiceless consonants, which are strong –more fortis. Following Lavoie (2001), in this chapter I will use the term *weakness* rather than lenition to avoid the diachronic connotations associated to the latter term.

I adopt the phonologically-based assumption that progressive weakening across segment types corresponds to a scalar increase in sonority (Hankamer and Aissen, 1974; Lavoie, 2001). The increase in sonority is a process of lenition in which a consonant with high sonority 'weakens' itself to resemble a vowel. The voicing scale is represented in (3.1).

(3.1) vowel ⟷ consonant
+ weak — - weak

I adopt the voicing hierarchy proposed by Zec (1995) as illustrated in (3.2). However, this hierarchy leaves language specific detail unstated.

(3.2) Sonority Hierarchy (after Zec, 1995)

Obstruent	↓	stronger (less sonorous)
Sonorant		
Vowel		weaker (more sonorous)

Zec's hierarchy presents voicing as a scale of weakening; consonants weaken by increasing in sonority from the less sonorous obstruents to the more sonorous vowels. The hierarchy is very general and requires language specific analyses to determine more detailed rankings. For example, in the cases of Spanish and English, Lavoie (2001) carried out a study

comparing the intensity of each segment. Regarding consonants, she found that within the Spanish liquids, the rhotic (/ɾ/) has lower intensity than the lateral /l/, whereas in English, there is no distinction between /r/ and /l/ in terms of intensity. Her overall results support the scaled down hierarchy of Zec (1995), as her results did not show differences between stops and fricatives but rather between voiced and voiceless sounds. Based on these findings, a phonetic hierarchy based on the degree of voicing in Spanish is presented in (3.3).

(3.3) Sonority Hierarchy (after Zec, 1995 and Lavoie, 2001)

Voiceless Obstruents
Voiced Obstruents
/r/
/l/
vowels

The implication of the assumption that weakening is a scalar increase in sonority (Hankamer and Aissen, 1974; Lavoie, 2001) is that in the consonant strength hierarchy, the strength of a consonant is in an inverse relation with its sonority level: the consonants with the lowest sonority level are the strongest consonants and those with higher sonority levels are the weakest ones. Therefore we should find that weaker consonants (and therefore weaker contexts) have more vowel-like features.

Regarding vowels, we adopt Vennemann's (1988) sonority hierarchy as presented in (3.4). According to this hierarchy, the least sonorous vowels are those with a more constricted air flow (high vowels) and the more sonorous are those with a less constricted airflow (lower vowels).

(3.4) Vowel strength hierarchy (Vennemann, 1988)

Strongest	high vowels (/i/ /u/)	Lowest Sonority
	mid vowels (/e/ //o/)	
Weakest	low vowels (/a/)	Highest sonority

Thus, the prediction that stems from the sonority hierarchy for consonants and vowels is that the rate of occurrence of EVs will increase in the context of weaker consonants and vowels. For consonants, it is expected that clusters with voiceless obstruents will create a strong context where there will be a lower occurrence of EVs, whereas in clusters with voiced obstruents the context is weak and a higher occurrence of EVs is expected.

As for vowels, we adopt Vennemann's (1988) hierarchy, according to which we can expect to find a higher occurrence of EVs in the context of low mid vowels /a/, and less occurrence of EVs in the context of the high vowels (/i/ /u/).

Following the sonority hierarchy it would be expected that the clusters formed with the low-intensity /ɾ/ would form a strong context and therefore there would be a lower occurrence of EVs, whereas clusters with the high intensity /l/ would form a weak context in which there would be a higher expected occurrence of EVs.

However, my experimental results will show that this is not the case: a higher rate of EVs is found in Cr clusters than in Cl clusters. I will argue that these results are not due to sonority discrepancies, but to the difference in duration of /ɾ/ with respect to /l/. The former is a short tap that requires EVs for recoverability despite the fact that it forms a more perceptible context (it

has lower sonority). In the case of /l/, its longer duration does not require an EV to ensure perceptual recoverability. I argue that there is a tradeoff between the occurrence of a high sonority /l/ and the perceptual need for EVs. That is to say that I argue for an interaction between the fortis or lenis characteristics of the singleton consonants, and other linguistic factors which influence the perceptibility of the segments within a cluster.

As discussed in Chapter 2, in terms of voicing, Blecua (2001) argued that voiced segments are inherently shorter, and that in the case of Cr clusters the tap is very short, so that more and longer EVs must appear in this context for lengthening compensation. Nonetheless, I argue that if this were the case, we would find a similar rate of EV occurrence in Cr and Cl clusters since both liquids (/l/ and /r/) are voiced.

Thus, I argue that in addition to the effect of the strength of the cluster's consonants, the nucleic vowel has also an effect in the strength of the consonant cluster. The vowel has a direct effect on the perceived identity of the liquid consonant (/r/ or /l/) since there are formant transitions between them (the vowel and the liquid consonant). I argue that a weak vowel will make the liquid also weaker, in which case more EVs will occur and more frequently. In contrast, in a strong consonant cluster fewer and shorter EVs will occur.

3.2.2. Consonant strength as duration and magnitude of gestures

From the point of view of Articulatory Phonology, speech articulation is represented by a sequence of articulatory gestures[16] that have different timing relations or properties, such as

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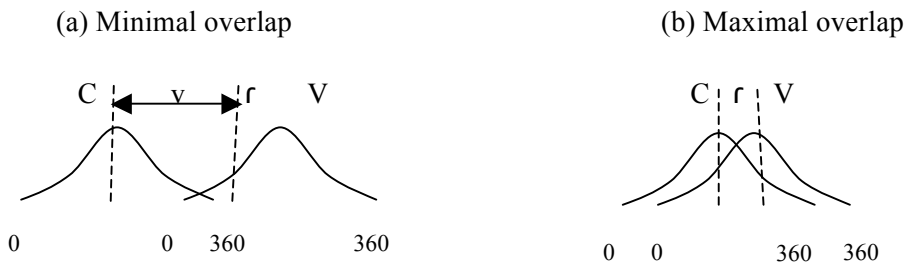
[16] We adopt, along with Browman and Goldstein (1992), the working definition of gestures as the characterizations of discrete, physical real events (movements of speech articulators) that unfold during the process of speech production. Those movements are represented by a bell shaped curve where (0) marks the initial position of the

stiffness or magnitude (contact of the articulators). In fast speech the gestures can be partially or completely overlapped by other gestures, and if a gesture is completely overlapped, it is not perceived by the listeners (Lavoie, 2001). A consonant cluster (Cr) with low gesture overlap and high exposure of the EV is presented in 3.5a (in cases with low gesture overlap perceptibility of the EV would be better), whereas 3.5b presents a high degree of gesture overlapping and obscuring of the EV.

(3.5). Patterns of gestural overlap for /Cɾ/ clusters



The degree of perceptibility of the segments is given by both the duration and magnitude of gestures (Browman And Goldstein, 1992). Lavoie (2001) argues that duration and magnitude are independent and thus they can reduce independently, especially in phonetic weakening of segments. Lavoie points out that "the decrease in a segment's duration cues the perception of voicing and the decrease in magnitude yields the perception of weakness", (20). Thus, a segment that is articulated with shorter duration than average is perceived as a voiced segment by the

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articulators, the peak represents the highest point of opening or closure, and then there is a return to the initial state (represented as the point 360 degree point).

listener. On the other hand, a segment that is "lazily" articulated (undershot articulation) is perceived as weak.

As for the factors that affect magnitude, Pierrehumbert and Talkin (1992) argue that accentuation increases gestural magnitude so that in a stressed position "vowels are more vocalic and consonants are more consonantal" (109).

In Spanish, stress is determined by a combination of pitch, duration and intensity[17]. It is generally accepted that in stressed syllables not only does the vowel nucleus undergo lengthening, but the consonantal segments do as well (Cuervo, 1867, 1972; Bolinger, 1961; Contreras, 1964; Navarro Tomas, 1964). The prediction in terms of consonant strength and duration is that there will be fewer EVs in the contexts of fortis consonants (stressed syllables) and there will be more EVs in the context of weak (lenis) consonants (unstressed syllables).

3.2.3. Place of Articulation

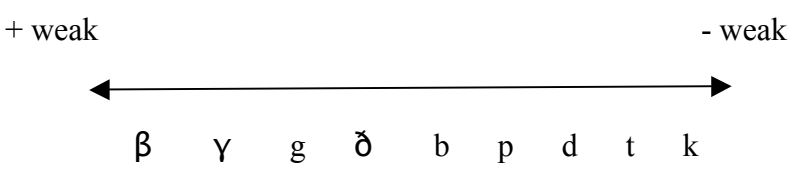
Lavoie (2001) conducted a study of the strength of Spanish and English sounds by their place of articulation. The author used Electropalatography, or dynamic palatography, in which a Palatometer provides contact information about the articulators over time. The strength of the consonants was measured in terms of the contact of the tongue with the soft palate at different positions (coronal and dorsal) and at the lips for labial sounds. The target segments were in the context of /o/ in different positions within the word and in stressed and non-stressed positions (i.e., ***t****oca* 'he/she touches; ***t****ocar* 'to touch, inf' ***p****osa* 'he puts down' ***p****osar* 'to put down, inf'; the

[17] There is controversy, however, as to whether the main cue to stress in Spanish is the pitch (Quilis, 1971, 1981; Gili Gaya, 1981; Llisterri et al., 2003) or the pitch and duration (Ortega-Llebaria and Prieto, 2007). This controversy is not relevant to the point in question and therefore it will not be developed.

target consonant is in bold font). She found that the "Spanish patterns differed almost exclusively by manner of articulation with no differences by position." (97). That is to say that dorsals ([k], [g] and [ɣ]), coronals ([t] [d] and [ð]), and labials [p], [b] and [β]) behaved in a similar way with no significant effect caused by the position within the word. However, the author points out that surprisingly the dento-alveolar approximant [ð] presented similar characteristics than stops even though it lacks a seal (which is typical only of the stop consonants and affricates), and that it shows more contact (between the tongue and the alveolar ridge) in stressed word-initial position than in other positions. She also notes that [t] has a tendency for more contact in non-stressed syllables (97). As with labials, the voiceless [p] presents more closure than the voiced [b] and [β] in any position. That is to say that in the articulation of [p], the articulators present full closure and more tension at the time of closure than in the articulation of [b] and [β].

We assume consonant fortition to be scalar in nature. We propose the scale in (3.6) based on the strength of the contact (as indicated by tension and area of contact) of the articulators involved (tongue and the soft palate) at the different positions (coronal and dorsal) and the lips (for labial sounds).

(3.6)



Hence, based on the scale of lenition presented in (3.6), the assumption regarding place of articulation is that among dorsals and coronals, the stronger context will be given by the

occlusives /k/ and /t/, which have more linguopalatal contact (area of contact) than the approximants /ɣ/ and /ð/. As for the labials, the more fortis segment is /p/. Thus, it is assumed that the strongest context among labials is in the vicinity of /p/. To sum up, it is expected that we will find a higher frequency of EVs in the context of weak consonants than in the context of strong consonants.

3.2.4. Predictions about prosodic position

It has long been accepted that the articulation and perception of segments is conditioned by their position within the word and their position with respect to lexical stress. For instance, Escure (1977) examined the positions in which lenition was more likely to occur. The author proposes the hierarchy in (3.7).

(3.7) Environment hierarchy (after Escure, 1977)

Word final	(weakest position)
Intervocalic position	
Word initial	(strongest position)

Research supports the claim that prosodic position is a factor in consonant strengthening or weakening. For instance, Pierrehumbert and Talkin (1992: 90) point out that in English the /t/ is aspirated in syllable initial position, it is glottalized when in syllable-final position, and it is unreleased and voiced throughout when flapped in an intervocalic falling stress position. They also point out that other unvoiced stops also have aspirated and glottalized variants depending on the word position in which they appear.

Several studies have shown that the position within the word has direct effects on the articulation of segments: word initial segments are generally clearer and less prone to processes such as assimilation and lenition, among others (McGlone et al., 1967; Fujimura and Sawashima, 1971; Kohler and Hardcastle, 1974; Benguerel, 1977; Fujimura, 1977; Hardcastle and Barry, 1985; Vaissière, 1988; Macchi, 1988; Krakow, 1989, 1993, and 1999; Byrd, 1994, and 1996; Browman and Goldstein, 1995; Keating, Wright and Zhang, 2001). Fougeron (1999) summarizes the findings of articulatory studies by pointing out that consonants and vowels in initial positions are more clearly perceptible because "the glottal opening gesture for consonants is longer and greater and vowels are glottalized or preceded by a glottal stop", (26). In addition, he points out that: "Labial muscular activity in initial consonants and vowels is greater. The velum is higher in initial oral and nasal consonants. The tongue is higher and linguopalatal pressure greater in consonants." (26). In addition to the articulatory properties of the segments based on their position, it has been shown that other perceptual cues are tied to position. Quené (1992) showed that in Dutch, not only are word-initial consonants longer than word-final consonants, but lengthening of the initial consonant can also serve as a perceptual cue to Dutch word boundaries, which is an important cue for word recognition and for phonotactic learning, because to "the extent that listeners have such signal-based cues, they do not need to rely only on statistics of phonological patterns and knowledge of the lexicon" (68).

Keating, Cho, Fougeron, and Hsu (2003) point out that some consonants seem more prone to exhibit word boundary effects than others. In a study of English segments, Keating et al. (2003) found that for /t/ /d/ and /ʧ/ the effects of position within the word are highly significant. They argue that those segments have more articulatory contact in initial position than in final position, which shows that the position in the word has a significant effect. Macchi (1988) and

Krakow (1999) claim that the effect of position within the word is weak with labial consonant articulations. Browman and Goldstein (1995), in a study of the peaks of consonant gestures (lip aperture, tongue tip height, tongue dorsum height), showed that boundary effects are much more noticeable for /t/ than for /p/ or /k/.

From a perceptual standpoint, Kohler (1992) argues that articulatory weakening, characteristic of word-final position, is listener motivated since the listeners focus their effort on segments that are "already highly perceptible (word-initial segments); final consonants are less important to listeners and therefore to speakers." (207). They argue that the reason for this shift in listeners' focus is due to the fact that lexical access is a left to right process, so that often by the end of the word the listener already knows what the word is.

In the case of Spanish, in terms of position within the word, Lavoie (2001) found that the longest voiceless stop consonants occur in stressed word-initial position, and the shortest consonants occur in medial unstressed position. Cole, Hualde, and Iskarous (1999) also found that position has an effect on the quality of a consonant. Specifically, they examined the variation of the velar voiced stop-spirant, [g] and [ɣ] respectively, in Castilian Spanish. The authors carried out an acoustic experiment to evaluate the influence of prosody and segmental context on the degree of spirantization of the velar segment in intervocalic contexts. Spirantization was evaluated in terms of acoustic energy; the results showed that stop-like utterances almost always occurred in utterance initial position, and much less variation in degree of constriction occurred in that context.

A propos of position within the word, it is predicted that the strongest context, and therefore the one with the lowest occurrence of EVs, will be in word-initial position, whereas the

weakest context will be in word-final position, where a higher rate of occurrence of EVs is expected.

In this study we use the term pre-tonic as equivalent to the word-initial position and post-tonic as equivalent to the word-final position due to the fact that the distribution of stress coincides with word position. That is to say that all the clusters in pre-tonic position are word-initial syllables and all the clusters in post-tonic position are in word-final syllables.

The assumptions with respect to production are summarized in Table 3.1.

Table 3.1.

Summary of assumptions per linguistic condition

Factor	more EVs (weakest position)	fewer EVs (strongest position)
Voicing	←	→
	vowel /l/ /r/	voiced obstruents voiceless obstruent
Word position (stress)	word final ←	→ word initial
Place of articulation	←	→
	β γ g ð b p d t k	
Type of vowel	←	→
	/a/ /e/ /o/ /i/ /u/	

The following section analyses the linguistic and prosodic factors and their effect on the rate of occurrence and duration of the EV.

3.3. Methodology

The data used in this analysis is the same data employed for the analysis of speech rate and EV (Chapter 4). For the sake of convenience I will present the main characteristics of the experimental design as follows:

3.3.1. Selection of words

The words used in this study were real words forming quasi minimal pairs, with the first word containing a cluster and the second word containing two full vowels (e.g., *prosa*, *porosa*). The clusters were analyzed in the context of all of the cardinal vowels /a, e, i, o, u/, and only tauto-syllabic clusters that are phonologically possible in Spanish were used: obstruents /p, t, k, b, d, g/ followed by the alveolar flap /r/ or the lateral /l/. Non-existent words were used where it was not possible to find a real word that formed a quasi minimal pair with the desired target. The distribution of the pairs according to place of articulation, type of liquid, voicing, stress and nucleic vowel is presented in (3.3). The number of pairs is indicated in parentheses.

(3.3) Distribution of pairs by condition.

1. bilabial (13), dental (13) or velar (11)
2. flap (24) or lateral (19)
3. voiced (18) or voiceless (25)
4. pre-tonic (13), tonic (18) or post-tonic (12)
5. /a/ (11), /i/ (10), /u/ (10), /e/ (6), and /o/ (6)

In addition to the 43 quasi-minimal pairs of words, 14 distractor pairs were included (Appendix 1 presents the complete list of quasi minimal pairs).

3.3.2. Participants

This exploratory study examines production from speakers from different Spanish-speaking countries[18]. All were adult native speakers of Spanish, including four females and four males. None of them reported having any speech or auditory difficulties. Their background information is presented in Table 3.2.

Table 3.2.

Background information for the eight participants.

Consultant	**Age**	**Gender**	**Country of origin**
C1F	33	F	Colombia
C2F	25	F	Mexico
C3M	31	M	Colombia
C4M	40	M	El Salvador
C5M	28	M	Peru
C6F	26	F	Spain
C7M	23	M	Mexico
C8F	19	F	Colombia

3.3.3. Recording procedure

The 100 test words were randomly divided into four groups of 25 words each in order to facilitate the reading task. Each word was embedded in a carrier sentence such as *repita __ normalmente* "repeat __ normally." The participants were asked to read the sentences two times in a natural manner. The data analysis comes from the second reading because it was considered more natural sounding than the first one, as assessed by two trained phoneticians.

[18] The low number of participants from each country did not allow us to analyze the country of origin as a factor.

3.3.4. Measurement procedure

The recordings were digitized and analyzed using Praat 4.6.40 software (Boersma and Weenink, 2010). The speech was low-pass filtered and digitized at a sampling rate of 22050 Hz and the measurements were made from wide-band spectrograms. The frequency of the vocalic formants was measured in the middle of each segment to avoid effects of the transitions between segments. Only instances where the epenthetic vowel could be clearly differentiated from the contiguous sounds were tallied and analyzed.

3.4. The occurrence of the epenthetic vowel

The occurrence of EVs in consonant clusters has been found to be especially variable (Blecua, 2001; Ramirez, 2001, 2006; Schmeiser, 2006). Variation in the occurrence of EVs happens among speakers and within speakers even in the same utterance. Visual identification of EVs in spectrograms proves to be difficult. In some cases, the spectrogram does not show a clear vocalic element, which is potentially due to overlap with contiguous elements. Another possibility is that an EV was simply not produced in that word, which unfortunately cannot be proved or disproved. Blecua (2001) argues that in some cases the epenthetic can be so short that it is confused with the burst of the liquid (117).

To examine the rate at which different variables contribute to the occurrence of the epenthetic vowel, an analysis of the data was conducted using Binary Logistic Regression. A logistic regression analysis produces a model that identifies the different factors that account for variability and proposes a ranking of the factors according to the predicted variability they contribute. That is to say that this logistic model identifies the variables that contribute significantly to the prediction of the occurrence vs. nonoccurrence of EVs, although it does not

weigh in the influence of individual subjects. To consider the effect of individual subjects, the data were analyzed with subjects as a covariate in the statistical model.

Section 3.4.1 presents the results of the model of occurrence vs. no occurrence of EV in Obstruent + liquid (/l/, /r/) clusters. Section 3.4.2. presents the results for the Cl clusters, and 3.4.3. presents the results for the Cr.

3.4.1. Occurrence of EVs in Obstruent + liquid (/r/, /l/)

The outcome of logistic model correctly classifies 55.7 % of the cases in which no epenthetic vowel occurred and 72.5 % of the cases in which it did. The overall percentage predicted is moderately good at 65.2%. (Table 3.3). Although this is far from a perfect model fit, it is still possible to determine which factors are significant and which are not.

Table 3.3.

The Observed and the Predicted Frequencies for the occurrence of the Epenthetic by Logistic Regression With the Cutoff of .5

		Predicted		
		Occurrence of the epenthetic		
Observed		No	Yes	% Correct
Occurrence of the epenthetic	No	64	51	55.7
	Yes	41	108	72.5
	Overall % correct			65.2

The Model Chi-square statistic shows that the independent variables, as a whole, significantly predict the occurrence or non-occurrence of the epenthetic vowel χ^2 (1, N = 264) = 30.084, p < .0001. Further analysis of the effects of individual factors was carried out.

3.4.2. Type of liquid

Table 3.4 presents the rate of occurrence and duration of the EV and NV by type of liquid. The logistic regression on the data (Table 3.5) shows that clusters with /l/ have an odds ratio of 0.217 with respect to the /r/, which is the baseline for the comparison (with a baseline value 1 or 100%). That is to say that Cl clusters have only a 22% probability of containing an EV compared to Cr clusters[19]. These results are in line with what was expected: the /l/ is longer and more sonorous than /r/ and therefore there is a tradeoff resulting in fewer occurrences of EVs.

Table 3.4.

Number of occurrences and average duration according to type of liquid

	No. of occurrences	**Duration**		
			EV	**Full**
/r/	75% (101/144)	M	19.06	87.60
		SD	6.56	23.15
/l/	40% (48/119)	M	21.61	79.35
		SD	13.59	22.95

[19] The baseline is decided by the coding used: /r/ = 1 and /l/ = 2

Table 3.5.

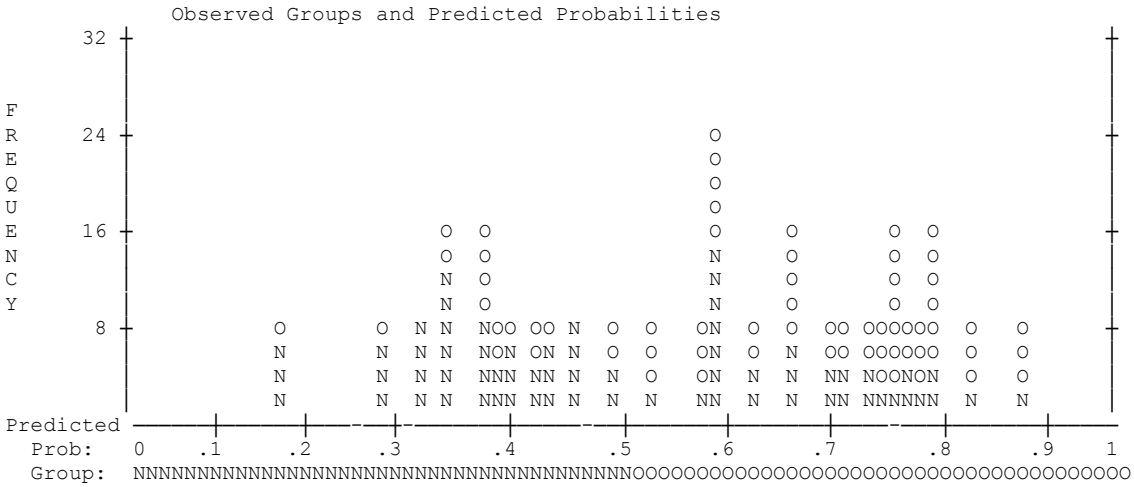
Logistic Regression Analysis of eight speakers' production of Obstruent + /l/, /r/ Spanish consonant clusters.

Predictor	B	SE β	Wald's χ^2	Df	*P*	e^{β} (odds ratio)
Constant	1.236	.528	5.489	1	.019	3.442
Type of liquid l r	-1.527	.308	24.584	1	**.000**	.217

The highly significant result for Type of liquid (p = .000) indicates that the occurrence of an epenthetic vowel is highly predictable by type of liquid (/l/, /r/).

The overall effectiveness of this model is illustrated in the classplot (Graph 2). Using this graph we can see that there is a higher probability of correctly predicting the occurrence of an epenthetic vowel, as opposed to predicting a non-occurrence. It can be observed in that more occurrences (O) are over the .5 predicted probability, whereas more non-occurrences (N) are below the .5 cutoff point.

Figure 3.1. Classplot of Predicted probability for occurrence and non-occurrence of epenthetic vowels.



Predicted Probability is of Membership for Occurrence. The Cutoff Value is .50
Symbols: N – Non-occurrence
O - Occurrence
Each Symbol Represents 2 Cases.

The results from the logistic model suggest that type of liquid is a major predictor of the occurrence of an epenthetic vowel. These results strongly suggest that each of the liquids (/l/ and /r/) have to be considered separately: they contribute to the occurrence of EVs at a different rate. Thus, a thorough analysis requires the examination of each cluster individually. The following section presents the results of the Obstruent + r and Obstruent + l as separate groups. The rate of occurrence and duration of the EV and the full vowel are presented in Table 6.

Table 3.6.

Number of occurrences and average duration according to linguistic condition

	Liquid	**No. occurrences**	**Duration**			
			EV		**Full Vowel**	
			M	**SD**	**M**	**SD**
Place of articulation						
Bilabial	l	14/48 (29%)	20.7	6.36	86.96	19.57
	r	33/48 (69%)	25.49	11.6	82.35	17.97
Coronal	l	12/24 (50%)	19.34	6.41	90.02	26.89
	r	36/48 (75%)	26.98	11.6	79.65	19.81
Dorsal	l	22/48 (46%)	20.71	6.35	81.87	18.13
	r	32/48 (67%)	28.93	11.53	86.11	23.11
Voicing						
Voiced	l	18/48 (37%)	19.22	6.92	86.96	19.57
	r	50/72 (69%)	29.11	11.09	82.35	11.17
Voiceless	l	30/72 (42%)	18.97	6.46	84.43	22.49
	r	51/72 (71%)	24.99	9.36	79.15	16.31
Stress						
Pre-tonic	l	15/40 (37%)	15.82	6.86	89.56	18.50
	r	29/48 (60%)	24.69	8.16	85.16	22.11
Tonic	l	14/40 (35%)	19.87	4.66	93.52	19.21
	r	40/48 (83%)	30.44	10.83	101.66	20.31
Pos-tonic	l	19/40 (47%)	21.03	6.17	84.07	19.19
	r	32/48 (67%)	28.93	11.53	79.90	24.86
Nucleic Vowel						
/a/	l	14/32 (44%)	18.46	5.16	92.63	19.84
	r	26/32 (81%)	26.67	10.74	86.55	19.97
/i/	l	16/32 (50%)	20/95	7.04	75.97	32.94
	r	10/16 (62%)	18.19	6.36	79.65	19.81
/u/	l	8/32 (25%)	22	4.98	84.83	13.73
	r	29/49 (59%)	28.93	11.53	86.19	20.19
/e/	l	8/24 (33%)	20.95	4.57	78.54	26.46
	r	19/32 (59%)	33.18	9.61	96.17	25.85
/o/	l	8/24 (33%)	21	8.60	80.26	24.34
	r	22/32(69%)	25.22	10.12	88.67	20.87

3.4.3. Frequency of Occurrence of EV in Cr

A logistic model of the occurrence of EV in the Cr clusters was calculated. As Table 7 illustrates, the model correctly predicts the occurrence of EVs at a rate of 95 %, whereas it predicts the non-occurrences at only 7%. The overall percent of successful prediction seems moderately good at 68.8%.

Table 3.7.

The Observed and the Predicted Frequencies for the occurrence of the Epenthetic occurrence in Obstruent + /r/ by Logistic Regression With a Cutoff of 0.50

Observed		Predicted		
		Occurrence of epenthetic		
		No	Yes	% Correct
Occurrence of epenthetic	No	3	40	7
	Yes	5	96	95
	Overall % correct			68.8

The results for the relative effect of each variable (Table 8) show Stress as the most powerful predictor of the occurrence of the EV in Cr clusters. This is shown by the significant effect of the pre-tonic (p = .025), which is the baseline and the indicator for the factor 'Stress'. Among the predictors within Stress, the post-tonic position is a strong predictor as compared to pre-tonic (which is the baseline for comparison with an assigned value of 1). The positive value of β (1.496) indicates that post-tonic stress is positively correlated with the occurrence of EVs (as compared with the baseline: the pre-tonic position). The results shown in the odds ratio column indicate that the odds of an EV occurring in a cluster in post-tonic position are 4.46 times greater than in the pre-tonic position, and that this predictor is statistically significant (p = .017). In the case of stressed (tonic) syllables, this context is negatively correlated. The odds of an EV

occurring in tonic position are 0.83 compared to the pre-tonic position. Thus, in working with logistic models, special attention is given to the odds ratio (e^{β}) value, and not too much to the significance level since the odds ratio indicates the contribution that a factor or a level within a factor makes to the model even if it is non significant. It is also important to note that the baseline for each factor (dorsal, pre-tonic, and vowel (a)) have empty cells in the Beta, Standard Error of Beta and the odds ratio columns. This is due to the fact that they are the baselines for comparison and therefore no value for comparison with themselves can be computed. In the case of factors with only two levels such as Voicing (voiced and voiceless) the value that appears is the result of the comparison with a baseline that does not appear in the table. We know which of the two levels is the baseline based on the coding of the data: the level that is assigned the lowest value is the baseline. For instance in the case of voicing, the data was coded 1 for voiceless and 2 for voiced, in which case the voiceless condition is the baseline for comparison.

The results for Stress support our prediction that EVs would occur more commonly in post-tonic position since, perceptually, it is the weakest. We can hypothesize that the motivating factor for the occurrence of EVs in the weakest context is perceptual recoverability in terms of production and perception.

Table 3.8.

Logistic Regression Analysis of eight speakers' production of Obstruent + /r/ clusters.

Predictor	β	SE β	Wald's χ^2	*df*	*P*	e^{β} (odds ratio)
Constant	.882	.726	1.475	1	.225	2.416
Place of articulation: dorsal			1.945	2	.378	
Place of articulation: coronal	-.942	.781	1.457	1	.227	.390
Place of articulation: bilabial	-.234	.625	.140	1	.708	.792
Stress: pre-tonic			7.377	2	**.025**	
Stress: tonic	-.186	.495	.142	1	.706	.830
Stress: post-tonic	1.496	.627	5.692	1	**.017**	4.462
Voicing	.356	.451	.625	1	.429	1.428
Vowel (a)			5.366	4	.252	
Vowel (e)	-.051	.738	.005	1	.944	.950
Vowel (i)	-1.062	.652	2.654	1	.103	.346
Vowel (o)	-.542	1.060	.261	1	.609	.582
Vowel (u)	.690	.647	1.138	1	.286	1.994

Test			Wald's χ^2	df	*p*	
Overall model evaluation						
Likehood ratio test			111090	1	**.043**	
Goodness-of-fit test Hosner & Lemeshow			4.197	7	.757	

-2 Log likelihood = 162.036, Cox & Snell R Square = .090, Nagelkerke R Square = .127
Note: The predictors order and manner of articulation are omitted from the table

Although the results from factors other than pre-tonic and post-tonic stress are not significant, they appear to be contributing to the model. Given the exploratory nature of this study, with only 8 speakers, it is worthwhile to examine the nonsignificant results as possible trends and a focus for future research.

The results for place of articulation (with dorsal being the baseline for comparison with the assigned value of 1) show that EVs have an odds ratio of .39 of occurring in coronal clusters as compared to dorsals. With bilabials, the odds are .79 of the likelihood of having an EV, as compared to the baseline condition. These results contrast with findings from previous studies (Ramirez, 2002, and 2006) in which a higher occurrence of EVs was found in the context of coronal consonants. A possible reason for the discrepancy is that in the previous studies the variables were analyzed individually, whereas the current study analyzes the effects of all the variables together in a regression model. That is to say that the current analysis does not look at the direct relationship of a factor (i.e., place of articulation) on the occurrence of the EV, but rather the effect of the factor along with all other factors (i.e., Voicing, Stress, Vowel, etc.) affecting the occurrence of the EV computed together. Moreover, previous work did not correct for multiple analyses on the same data in different combinations (e.g. with a Bonferroni correction). The different results are likely due to this new and more sophisticated analysis, which incorporates safeguards against spurious significance. Since the sample size for this study was based on previous findings, it is necessary to note that the new analysis in fact requires more data to obtain more statistical power and therefore to confirm or refute these findings.

3.4.3.1. Place of articulation for Cr

As noted previously, Lavoie (2001) found that there are no significant differences in sonority/strength due to place of articulation. I argue that for place of articulation, articulatory principles take precedence over perceptual recoverability. Furthermore, I argue that EVs are more likely to occur in dorsal position because the tongue (the active articulator) requires more displacement from the dorsum area (for the articulation of the consonant) to the tip of the tongue for the articulation of /r/. The longer movement allows the gestures to complete themselves and

the degree of overlap decreases, allowing EVs to surface. In the case of clusters with coronal consonants, there is a short movement of the tongue (since the consonant and the /r/ are both articulated with the tip). The relatively short movement of the articulator increases the overlap of gestures, and thus the EV is masked.

Thus, the ideal conditions for the occurrence and the duration of the EV is an interaction of factors: the occurrence of more EVs is more likely in those contexts in which both consonants are not strong enough, and the ideal conditions for longer EVs are those contexts in which, in addition to two consonants that are not strong enough, there is little overlapping of the consonants.

3.4.3.2. Voicing

In terms of voicing, the results from this exploratory study show that voiced clusters have a 42% (1.428) higher probability of having an EV compared to voiceless clusters (the baseline). The results conform to our assumption regarding voicing: more EVs are likely to occur in the context of a voiced consonant since it creates a weaker context as opposed to the voiceless consonants. Notice, though, that these results can also be interpreted as support for the assumption that more EVs occur in the context of voiced segments to compensate for the inherently short duration of the segments. Further discussion of the effects of voicing is presented in the review of the results for Cl clusters in section 3.1.3.

3.4.3.3. Nucleic Vowelicing

The results with respect to the nucleic vowel show that when the cluster occurs in the context of the high back vowel /u/, it is 99% (1.994) more likely to encounter an EV than when it is in the context of the low mid vowel /a/ (the baseline for comparison set at 1). In contrast, the

context of the /i/ vowel presents the lowest probability of presenting an EV (.34) when compared to /a/. The distinction in terms of nucleic vowels seems to be given by the difference in the frontness or backness of the vowels instead of the high-low difference that was initially assumed. Figure 3 shows the distribution of the occurrence of EVs by type of vowel.

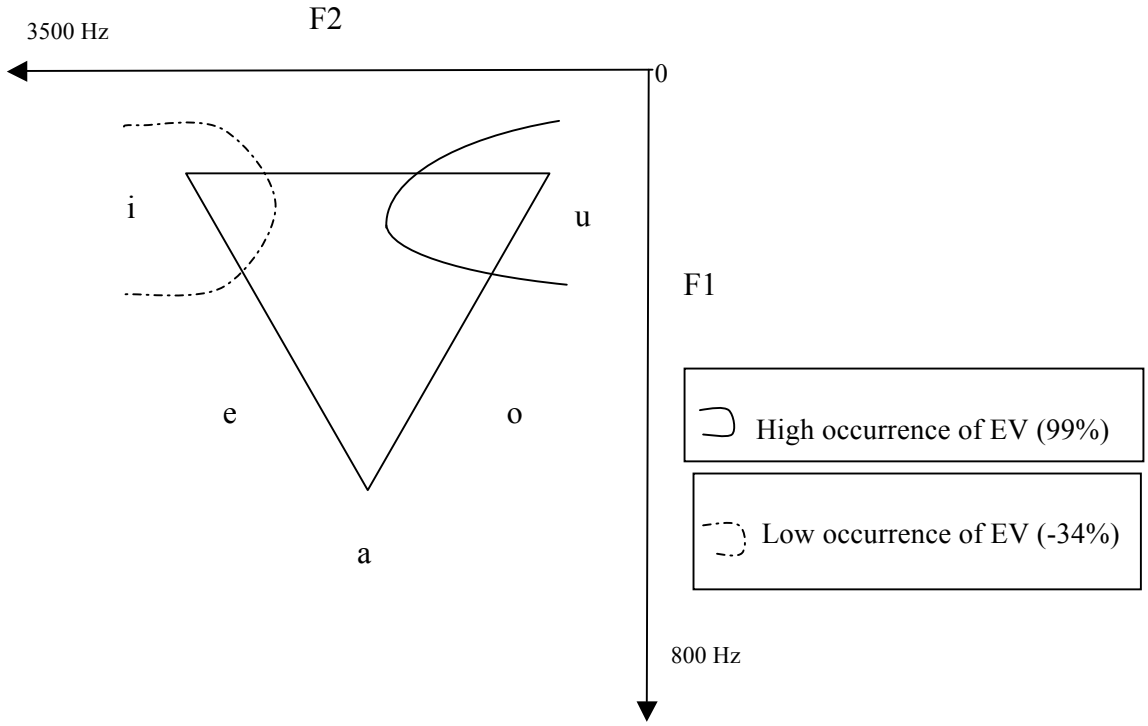


Figure 3.2. Vowel distribution by frequency of formants.

Figure 3.2 illustrates the distribution of the vowels by the frequency of their formants (F1 and F2). F1 usually varies from 300 Hz to 1000 Hz. The lower it is, the closer the tongue is to the roof of the mouth. The vowel /i/ (as in the word 'beet') has one of the lowest F1 values - about 300 Hz. In contrast, the vowel /a/ (as in the word 'bought') has the highest F1 value - about 950 Hz.

F2 measurements can vary from 850 Hz to 2500 Hz, as they are proportional to the frontness or backness of the highest part of the tongue during the production of the vowel. In addition, lip rounding causes a lower F2 compared with unrounded lips. For example, /i:/ as in

the word 'beet' has an F2 value of 2200 Hz, the highest F2 of any vowel. In the production of this vowel the tongue tip is quite far forward and the lips are unrounded. At the opposite extreme, /u/ as in the word 'boot' has an F2 measurement of 850 Hz; in this vowel the tongue tip is very far back, and the lips are rounded.

The tentative findings for type of vowel can be formalized using the OT framework (McCarthy and Prince, 1995). The main goal for using OT is to attempt to present a principled account of the interaction of phonetic information such as the Vowel factor and phonological constraints. This seems to be a more promising option for providing an account of phenomena such as EV. A purely phonological account is ruled out because it would require constraints that would act in proxy segments (a third segment). That is to say that it requires constraints that specify that when the segment or segments in a cluster are lenis, a new segment (in this case EV) would occur. This type of constraint could not be stipulated because it would be *ad hoc*, and thus, not generalisable to all cases. For further discussion see Côté 2000.

The set and ranking of the constraints that I propose is the following (I adopt Côté's 2000 markedness constraint):

C ◄► V: 'A consonant is adjacent to a vowel'.

That is to say, epenthesis occurs to ensure that every consonant is adjacent to a vowel. Formally, epenthesis is derived by ranking C ◄► V above IDEN, the faithfulness constraint that requires that the structure be maintained at the underlying and surface forms. IDEN is violated with the insertion of an EV. Tableau 1 shows the analyses for the syllables g*ri, gre, gra, gro,* and *gru* simultaneously. Within each syllable we consider the two possible candidates: g + r + vowel and g + EV+ r + vowel. Note that the underlying form, represented with a backslash (i.e., /gri/), is

the same for the two surface forms (represented in square brackets) since the EV occurs only at the surface (phonetic level) with no representation at the underlying level.

Tableau 5.

	C◄►V	F2	F1	IDEN
/gri/ [gri]	*			
/gri/ [gᵊri]		**		*
/gre/ [gre]	*			
/gre/ [gᵊre]		*	*	*
/gra/ [gra]	*			
/gra/ [gᵊra]		*		*
/gro/ [gro]	*			
/gro/ [gᵊro]			*	*
/gru/ [gru]	*			
/gru/ [gᵊru]			*	*

From Tableau 5, we can see that the most successful candidate (the one most likely to occur) is the one that presents the fewest violations of the constraint (marked with *). Thus, for the syllable *gri,* the tongue presents little vertical displacement (F1) and little horizontal displacement from the point of articulation of /r/ to the vowel (/i/). These violations are marked for F2 and F1 respectively. In addition to the violation of the low ranked IDEN, the candidate *gri* causes three violations. For g*ru*, there is the highest level of horizontal displacement of the language from the point of articulation of the /r/ to the point of articulation of the vowel (/u/). Therefore, there is no violation of the F2 constraint, but it presents little or no vertical displacement, causing a violation of F1. In total, *gᵊru* presents two violations, which makes it more likely to emerge than *gri*, which presents three.

3.4.4. Frequency of occurrence of EVs in Cl

The model proposed by the logistic regression is summarized in Table 9. The overall percentage correctly predicted is also moderately good at 66.7%. But, in contrast with the previous models, this one predicts the non-occurrence of EV at a higher accuracy (83.3%) compared to occurrence (41.7%).

Table 3.9.

The Observed and the Predicted Frequencies for Epenthetic occurrence in Obstruent + /l/ by Logistic Regression With a Cutoff of 0.50

		Predicted		
		Occurrence of epenthetic		
Observed		No	Yes	% Correct
Occurrence of epenthetic	No	60	12	83.3
	Yes	28	20	41.7
	Overall % correct			66.7

The analysis of the relative effect of each variable (Table 3.10) shows that the factor Place of articulation is a significant predictor, as shown by the significant value of the baseline category: dorsal (p = .003). Specifically, in the clusters formed with a labial consonant (/pl/ and /bl/), the odds of an EV occurring are 4.2 times higher than in the dorsal clusters (cl/ and /gl/), whereas in the cluster with the coronal (/tl/) the odds of occurrence are 67% (1.67) higher than in dorsals.

Table 3.10.

Logistic Regression Analysis of eight speakers' production of Obstruent + /l/ clusters

Predictor	B	SE β	Wald's χ^2	*Df*	*P*	e^{β} (odds ratio)
Constant	-0.134	.599	0.050	1	.824	0.875
Place of articulation: dorsal			6.693	2	**.003**	
Place of articulation: coronal	0.514	0.665	0.597	1	0.440	1.672
Place of articulation: bilabial	1.433	0.616	5.413	1	**0.020**	4.191
Stress: pre-tonic			1.610	2	0.447	
Stress: tonic	-0.644	0.648	0.985	1	0.321	0.525
Stress: post-tonic	-0.618	0.756	0.668	1	0.414	0.539
Voicing	0.321	0.568	0.319	1	0.572	1.378
Vowel (a)			3.350	4	0.501	
Vowel (e)	-0.667	1.119	0.355	1	0.551	0.513
Vowel (i)	-0.497	1.009	0.243	1	0.622	0.608
Vowel (o)	0.018	0.971	0.000	1	0.985	1.018
Vowel (u)	-1.333	0.888	2.253	1	0.133	0.264

Test			Wald's χ^2	Df	*P*	
Overall model evaluation						
Likehood ratio test			152.755	1	**.000**	
Goodness-of-fitt test Hosner & Lemeshow			6.510	6	**.369**	

-2 Log likelihood = 161.695; Cox & Snell R Square = .094; Nagelkerke R Square = .126
Overall model evaluation tests such as the Score test and Wald test were not reported in SPSS.

This exploratory analysis shows that the other strong contributor to the occurrence of EVs is voicing: the clusters with voiced consonants have a 37% (1.37) higher chance of presenting an EV than those with voiceless consonants. These results seem to support the assumption that EVs are more likely to occur in the weakest context (after voiced consonants).

In terms of voicing, both the Cr and Cl clusters yield the same results: the odds are greater for EVs to occur in the context of a voiced consonant than a voiceless consonant. These results argue against the hypothesis that EVs occur more often in Cr clusters with voiced

consonants (C) simply because the EV compensates in lengthening for the inherently shorter consonants. If that were that the case, we would expect to find a different trend in Cl clusters where the /l/ has a longer duration than the /r/. Instead, we argue that the results support the assumption that the EV occurs in the weakest context, motivated by perceptual recoverability.

As for stress, there are higher odds that an EV will occur in the pre-tonic position, at least descriptively speaking. Tonic position and post-tonic position present lower odds (53% and 54% respectively) compared to the pre-tonic position. These results do not support our assumption that EVs are more likely to occur in the weakest context – the post-tonic position.

As discussed above, Cr clusters had more EVs than Cl clusters; we argue that the main difference resides in the duration of the liquid. Since the /l/ is long, there is a trade-off with the EV, whereas there is no such trade-off with the short tap /r/. The results for stress confirm the fundamental differences between these clusters. Since EVs in Cl clusters do not play such an important role in perceptual recoverability, they follow the production and perceptual tendencies established previously: segments are more clearly articulated in the word-initial position. This more careful articulation includes a more frequent occurrence of EVs.

Regarding the type of vowel, there is a lower chance for an EV to occur in the context of the vowel /u/ (.26) when compared to the baseline /a/, and this is marginally significant. Similarly, in the context of /e/ and /i/ we also see lower odds of finding an EV (51% and 60% respectively). Figure 3.3 illustrates the percentage of occurrence of each vowel.

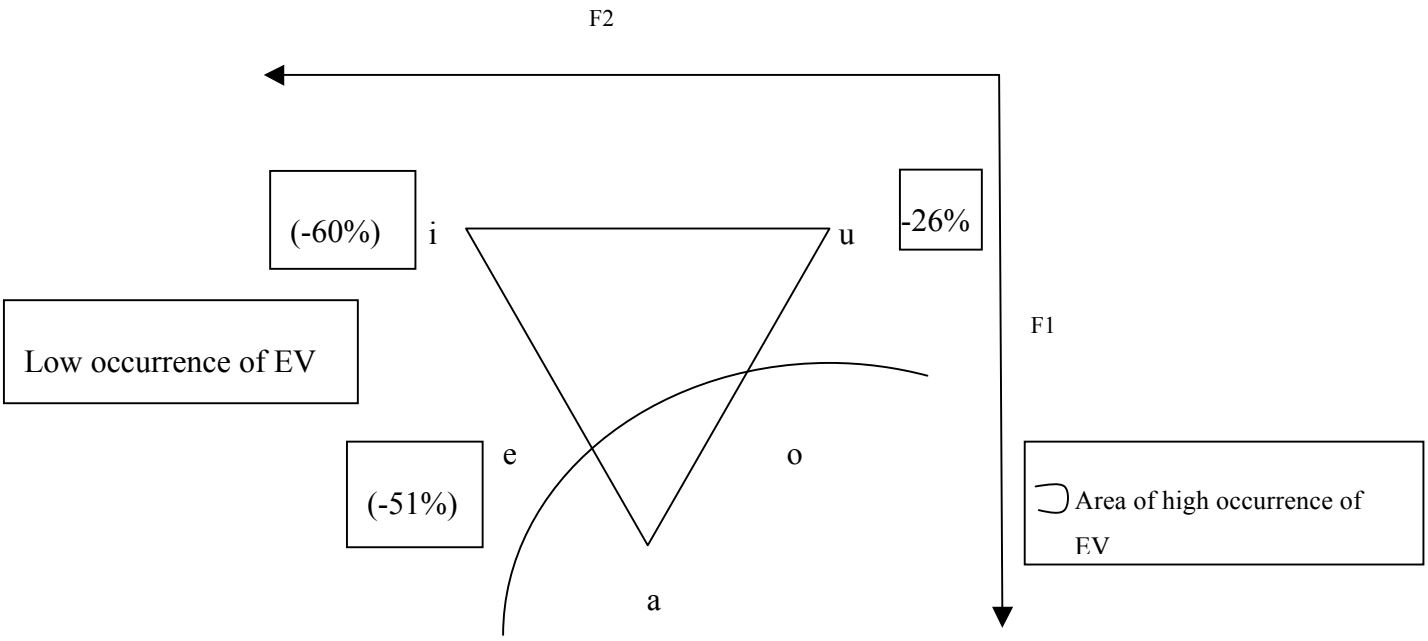


Figure 3.3. Percentage of occurrence of EV by vowel for Cl.

In contrast to the Cr clusters, the vowels in Cl clusters show F1 as ranking higher than F2. That is to say that the syllables with vowels that differentiate themselves from the /l/ along the vertical axis are more likely to contain an EV. In the case of *blu*, (Tableau 6) there is little displacement along the vertical axis (F1). This violation is marked with a double asterisk since it is a stronger violation that those violations to F2 displacement. In total, *blu* presents three violations, whereas *bla* presents only two (F2 and IDEN); therefore *bla* is preferred over the other candidates.

Tableau 6.

Contraints F1, F2 and IDEN

	C ◄►V	F1	F2	IDEN
/bli/ [bli]	*			
/bli/ [bᵊli]		**	*	*
/ble/ [ble]	*			
/ble/ [bᵊle]		*	*	*
/bla/ [bla]	*			
/bla/ [bᵊla]			*	*
/blo/ [blo]	*			
/blo/ [bᵊlo]			*	*
/blu/ [blu]	*			
/blu/ [bᵊlu]		**		*

To sum up, the results in Cr clusters contrast with those in Cl clusters. In syllables with /u/ the Cr clusters are more likely to present an EV, whereas in Cl clusters they are the least likely to surface. These results may be explained by the articulatory dynamics of the consonant clusters. In the case of Cr clusters, movement between the first consonant and /r/ and /u/ involves a change in the overall position of the tongue body, while there is virtually no change in tongue body position for movements between /r/ and /i/ where EVs are less likely to be found.

I suggest that the effect of adjacent vowel quality on the production of EVs is a consequence of the dynamics of tongue body movement. Movement of the tongue body is necessary to produce a consonant closing gesture, whether the movement is in the horizontal dimension, the vertical dimension, or both. In the pronunciation of the cluster, the tongue body position is changed in the movement between the consonant and /r/, and the adjacent vowel, as part of the articulation plan. Thus in the pronunciation of /r/ and /u/, the tongue has to be displaced rapidly from the articulation of the /r/ with the apex of the tongue and the /u/ with the dorsum of the tongue. The displacement of the tongue between /r/and /u/ is longer than between

/r/ and /i/, since the latter is articulated with the front of the tongue. The longer displacement requires that the closing gesture for /r/ be more fully realized, and thus the EV occurs so that the segments can be differentiated from each other. However, when no overall change in the tongue body position is required, as in the movement between /r/ and adjacent /i/, the closing gesture for /r/ is not effectively realized, and it is likely that we will see a higher level of overlapping of the gestures.

This account of vowel quality as a conditioning factor in the occurrence of EVs suggests that different vowel quality effects will be found for the labial, coronal and dorsal consonants due to differences in the dynamics of the CvCV gestures involved. However, the few data points per place of articulation do not provide fully reliable results. This will be a matter for future research.

Another possible approach to explaining the occurrence of EV would be the use of perceptually-ranked faithfulness constraints (Fleming, 2003). However, the ranking constraints for cases of consonant deletion, according to the strength of the cues to the presence of a consonant, can only allow deletion of poorly-cued consonants, but it cannot motivate epenthesis to improve the cues to a consonant (see Fleming 2003 for discussion).

3.5. Conclusions

The statistical models proposed in this study show that the occurrence or non-occurrence of the epenthetic vowel is greatly affected by the type of liquid (/l/, /r/) in the cluster. Despite the fact that both are coronal liquid sounds, they behave very differently. Further analysis of each type of cluster shows that different factors have different weights.

Table 3.11.

Results by linguistic and prosodic predictors

Predictor	Cr clusters	Cl clusters
Voicing	voiced > voiceless	voiced > voiceless
Position within the word (Stress)	pos-tonic > tonic > pre-tonic	pre-tonic > tonic/pos-tonic
Place of articulation	dorsal > bilabial > coronal	bilabial > coronal > dorsal
Type of vowel	u a e o i +EV ⟷ -EV	a o i e u +EV ⟷ -EV

A summary of the results is presented in Table 11. The initial predictions with respect to the occurrence of EVs in Cr clusters are supported for voicing and stress. Regarding the Cl clusters, the initial assumptions hold for factors such as type of vowel and position within the word.

The results of the type of vowel based on the effects of frontness vs. backness of the vowels were an unexpected finding, whether we consider the EV to be perceptually motivated or employed to avoid gestural overlap. The hypothesis about perceptual motivation requires that the EV occur in the most perceptually weak context. Since the mid-low vowel /a/ is the weakest in perceptual terms, we would expect EVs to occur at a higher rate. Our data reveal that this is not the case. Similarly, since the high vowels /u/ and /i/ create the strongest context we would expect to find fewer occurrences of EVs in their context. The results partially support this assumption. The highest probability of occurrence of EVs is in the context of the high-back vowel /u/ whereas the lowest probability is in the context of high-front vowel /i/. The behaviour of EVs with respect to NVs seems to be mainly determined by articulatory factors, specifically the displacement of the tongue (the active articulator). In cases where there is more displacement of the tongue, the articulatory gestures are clearly completed, while in cases of short displacement,

such as in homorganic coarticulations, there is a higher degree of overlapping of the gestures; thus, the EV tends to be obscured by the neighboring segments.

Chapter 4

The Effect of Speech Rate on EV

This chapter examines the effects of speech rate on epenthetic vowels. The first part presents a review of the effects of speech rate on vowels. The second part presents an experiment on the effects of speech rate on EVs. The third part presents the results and discussion, and the fourth part presents the conclusions.

The aim of this experiment is to identify whether the rate of speech has an effect on the frequency of occurrence or the duration of EVs, or whether it combines with linguistic and prosodic factors as part of an articulatory plan.

I argue that the EVs adapt to changes in speech rate as part of the articulatory plan in which the goal is perceptual recoverability by the listener. Thus I argue that at faster speech rates (a less perceptually recoverable context), EVs are more frequent and of longer duration to ensure recoverability of the cluster. Notice, however, that this goes against the simplest hypothesis, which would be that as speech rates increase, all segments, including EVs, would be shorter in duration.

The perceptual motivation for EVs requires that at a low speech rate, when word duration is longer, the segments are more clearly articulated, and the articulatory gestures have less overlap. Thus, I argue that in this case shorter EVs (or none at all) are required to enhance the perceptibility of the consonant cluster. In contrast, at a high speech rate, when word duration is shorter, the segments are less clearly articulated. The articulatory gestures overlap, creating a less

perceptually recoverable context. I argue that in this context there will be more and longer EVs to enhance the recoverability of the consonant cluster.

The occurrence of more and longer EVs would demonstrate the role of EVs in boosting the perceptual salience of consonant clusters under adverse listening conditions. It would also indicate that the speaker's use of EVs is sensitive to their effect on cluster recoverability.

4.1. The role of speech rate on vowels

One of the main factors that introduces variability in speech is speech rate (tempo) (Miller, 1981). Research has shown that within the speech signal the most elastic part of a syllable are its vowels, especially its vowel nucleus (Gay, 1978; Kozchevnikova and Chistovich, 1965), and that most changes due to speech rate (utterance duration) occur in the vowels (Miller, 1981:49). A faster speech rate affects vowels mainly in two forms: it makes them shorter, and it changes the frequency of their formants, generally resulting in undershooting of their targets. It is also accepted that the duration of the vowels and the absolute and relative values of the formants of a vowel vary systematically from speaker to speaker and even within speakers (Joos, 1958).

The effect of speech rate on vowel duration has been attested to experimentally. In one of the classic studies, Lindblom (1963) studied the effect of a change in tempo and stress on eight Swedish vowels in a consonant-vowel-consonant (CVC) context. He found that as the duration of the syllable decreased, as well as in most of the unstressed contexts, the vowel decreased in duration and the values of the vowels formants (F1, F2, and F3) undershot their targets, giving them more of a schwa-like character (vowel reduction). Lindblom (1963: 1780) proposed that "the pattern of neural commands to a given vowel remains constant across changes in stress and rate (and consonant context); however, due to physiological constraints, the vowel targets cannot

be fully realized during fast speech, resulting in the undershoot of target values". However, in a similar study on English vowels, Verbrugge and Shankweiler (1977) did not find vowel reduction (i.e., alteration in formant frequencies), but they did find a reduction in vowel duration. Furthermore, Stevens and House (1953) found that as a product of speech rate, vowels embedded between consonants often fail to reach their formant frequency targets and end up getting assimilated toward the values of the surrounding consonants. In short, there are consistent results showing that at a higher speech rate vowels have shorter durations. However, the results are inconsistent regarding the effect of speech rate on vowel reduction.

It is accepted that variability in the speech signal means that the listener needs to adjust not just to variation among speakers, but also to variation within the same speaker (Bard, Anderson, Sotillo, Aylett, Doherty-Sneddon, and Newlands, 2000). Even though speech rate was not controlled in the present study, as the stimuli were not collected at different speech rates, it is crucial to identify whether the rate of speech has an effect on the frequency of occurrence or the duration of EVs, or whether it combines with linguistic and prosodic factors as part of an articulatory plan. To this end, an analysis was carried out to identify whether the EV and the NV are affected in similar ways or not by the speech rate. This information will shed light on whether they are part of the same articulatory plan or not.

In order to study the effect of speech rate on the duration of the epenthetic vowel in Cr and Cl clusters, an experiment was set up to study the proportion of the duration that the EV takes up with respect to the word's total duration. The null hypothesis is that speech rate affects the duration of the vowels (both EVs and NVs) in the same manner. That is to say, a positive correlation is expected in which shorter NV and EVs will both occur at a higher speech rate. Similarly, at low speech rates, we expect to find long NVs and EVs. Furthermore, we can also

expect that we will find that the effects of speech rate on clusters formed by C + r with respect to C + l clusters will be different. The main differences are expected because of dissimilarities in duration since [l] (*M* = 68.52, *SD* = 17.46) is significantly longer than [r] (*M* = 23.45, *SD* = 6.05. In addition, the difference in sonority could be a factor: [r] is less sonorous than [l]; thus, we expect that the former would be more recoverable.

4.2. Methodology

4.2.1. Selection of words

The words used in this study were (mainly) real words forming quasi minimal pairs, with the first word containing a cluster and the second word containing two full vowels (e.g., *prosa*, *porosa*). The clusters were analyzed in the context of all of the cardinal vowels /a, e, i, o, u/, and only tautosyllabic clusters that are phonologically possible in Spanish were used: obstruents /p, t, k, b, d, g/ followed by the alveolar flap /r/ or the lateral /l/. Non-existent words were used where it was not possible to find a real word that had the properties needed. The distribution of the pairs according to place of articulation, type of liquid, voicing, stress and nucleic vowel is presented in (4.3). The number between parentheses represents the number of pairs.

(4.3) Distribution of pairs by condition.

1. bilabial (13), dental (13) or velar (11)
2. flap (24) or lateral (19)
3. voiced (18) or voiceless (25)
4. pre-tonic (13), tonic (18) or post-tonic (12)
5. /a/ (11), /i/ (10), /u/ (10), /e/ (6), and /o/ (6)

In addition to the 43 quasi-minimal pairs of words, 14 distractor pairs were included (Appendix 1 presents the complete list of quasi minimal pairs).

4.2.2. Participants

The data used for this analysis are the same used in Chapter 3. The participants are eight native speakers of Spanish, four females and four males.

4.2.3. Recording procedure

The 100 test words were randomly divided into four groups of 25 words each in order to facilitate the reading task. Each word was embedded in a carrier sentence such as *repita __ normalmente* "repeat __ normally." The participants were asked to read the sentences two times in a natural manner. The data analysis comes from the second reading because it was considered more natural sounding than the first one. The participants were recorded using a Sony cassette recorder, model TCM-453V, and a multidirectional lapel microphone. The cassettes were Type 1 normal bias.

4.2.4. Measurement procedure

The recordings were digitized and analyzed using Praat 4.6.40 software (Boersma and Weenink, 2010). The speech was low-pass filtered and digitized at a sampling rate of 22050 Hz and the measurements were made from wide-band spectrograms. The frequency of the vocalic formants was measured in the middle of each segment to avoid effects of the transitions between segments. Only instances where the epenthetic vowel could be clearly differentiated from the contiguous sounds were tallied and analyzed.

4.2.5. Analysis procedure

For the study of the effect of speech rate on the duration of vowels two analyses were conducted. First, as a measurement of a more global speech rate, the duration of the carrier sentence (Repita ____ normalmente) minus the duration of the target word was compared against the duration of the EV. In the second analysis, the duration of the EV in milliseconds was converted to a proportion with respect to the word's duration. For instance, an EV of 25 ms in a word lasting 400 ms has a proportion of 0.0625 (25/400). Using the proportion that the EV occupies in the overall word allows us to compare the EV and the word's duration. In this case, the word's duration was taken as an indicator of speech rate. The cases in which no EV was identified were not considered; only cases with an EV of at least 10ms were considered for analysis.

4.2.6. Experimental design

For the study of the correlation between vowel proportion and word duration, the analysis was conducted by items rather than by participants.[20] Since all speakers did not have an EV in the same words, their data were pooled and the analysis was restricted to words containing EVs.

In addition, an analysis was conducted on the correlation between the duration of the EV and the nucleic vowel in the same syllable and the full vowel in the previous syllable. For example, in tokens such as *escudriñar* [eskud^{i}riñar], "to pry", the EV in the syllable *d^{i}ri* was measured against the [i] in the same syllable and against the [u] from the previous syllable (cu).

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[20] "Participant" was not analyzed as a condition because in a pilot analysis interspeaker variability did not produce new information other than the well-attested data on variability among speakers.

4.3. Results and discussion

4.3.1. Duration of the epenthetic vowel

Figure (4.1) shows the spectrogram of the word *grupo* "group", as produced by speaker C1F. It is clear that the epenthetic vowel is an independent element and does not constitute part of the burst of the occlusive or part of the following liquid.

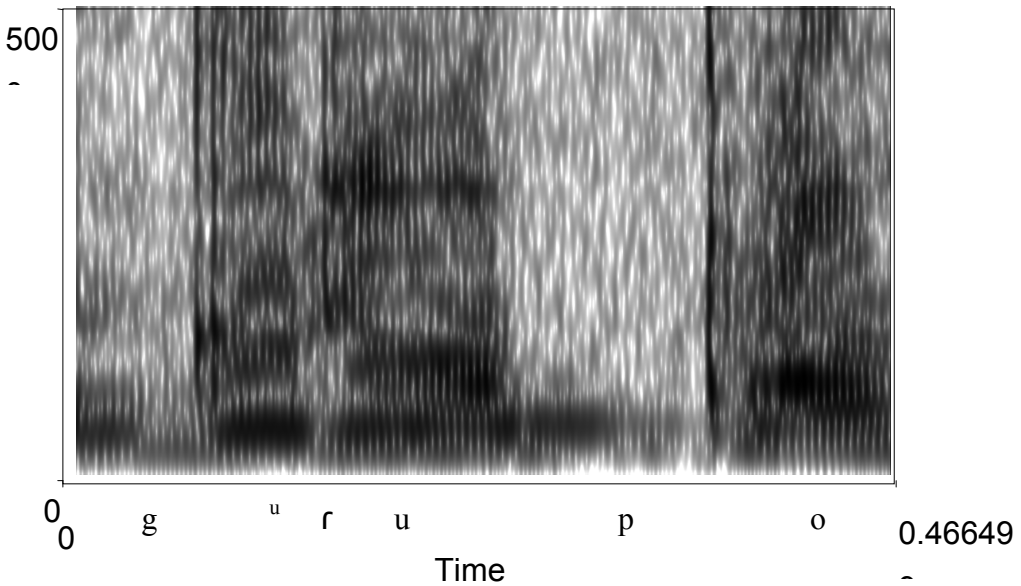

Figure 4.1. Spectrogram of *grupo* ‘group’ as produced by a native Spanish speaker

In the quasi-minimal pairs (i.e., *platal* ‘silvery’ and *palatal* ‘palatal’), the average duration of the EV (such the one in the clusters *p^{a}la* in the word *p^{a}latal*) and the full vowel (such the vowel in the syllable *pa* in *palatal*) is presented in Table 4.1. In addition, Table 4.2 presents the average duration of the full vowels of the non-cluster words before and after the liquids, that is, the measurements of the vowels before and after the lateral /l/ in cases such as the first and

second /a/ in *palanca* 'lever', and the /a/ before and after the flap (/r/) such as in *barado* 'broken down'.

The duration of the epenthetic constituted an average 27% of the duration of the full vowel. The vowel before a liquid was, on average, 27% shorter than the vowel after a liquid because the vowel in pre-liquid position was always in a non-stressed position, whereas the vowel in post-liquid position occurred in both stressed and non-stressed positions.

Table 4.1.

Average duration of epenthetic vowel versus full vowel for cluster words

	M	**SD**
Epenthetic vowel	25.08	10.23
Full vowel	92.23	24.22

EV = Epenthetic vowel, M = mean (measured in ms), SD = Standard deviation

Table 4.2.

Average duration of a full vowel before and after a liquid in non-cluster words

	M	**SD**
Full vowel before liquid	76.95	25.25
Full vowel after liquid	92.26	24.22

As a measure of speech rate, I measured the duration of the carrier sentence '*repita ___ normalmente'* minus the duration of the target word. Then its correlation with the EV and NV was calculated. The results show that the variables carrier sentence (M = 525.5, SD = 373.3) and the EV duration (M = 17.2, SD = 14) were not correlated, $r(168) = .122$, $p = .114$. Also, the duration of the carrier sentence (minus the target word) was not correlated to duration of the nucleic vowel (NV) (M = 92.49, SD = 24.68) $r(172) = .020$, $p = .794$.

Close inspection of the recordings shows that in many cases the speakers make a small pause before the target word. Thus, the duration of the carrier sentence maybe affected, and the may not be the best indicator of speech rate.

For further analysis of the effect of speech rate, I examined the effect of speech rate at the word level, by linguistic condition. The linguistic conditions analyzed were type of liquid, voicing, stress, manner of articulation, and place of articulation. The distribution of the items by condition is presented in Table 4.3.

Table 4.3.

Distribution of duration of epenthetic vowels (EV) and nucleic vowels (NV) by linguistic condition

		EV Mean duration (SD)	NV Mean duration (SD)	N
Type of liquid	/r/	18.8 (15.2)	90.7 (24.1)	111
	/l/	6.3 (9.7)	54.5 (45.3)	141
Voicing	Voiced	12.4 (15.2)	71.4(38.6)	128
	Voiceless	12.8 (13.1)	66.6 (44.2)	125
Stress	Pre-tonic	9.7 (12.2)	70.0 (38.5)	87
	Tonic	15.5 (15.9)	68.7 (44.5)	84
	Post-tonic	12.5 (13.7)	72.9 (41.5)	81
Manner of articulation	Occlusive	12.8 (13.1)	69.6(44.2)	125
	Approximant	12.4(15.2)	71.4 (38.6)	130
Place of articulation	Dorsal	14.4 (14.9)	74.4 (39.9)	89
	Coronal	12.5 (13.8)	68.8 (43.5)	82
	Labial	10.9 (13.6)	67.2 (41.0)	86

The correlation between mean word duration (M = 525.5, SD = 93.2) and mean EV proportion (M = .0443, SD = .016) was highly significant $r(40) = -.379$ $p = .016$. The negative correlation indicates that the longer the duration of the word, the shorter the EV's duration, and *vice versa*: the shorter the word duration the longer the EV.

In order to identify the amount of variance in an EV's duration that is accounted for by the word's duration, R-squared was calculated. The results of squaring r $(-.379)^2 = .14$. Thus, the analysis of speech rate and the EV's proportion shows that word duration (speech rate) accounts for 14% of the variability in the duration of EVs.

The results showing that shorter EVs occur with longer word duration is consistent with the perceptual motivation for EVs. I argue that at a low speech rate, when word duration is longer, the segments are more clearly articulated, and the articulatory gestures have less overlap. Thus, shorter or no EVs are required to enhance perceptibility, whereas at high speech rates there is overlapping of segments; and therefore, more and longer EVs occur to enhance perceptibility.

Further analysis of the correlation between the EV's proportion and the neighbouring vowels in each type of cluster was performed. The results are presented by consonant cluster.

4.3.2. Cr clusters

For the Cr clusters there was no significant correlation either between the proportion of EV with respect to the whole word ($M = .0508$, $SD = .027$) and the nucleic vowel in the same syllable ($M = 95.97$, $SD = 25.4$, $N = 107$) [$r(122) = .133$ $p = .173$], or between the EV's proportion and the nucleic vowel in the previous syllable ($M = 78.35$, $SD = 25.07$ $N = 108$) [$r(108) = -.025$ $p = .801$]. There was also no correlation between the EV proportion ($M = .018$, $SD = .0197$ $N = 95$) and the word's duration ($M = 540$, $SD = 120.6$ $N = 95$)) [$r(95) = -.067$ $p = .521$]. The lack of correlation with any NV suggests that the EV is not a part of the same articulatory plan as the NV.

4.3.3. Cl Clusters

The results for the Cl cluster show a significant correlation between EV proportion (*M* = .0361, *SD* = .019) and word duration (*M* = 555.3, *SD* = 115.1) $r(46) = -.444$ $p = .002$. There was also a significant correlation between EV proportion and the nucleic vowel duration of the previous syllable (*M* = 72.15, *SD* = 26.29) $r(43) = -.348$ $p = .022$. However, there was no significant correlation between the EV's proportion and the tautosyllabic nucleic vowel (*M* = 85.8, *SD* = 23.6, $r(42) = -.066$ $p = .678$). The inverse correlation between EVs and word duration (-.348) indicates that in faster speech (i.e. when word duration is shorter), EVs become relatively longer. In the same way, when the previous nucleic vowel is shorter due to fast speech, EVs are relatively longer. These changes in EV duration appear in response to changes in speech rate. This suggests that perceptual recoverability is the determining factor: in fast speech the elements become shorter and more difficult to recover, thus the EV becomes longer to ensure the cluster's perceptibility. Nevertheless, the lack of statistical correlation between the EV and the NV in the same syllable suggests that they are not part of the same articulatory plan.

It is not clear why there is a negative correlation between the EVs and the full vowels in Cl clusters, but no correlation in Cr ones. It could be argued that the negative correlation in the Cl clusters is due to the higher sonority of the /l/ with respect to the /r/. The Cl cluster, due to the higher sonority of the /l/, would require a longer EV to be perceptually recoverable during speech at a high rates, whereas the less sonorous Cr clusters are less affected by the speech rate. Another possible factor could be the length of the liquid. I argue that at higher speech rate, the /l/ tends to become shorter and thus the cluster becomes harder to perceive. Under such conditions of poor recoverability, more and longer EVs occur to ensure the recoverability of the cluster. In contrast, at slower speech rates the /r/ maintains its duration and therefore fewer and shorter EVs

occur, since the recoverability of the cluster is high. In the case of Cr clusters, the /r/ is a short segment (average 17ms) that does not get substantially shorter in high speech rate because it could fall under the perceptibility threshold. I argue that in the case of Cr clusters, speech rate is not the main factor affecting the occurrence and duration of EVs.

Further analysis of the correlation between the EV's duration (not its proportion) and the duration of the NV was performed. The results (presented in Table 4.4) show no correlation between EV and NV by linguistic condition (type of liquid, (/r/ or /l/), voicing, manner or place of articulation). The only significant correlation was between the duration of the EV and the NV in stressed syllables r(84) = .236, p = .030. This correlation is explained by the effects of lexical stress, which is expressed by lengthening of the segments of the stressed syllable. Thus, the NV as well as the EV behave similarly in that they both undergo lengthening under lexical stress.

Table 4.4.

Re*sults for correlation analysis of the duration of EV and NV by linguistic contexts*

		EV Mean (SD)	NV Mean (SD)	N	Pearson correlation	*P*
Type of liquid	/r/	18.8 (15.2)	90.7 (24.1)	111	.144	.129
	/l/	6.3 (9.7)	54.5 (45.3)	141	.125	.764
Voicing	Voiced	12.4 (15.2)	71.4(38.6)	128	.128	.147
	Voiceless	12.8 (13.1)	66.6 (44.2)	125	.166	.063
Stress	Pre-tonic	9.7 (12.2)	70.0 (38.5)	87	.191	.073
	Tonic	15.5 (15.9)	68.7 (44.5)	84	.236	.030*
	Post-tonic	12.5 (13.7)	72.9 (41.5)	81	.000	.998
Manner of articulation	Occlusive	12.8 (13.1)	69.6(44.2)	125	.166	.063
	Approximant	12.4(15.2)	71.4 (38.6)	130	.128	.147
Place of articulation	Dorsal	14.4 (14.9)	74.4 (39.9)	89	.150	.161
	Coronal	12.5 (13.8)	68.8 (43.5)	82	.127	.257
	Labial	10.9 (13.6)	67.2 (41.0)	86	.134	.219

* Significant at the .05 level.

An analysis of the correlation between the duration of the vowels that occur before and after the liquid was performed (i.e. the two 'o' vowels in *porosa* [po.'ro.sa]. The results of this analysis show that the relationship is not statistically significant ($r(216) = .517$ $p = .48$). The lack of correlation can be attributed to the effect of syllabic border. That is to say that while the vowel in the stressed syllable (*ro*) is lengthened by the effects of stress, the vowel in the unstressed syllable *po* is not affected.

4.4. Conclusions for speech rate

The results show that speech rate accounts for 14% of the variation in an EV's length, and that this is a statistically significant finding. As predicted, Cl and Cr clusters behave differently with regard to changes in the speech rate. In Cl clusters, the EV is negatively correlated to speech rate (word length) and the NV in the previous syllable.

I argue that the lengthening of EVs at higher speech rates is a compensatory mechanism to ensure perceptibility. At high rates of speech there is a shortening and overlapping of segments; hence, the lengthening of the EV contributes to the distinguishability of the neighboring segments and, in the end, to the perceptibility of the cluster.

Nonetheless, in Cr clusters there was no correlation between EVs and speech rate nor between EVs and NVs. The lack of correlation can be attributed to the difference in duration and sonority of /l/ with respect to /r/. The former is longer, and it takes up a greater portion of the syllable, and thus it can trade off with the EV, whereas the latter is a short tap that does not take up a great part of the syllable and, thus, cannot trade off with the EV.

The fact that speech rate affects Cl clusters but not Cr clusters can be explained by the difference in the inherent duration of the liquids. At high speech rates, the /l/ becomes shorter,

which makes the cluster difficult to perceive. In such cases, more and longer EVs occur. In contrast, the /r/, which is inherently short, is not shortened substantially because it could fall under the perceptual threshold. Thus, the speech duration is a main factor affecting EVs in CL clusters but not in Cr clusters.

In terms of linguistic factors, the results by linguistic condition show that there is no correlation between the duration of NVs and EVs, except in stressed syllables. The effect of stress is expected since it is manifested by lengthening of all the segments of the syllable where it occurs, and in particular, the nuclear vowel.

Although the analysis of speech rate was not in the original experimental design, the results presented in this chapter show its contribution to the variability in the occurrence of EV. Further research is necessary on the effect of speech rate, in normal conversational conditions, on the EV. The controlled condition of sentence or paragraph reading, which are more commonly used in phonetic studies, seem to create a reading-list effect in which the participants use an affected speech rate. The reading-list effect is a set of prosodic characteristics more commonly used when reading a list than when having a normal conversation between native speakers of the target language.

Chapter 5

Speech Perception: Perceptual Identification of EVs

In this study, speech perception is operationalized as identification and discrimination. This chapter examines the results of a study on perceptual identification of EVs, and Chapter 6 presents the results of a discrimination experiment. The perceptual analysis builds on the production results for clusters Cr and Cl presented in the previous chapters, and it seeks to examine the cues from segmental and prosodic factors that may be more salient to listeners. It analyzes the perceptual cues individually and in their interacting with other factors. The chapter is organized as follows: Section 1. presents an overview of the relevant research in perception, Section 2. presents the current study and section 3 presents the discussion of the findings.

5.1. Overview

It is generally accepted that some linguistic contexts are more easily perceived than others. For instance, the context Consonant + Vowel (CV) as in *ga* yields a higher degree of correct perception than contexts such as Consonant + Consonant + Vowel (CCV) as in *gra,* Kenstowicz, (1994). As presented in Chapter 1, in the case of the CCV clusters in Spanish, an epenthetic vowel (EV) occurs between the two consonants of the clusters. Thus, in order to measure the rate at which Spanish and English native speakers can identify the epenthetic vocalic sound embedded between consonants (in the contexts of the type C^vC), a variation on the identification test was carried out. This identification task compares the listeners' identification accuracy between the two possible vowel sounds that could occur between consonants: an intrusive (epenthetic) vowel and a full or nucleic vowel. That is to say, they listen to the contexts

formed by Consonant + Epenthetic Vowel + Consonant (C^vC), and Consonant + Nucleic Vowel + Consonant (CVC) such as in [p^orosa] 'prose' and [porosa] 'porous' respectively.

Perceptually, the phonetic salience of the different segments has been examined under different paradigms. From a phonological standpoint, Jun (1995) proposed a ranking of the places of articulation based on the robustness of the cues and a better cue package. Thus, for unreleased stop consonants, the author proposes the universal salience ranking in (5.1).

(5.1) Perceptual salience

dorsal > labial > coronal

The robustness of the acoustic cues is derived from the observation that in some vowel contexts, the coronals have short transitional cues: the tongue gesture is short and rapid, while labials and dorsals have long transitional cues: the tongue gesture is long and slow. Jun argues that the longer the transitions, the more information is provided. Thus, coronals are considered to be less robust than labials and dorsals. Hume, Johnson, Seo, and Tserdanelis (1999) point out that dorsals are more salient because they contain additional acoustic cues for place of articulation from the convergence of F2 and F3 from neighboring vowels.

Based on phonological markedness, Iverson and Lee (1994) concur on the proposed hierarchy: dorsal > labial > coronal. They argue that within the context of feature geometry, the greater the representational structure the more marked the segment. Therefore, coronals, which have more representational structure, are the most marked segments, and dorsal, with the least representational structure, are the least marked.

In contrast to a markedness-based account, Hume (2004) points out that markedness is itself a vague term that, instead, can be better explained in terms of predictability and experience.

The author points out that "an element that is predictable within the system is less crucial to successful communication," (3). The predictability is determined by factors such as perceptual salience, articulatory simplicity, and the speaker/hearer experience. As for the role of experience, Hume points out that the "more experience that one has with an element, the greater the expectation that that element will occur," (3). On the other hand, in a language, articulatory simple elements tend to occur more frequently than elements with complex articulation, which gives listeners more experience with those sounds.

Hume points out that a speaker/listener is biased towards the predictable. Thus, in cases of vocalic epenthesis in English, it is to be expected that the vowel epenthesized in a schwa (since it is the most common vowel in the language and not because it is the weakest vowel phonetically), should therefore be the least salient. Furthermore, substitution errors in the language would be determined by the more common sounds as opposed to the more marked ones. For instance, in the acquisition of Japanese as L1, children make more pronunciation errors by pronouncing /t/ as /k/, while in English, children more frequently substitute /k/ with /t/. This is explained by the fact that in Japanese /k/ is more frequent than /t/. In contrast, in English /t/ is more frequent than /k/. Hume (2004) also points out that elements with "good cues, like the CV syllable, are more likely to be predictable than ones with poor cues because the former tends to occur more frequently in the language" (13).

In light of this discussion, the purpose of this experiment is to investigate whether one of the elements in the CvC-CVC asymmetries can be characterized as the pattern that is identified correctly at a higher rate. That is to say, we try to identify the pattern in which the similarity between input (perception) and output (production) can be maximized. To carry out this task we use a matching identification task in which listeners are required to match a spoken word to a

written counterpart. Subjects' judgments in this task provide the kind of evidence necessary to test the claim that the epenthetic vowel is not perceived as a full vowel and that the CvCV construction is as salient to listeners as the CVC construction is. Equally important, it sheds light on the cues that listeners obtain from the segmental factors (place and manner of articulation, voicing) and prosodic factors (stress). These factors are a subset of all the possible factors that have been identified as the most salient in production. A secondary objective is to provide an indirect test of the salience ranking in (5.1).

5.2. The experiment

5.2.1. The identification task

Experiments based on the recognition of single words are commonly used in psycholinguistic studies. Since the words are presented in isolation, out of a larger context, this may be considered an artificial task. There are two main reasons for the use of these experiments. First, by reducing the task to recognizing one word, the interpretation of the results is simplified. It is assumed that the factors affecting the recognition of a single word are the same as those affecting a string of words. By using one word the effect of context is minimized; it is assumed that the context of many words operates, to a large extent, in the same way as the context of one word (Harley, 1995). Second, these experiments are relatively easy to perform; all that is needed is a computer and a program to display the stimulus and the possible answers.

In order to identify the rate at which Spanish and English native speakers can identify vocalic sounds embedded between consonants, a variation of an identification task was designed.

5.2.2. Participants

This study involved two language groups: English native speakers learning Spanish in an academic setting (L2 group) and Spanish native speakers (L1 group). In turn, the English native speakers were drawn from three levels of proficiency: beginners, intermediate, and advanced learners; their distribution was determined by the class in which they were registered. The L2 groups were all college students from a university in the USA. One of the criteria for inclusion in the study was that all the participants who were native speakers of English must have started learning Spanish after the age of eight years, the purported critical period for phonological acquisition (Scovel, 1988). In this sense, they would not be considered bilinguals. As for the Spanish native speakers, they are all monolingual students from a university in Bogotá, Colombia.

The distribution of participants by group and average age at which they started leaning the language is presented in Table 5.1 below.

Table 5.1.

Distribution of participants per group

Group	Number of participants	Females	Males	Mean age/Standard Deviation	Mean initial age of first instruction in Spanish
Beginners	41	21	20	19.3 (sd 12.6)	17.1 (sd 6.8)
Intermediate	51	32	19	22.1 (sd 5.0)	17.4 (sd 6.2)
Advanced	49	27	22	20.9 (sd 3.8)	17.3 (sd 6.4)
Native Spanish speakers	38	22	16	18.0 (sd 3.6)	

5.2.3. Experimental Design

5.2.3.1. Task

The experimental paradigm used for this task was an identification test. The stimuli were non-words recorded by a native Spanish speaker. The participants' task was to match the non-word heard to one of four possible written items. Within the four items there was only one that matched the stimulus. This version of an identification task was chosen because it provides information on the possible factors or elements that are more salient to the listener. This information can be inferred by the item chosen, which varies in one segment or feature from the intended target.

This variation of the identification task is cognitively demanding because it requires the participants to store the spoken stimulus in their short-term memory while they read several candidates and finally match the stimulus to one of the written choices. It can be argued that possible errors may not be attributed to the misidentification of the stimulus itself but to the difficulty or possibility of error in decoding the spelling. However, the spelling of Spanish words is highly phonetic – most of the graphemes represent only one sound and a sound is represented mostly by one grapheme. The high degree of affinity between grapheme and sound makes the matching a less demanding task. The few exceptions in which a grapheme can represent more than one sound are studied very early in Spanish classes. Keeping this in mind, the experiment was conducted after the tenth week of classes so that beginners would have had the opportunity to learn spelling rules and decode written Spanish.

The identification task is made up of two complex tasks in language recognition. First, it requires that the subjects retain (in their working memory) the word that he/she hears while

he/she reads written candidates that match the spoken word. The possibility of priming in the perception of the spoken word is increased by the possibility of visual priming at the moment of recognizing the written word. In this identification test non-words were selected to avoid the possible influence of words with similar or opposite meanings (see *semantic priming* as discussed in Meyer and Schvaneveldt, (1971); Trofimovich, P. and McDonough, K. (2011)). In addition, non-words are reported to trigger processing at the phonetic level, thus avoiding lexical priming (Werker and Logan, 1985). That is to say that using non-words avoids the possibility that listeners may recognize the word and not concentrate on listening to all the segments of the word. Furthermore, choosing non-words controls for possible effects of word frequency. Using real words was ruled out since the effect of frequency is high, not just between high frequency words and low frequency words but also between highly frequent words and slightly less frequent words (Whaley, 1978; Foster and Chamber, 1973). In addition, the selection of nonwords controls for the possible effect of *familiarity* with the word (Gernsbacher, 1984), and the age of acquisition, which arguably have more of an effect on word recognition than frequency (See Brown and Watson, 1987; and Morrison, Ellis, and Quinlan, 1992). Furthermore, the use of nonwords allowed us to tap into the phonetic (pre-lexical) processing level (Brannen, 2002).

For this test a total of 234 items were presented: 78 test items and 156 distractors. The words were presented in a randomized order to each student to control for any possible effect originating from the order in which the items are presented. The listeners heard the test words once with no breaks. The total duration of the task was approximately sixteen minutes. At the beginning and at the end of the test, participants completed a Consent-to-Participate form and a survey about their linguistic background and usage respectively.

The participants were tested in a computer lab using Yamaha MH500 headphones. Before the test, the participants were given instructions about the procedure for the task. To ensure that the procedure was understood, a short training session with two tokens was carried out. As part of the instructions, the participants were told that they would be hearing Spanish words. A reason we tell the participants this is to bias them towards using their L2 Spanish grammar instead of their L1 English grammar. The assumption adopted is that grammars may be separate/modular. Another reason to tell the participants that they would be hearing Spanish words was to activate their linguistic mode and not their general auditory mode of perception, which also assumes modularity of cognitive and perceptual functions (Brannen, 2002).

5.2.3.2. Stimuli

Stimulus Quality

The test items selected are phonologically and morphologically possible in Spanish; that is to say, they do not contain illegal strings of phonemes or graphemes which may be rejected by the participants (Dupoux et al. 1999; Kabak 2003). The items were examined by three Spanish speakers (two of them with linguistic training) and two English native speakers (both with linguistic training) to ensure that they did not contain apparent cognates in any of the two languages.

The test items were designed to contain the $C_1{}^{v}C_2$ and C_1VC_2 structures in quasi-minimal pairs such as *aterela* and *atrela.* The distribution and items were designed to contain the following segmental and prosodic factors:

(i) place of articulation of the first consonant (C_1): bilabial, dental or velar;

(ii) manner of articulation of the first consonant of the cluster (C_1): occlusives /p, t, k, b, d, g/ or the spirantized allophones [β, ð, γ].

(iii) position of the cluster with respect to the stressed syllable: pre-tonic, tonic or post-tonic position.

(iv) type of liquid that occurs as the second consonantal element in the cluster (C_2): *l* or *r*.

(v) the presence of an epenthetic or a full nucleic vowel between the consonants.

5.2.3.3. Recording of stimuli

The stimuli were recorded by a native Spanish speaker who was a professional in the health sciences with no training in phonology or phonetics. She was 27 years old and was in Canada studying English as a Second Language, and at the time of the recording she had been living in the country for 3 weeks.

In this study we opted for naturally produced stimuli over computer-synthesized signals to promote a linguistic rather than a general auditory mode of listening (See Brannen, 2002 for further discussion). In addition, using speech produced by a person avoids possible effects for "frame of reference" that listeners encounter when listening to speech signals produced by a person or a machine, (Johnson, 2005:365).

The stimulus words were embedded in the following carrier sentences: *diga* _____ *nuevamente* ' say _____ again' and *repitan* _____ *otra vez* 'repeat once again' to ensure that the stimuli were as natural as possible in order to avoid a list effect. The speaker read the sentences three times in different orders. Two trained phoneticians heard the different versions and selected the reading that sounded most natural. The sound signal was digitized and low pass

filtered at 22000 Hz. using Adobe Audition 1.5™. The target nonwords were then extracted from the carrier sentence and randomly organized into a sequence to create the listening part of the task.

The written part of the identification task was composed of four possible items from which the subjects had to choose the matching word. In addition to the target word, the list contained its quasi-minimal counterpart (i.e. *tempre – témpere*). The other two distractors varied in one of the segments. For instance, for the target item *tempre*, the distractors *tembre* and *tembere* were included, and if the wrong item was chosen, the researcher could observe what segment was more salient to the listener. For instance, if the item *tembre* was chosen, it could be inferred that the difference in VOT between the /p/ and /b/ could be a source of error in perception and not necessarily due to the presence/absence of the vowel embedded between the consonants. The test words and the distractors are presented in Appendix 2.

5.3. Results

Each participant's response was coded '1' if correct or '0' if incorrect, and each participant's accuracy rate was calculated for each contrast. Statistical analysis was done using generalized estimating equations (GEE), which is the equivalent of the mixed logit models. The results are presented as follows. Section 3.1 presents the results for all the clusters by type of liquids (*l, r*). 3.2 presents the results for the Cr clusters, and section 3.3 for Cl clusters. Section 3.4 presents the results by group.

An exploratory analysis was conducted on the general Obstruent + liquid type of clusters. That is to say that Cr and Cl clusters were examined together in order to examine the salience of the type of liquid.

5.3.1. Main effect models with odds ratios

5.3.1.1. Type of liquid /l/ or /r/

There was no significant main effect for the type of liquid that occurred as the second consonant in the cluster /l/ or /r/, ($p = 0.9$). These results contrast with the results from the discrimination experiment where the clusters with /l/ are perceived correctly at a higher rate. Furthermore, the results suggest that the perceptual tasks of discrimination and identification differ in nature. The differences may be based on the different levels of effort required by each task, the former being the most demanding since it requires that the listener remember a segment while waiting to hear the next one for comparison reasons. In contrast, the latter requires that the listener remember a word and then try to match it to one of four possible written items. Differences in the degree of attention required by the listener could explain the difference in results between the two sets of data.

5.3.1.2. Main effect for Group

The odds ratio results (Table 5.2) show that there was a significant main effect for Group: advanced learners of Spanish and Spanish native speakers have a statistically significantly higher chance (37% and 60% respectively) of correctly identifying the words as opposed to intermediate learners (11%) – beginning learners being the baseline for comparison (beginners = 1). These results confirm that longer exposure to the L2, in this case in the classroom setting, contributes to acquiring a more native-like perceptual performance with respect to consonant clusters. It is not highly probable that the proficiency level has a direct effect since, in this experiment, there was no tapping into the explicit knowledge about EVs. Instead, it required accurate matching between the acoustic signal and the written item. While it is true that

familiarity with written Spanish would have given an advantage to more advanced learners, all the tokens were non-words, which minimized the effect of familiarity regarding the test tokens.

Table 5.2.

Results of perception of consonant clusters by group

	C^vrV			
	e^{β} (odds ratio)	SE β	Z	*P*
Beginning learners	1			
Intermediate learners	1.11	.13	0.30	0.76
Advanced learners	1.37	.19	2.32	0.02
Spanish Native speakers	1.60	.24	3.08	0.00

5.3.1.3. The C^vCV and CVC sequences

Table 5.3.

Results of identification of clusters and non-clusters by type of liquid

	Epenthetic vowel (C^vCV)				Full vowel (CVC)			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	z	*P*
Cl	1				1			
Cr	1.97	.16	7.95	0.000	.43	.04	-8.93	0.000

The results for cluster (C^vCV sequence) and full vowel (CVC sequence), Table 5.3, show that Cr consonant clusters are 97% more likely to be correctly identified than the Cl clusters (the baseline for comparison). In contrast, the sequences with full vowels and /r/ (CVrV) only have a 43% chance of being correctly identified in comparison with the sequence formed with /l/ (CVlV), the baseline for comparison. These results show that, in perceptual terms, the CVr sequence is a weak context whereas the Cl cluster is a strong context. These results support our hypothesis regarding the influence of the liquid. In the case of the liquid /l/, which is longer than /r/, little or no enhancement it required for perception. In the case of the clusters, the data show

the opposite effect, namely that the Cr clusters are perceived correctly at a higher rate than the Cl clusters, which argues for the hypothesis that the role of the EV is to enhance perceptibility.

5.3.2. Interactions

The results for the interaction voicing x stress (Table 5.4) show that for the EV (in consonant clusters C^vCV), the interaction of voiceless x post-tonic position is 4.8 times more likely to be identified correctly when compared to the voiceless x pre-tonic position (the baseline for comparison). In contrast, in the sequence with the full vowel (CVC), the voiceless x post-tonic interaction is only 56% more likely to be identified correctly. The higher odds of correctly identifying the consonant cluster rather than the CVC sequence in the voiceless x post-tonic interaction supports the hypothesis that EVs enhance perceptibility in the weaker context (the post-tonic position, which has .4 odds of being correctly identified). The interaction of the voiceless segment, which is a perceptually strong context with the post-tonic position seems to enhance perceptability more than in the CVC sequence.

Table 5.4.

Table of interactions voicing X stress for epenthetic and full vowel

	Epenthetic vowel (C^vCV)				Full vowel (CVC)			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Voiceless X pre-tonic	1				1			
Voiceless X tonic	.46	.12	-2.92	0.003	.85	.21	-0.62	.533
Voiceless X post-tonic	4.80	1.15	6.53	0.000	1.56	.36	1.92	.055

The analysis of the interaction of voicing and vowels (Table 5.5) shows that the interaction voiceless x /e/ is the most likely be identified correctly (23.3 times more often), whereas the interaction of voiceless and high vowels (/i/ and /u/) is the least likely to be correctly identified

(36% and 67% as often, respectively). In the case of the full vowel, as expected, the context of /a/ is the least perceptible of all vowels.

Table 5.5.

Odds ratios of interaction voicing x nucleic vowel in clusters and non-clusters

	Epenthetic vowel ($C^{V}CV$)				Full vowel (CVCV)			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Voiceless X a	1				1			
Voiceless X e	23.3	9.2	8	0.000	2.9	.97	3.27	0.001
Voiceless X o	1.93	.8	1.58	0.11	11.5	4.6	6.13	0.000
Voiceless X i	.36	.14	-2.61	0.009	11.1	3.7	7.15	0.000
Voiceless X u	.67	.25	-1.05	0.29	16.2	4.8	9.38	0.000

The interaction voiceless x place of articulation (Table 5.6) is highly significant. The results for the clusters and the CVCV sequences show opposite tendencies. Whereas the interaction of voiceless x bilabial is the most likely to yield correct identification (6.7 times more as compared to voiceless x dorsal, the baseline), the same interaction in the CVCV sequence is the least likely to yield correct identification (odds of .18 as compared to voiceless x dorsal).

The data show that the weakest context (coronal) is perceptually enhanced when it interacts with a voiceless consonant only for clusters and not for the CVCV sequence. The enhancement of perceptibility can be viewed as a function of the EV.

Table 5.6.

Odds ratios of interaction voicing X place of articulation in clusters and non-clusters

	Epenthetic vowel ($C^{v}CV$)				Full vowel (CVC)			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Voiceless X dorsal	1				1			
Voiceless X coronal	4.63	1.08	6.54	0.000	.77	.18	-1.06	.29
Voiceless X labial	6.76	1.68	7.68	0.000	.18	.03	-8.06	.000

Followup analysis of the effects of stress, voicing, place of articulation and nucleic vowel by each type of cluster type (Table 5.7).

Table 5.7

Results of discrimination between clusters and non-clusters by type of liquid

	Cl				Cr			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	z	*P*
Pre-tonic	1				1			
Tonic	1.54	.20	3.34	**.001**	1.16	.15	1.14	0.25
Post-tonic	1.29	.25	1.34	.179	**.53**	.05	-5.61	**0.000**
Voiced	1				1			
Voiceless	.96	.14	-0.23	.816	**4.42**	.47	13.79	0.000
Dorsal	1				1			
Coronal	.27	.03	-9.57	.000	.68	.06	-4.01	.000
Labial	1.39	.20	2.23	.026	1.29	.15	2.16	.031
a	1				1			
e	1.08	.18	.49	0.62	.21	.03	-9.78	.000
o	1.06	.20	.32	.74	.24	.06	-5.71	.000
i	1.20	.27	0.80	.42	.68	.09	-2.73	.006
u	2.74	.65	4.25	0.000	.19	.03	-10.32	.000

Although the interaction voicing x stress (Table 5.4) shows that the clusters in post-tonic position are 4.8 times more likely to be correctly identified, the main effects show differences between Cr and Cl clusters based on their position within the word. The results in Table 5.7 show that the tonic syllable provides the context with the highest degree of accurate differentiation for both Cl and Cr clusters (54% and 16% more than in the pre-tonic position respectively). However, for the Cl cluster, contrary to what was expected, the pre-tonic position shows a lower percentage of correct identification, while the Cr clusters in the post-tonic position present the lowest probabilities (53% compared to pre-tonic).

For the Cr clusters, as expected, the end-of-word position is the weakest from both the articulatory and perceptual standpoints. As discussed in Chapter 2, there is a laxer pronunciation in this position (in articulatory terms) and listeners pay less attention to these clusters because it is highly probable that they have already recognized the word based on the first parts of the utterance. In the case of Cl clusters, they do not present low perceptibility rates in word-final position. I argue that this consistency in rates of perceptibility can be explained by the length of the /l/, which creates a perceptually strong context.

Regarding voicing, Cl clusters do not present statistical differences when formed with a voiced as opposed to a voiceless consonant. In contrast, the Cr clusters are 4.1 times more likely to be correctly identified after a voiceless consonant. These results support the hypothesis that the /l/ forms a perceptually strong context. They also confirm that, in clusters with /r/, the context of a voiceless consonant is perceptually stronger than the context of a voiced consonant.

With regard to place or articulation, both Cl and Cr clusters are less likely to be identified correctly after a coronal consonant, with 27% and 68% success respectively, as compared to dorsals (which are the baseline for comparison), whereas they are the most likely to be correctly identified after a labial (39% and 29% more than dorsals). These results partially support Jun's (1995) ranking of places of articulation, which lists the coronals as the weakest context. However, the results show that in Spanish, the labials are the strongest context perceptually and not the dorsals as proposed by Jun. A possible explanation is that the use of two distinct active articulators in the pronunciation of the two segments of the cluster (lips and tip of the tongue) produces a more perceptually salient cluster than when both segments are pronounced with one articulator (the tongue).

In terms of vowels, Cr clusters are more likely to be correctly identified with the /a/ than with any other vowels, whereas the Cl clusters are least likely to be recognized with /a/. This discrepancy is surprising since the articulation of the liquids is similar: both are alveolar. Further research needs to be done on the role of the nucleic vowel.

5.3.3. Type of vowel

The analysis by type of vowel in Table 5.8 shows that clusters in word-final position are the least likely to be correctly identified (43%) when compared to the word-initial position. The sequences with full vowels are most likely to be correctly identified.

Table 5.8.

Results of discrimination by type of vowel: EV, NV

	Epenthetic vowel (C^vCV)				Full vowel (CVC)			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Pre-tonic	1				1			
Tonic	1.04	.12	0.38	.7	3.45	.40	10.47	0.000
Post-tonic	.43	.05	-7.19	.000	**1.64**	.16	5.04	**0.000**
Voiced	1				1			
Voiceless	2.12	.19	8.07	.000	**1.23**	.10	2.37	0.018
Dorsal	1				1			
Coronal	**.20**	.024	-13.51	**.000**	1.05	.11	.54	.58
Labial	1.11	.13	0.88	.037	.91	.09	-0.83	.40
a	1				1			
e	.15	.025	-11.16	.000	1.77	.25	3.92	.000
o	.24	.044	-7.75	.000	3.54	.67	6.64	.000
i	.31	.051	-7.06	.000	1.59	.21	3.55	.006
u	.27	.049	-7.23	.000	1.09	.13	0.76	.45

In terms of voicing, as with previous results, both clusters and sequences with full vowels are more likely to be identified correctly after a voiceless consonant, which is a perceptually strong context.

Regarding place of articulation, consonant clusters are the least likely to be identified correctly after a coronal: there is only a 20% rate of success when compared to a dorsal. In contrast, there is no significant difference for place of articulation for sequences with full vowels. These results support the hypothesis that consonant clusters are perceptually weaker contexts than sequences with full vowels. In addition, they partially support Jun's (1995) hierarchy according to which coronals are the weakest context. However, the results show that the labials and not the dorsals (as predicted in Jun's hierarchy) present the strongest perceptual context.

Regarding the nucleic vowels, the results show that the consonant clusters are more likely to be perceived correctly within the context of /a/ whereas with full vowels the /a/ (and /u/) were the least likely to be perceived correctly. These results along with those in Table 5.4 suggest that in the weakest context (in this case the consonant cluster) the mid-low vowel /a/ is the best possible context for successful perception. This may be due to the fact that /a/ is the vowel in which there is the greatest opening of the oral cavity and the least air obstruction.

5.4. Discussion and Conclusions

The results of the identification task are relevant in two ways: first, they shed light on which cues which the listener uses for discrimination purposes. Second, they provide information as to the effect of L1 and L2 proficiency in the use of those cues. A summary of the results is presented in Table 5.9. They are presented in a ranking based on their salience for perceptual discrimination.

Table 5.9.

Significant main affects across consonant clusters

Factor	Cluster	
	Cr	**Cl**
Place of articulation	labial > dorsal > coronal	labial > dorsal > coronal
Voicing	voiceless > voiced	voiced/voiceless[a]
Stress	tonic/pre-tonic > post-tonic	tonic > post-tonic/pre-tonic
Type of vowel	epenthetic > full	epenthetic/full[a]
Group	L1 > beginner/intermediate	L1 > beginner/intermediate

[a] No significant difference was found.

With respect to place of articulation, listeners identify clusters formed with a labial consonant at a significantly higher rate than they can identify both dorsal and coronal ones. These results partially conform to Jun's (1995) universal salience ranking for stops in which the coronal consonants are the context with least perceptibility. However, the results show that for the Spanish consonant clusters, the context of the labials is more perceptually salient than the dorsals, and not dorsals over labials as proposed in Jun's hierarchy. These results suggest that the use of two active articulators in the pronunciation of the cluster enhances its perceptual recoverability.

With regard to voicing, the results show different results for Cr and Cl clusters. While the former presents higher rates of successful perceptual identification for voiceless segments, for the latter there was no significant difference between voiced and voiceless segments. These results contrast with the findings in the production study in which, for the Cr clusters, there are more chances to find EVs in the context of a voiced consonant. But, as the results from the perceptual identification show, consonant clusters – and therefore EVs - have a greater chance of

being correctly identified in the context of a voiceless consonant. The results suggest that, although the context of a voiced consonant is enhanced by the occurrence of an EV, it is not enough to surpass the salience of the voiceless consonant, which is a perceptually strong context. In the case of Cl clusters, the cues from voicing do not seem to influence the recoverability of the cluster. As pointed out above, we argue that the longer duration of /l/ as compared to /r/ ensures the recoverability of the cluster.

In terms of lexical stress, the interaction voicing x stress shows that correct identificaion is 4.8 times as likely to occur in post-tonic position. However, further analysis of lexical stress shows that perceptual identification of the clusters follows the pattern of speech perception: the most perceptually salient parts of the word are those in word-initial position and stressed position (tonic). The Cr clusters present the greatest odds of being correctly identified in tonic or pre-tonic position over the post-tonic position, whereas the Cl clusters are more likely to be correctly identified in the tonic position. The results support the generally accepted notion that edge effects make word final position the articulatorily weakest and the least focused upon by listeners. Nonetheless, since more EVs occur in post-tonic position, it can be argued that listeners confound more clusters and non-clusters in post-tonic position because they perceive more EVs in clusters and confound them with full vowels. Thus, listeners, especially L2 learners, seem to perceive EVs at such rate that there is only 53% accurate identification of Cr clusters in post-tonic position. In the case of the Cl clusters, EVs do not to create the same degree of erroneous identification in post-tonic position: clusters and non-clusters are identified correctly 29% more than in pre-tonic positon.

In terms of Group, the results for native speakers are significantly different from the results for beginner and intermediate L2 learners, but are not significantly different from the

results for advanced learners. This shows that the learners undergo a change in their mode of L2 listening.

I argue that the initial mode of listening in adults is language specific, and that it is characterized by listening strategies. Those strategies consist of the attention to specific cues in the signal and the ranking of those cues. In the process of language learning, learners undergo a transition in their perceptual mode from their L1 mode of listening toward an L2-like mode of perception.

The identification of the most salient cues is not straightforward. Listening strategies for identification in general seem to be a composite process in which the different segmental and prosodic variables contribute to a cue weighting process. This process is inferred from the interaction of effects at the highest levels for both Cr and Cl clusters (Tables 5.4 and 5.5). In addition, cues used for identification vary from cluster to cluster: the analysis of the type of liquid that appears in consonant clusters shows that the variables Place of articulation and Type of vowel are significant for all groups when identifying the Cr clusters. However, when identifying the Cl clusters, Voicing and Type of vowel are not significantly different. This disparity confirms that although, from an articulatory point of view, /r/ and /l/ share a place and manner of articulation (alveolar liquids), from a perceptual point of view, listeners rely on different clues. Similarly, for the Cr clusters, all groups show significant results for Voicing, but for Cl clusters the results are only significant for advanced L2 learners. Thus, the fact that some variables are more consistently significant statistically across the L2 levels than others leads us to propose the ranking in (5.2) for the identification of Cl clusters. The dashed line represents exchangeable positions.

(5.2) Ranking of perceptual cues for identification of Cl clusters

Place of articulation > Stress ¦ Type of vowel ¦ Voicing

On the other hand, the identification of the Cr cluster follows the ranking proposed in (5.3).

(5.3) Ranking of perceptual cues for identification of Cr clusters

Voicing > Place of articulation > Stress > Type of vowel

From the rankings in (5.2) and (5.3) it can be concluded that the type of vowel is especially important for the clear identification of Cl clusters more so than for the Cr ones.

Regarding native Spanish speakers, the perceptual cues seem to be obtained from all the variables for identification of Cr whereas for Cl they rely on all except Voicing.

The different results among groups suggest that L2 learners use different clues and that during the learning process those clues can be used and then discarded or used only at the higher stages of the learning process. These different strategies suggest a dynamic system of perceptual clues. The main characteristic of L2 perceptual strategies can be characterized in terms of stages in the development of L1-like perception. I infer that at the initial stage of the L2 learning process the L2 learners use perceptual strategies from their L1, and during the learning process there is a rearrangement in the hierarchy of perceptual cues. At advanced levels, L2 learners seem to use perceptual cues in a fashion similar to native speakers. However, more research is require to establish whether complete native-like perception is attainable.

Chapter 6
Perceptual Discrimination of EV

This chapter examines the perceptual salience of the epenthetic vowel in Spanish clusters (Obstruent + liquid) as perceived by native Spanish speakers and native English speakers learning Spanish. This chapter presents the results corresponding to a standard sound discrimination task. The chapter is organized as follows. Section 1 reviews some relevant theoretical issues; section 2 presents the methodological aspects of the experiment; Section 3 presents the results, and section 4 covers the analysis and conclusions.

6.1. Perceptual characterization of EV

The EV in Spanish has been characterized as a mid-central vowel. When it is mapped against the Spanish vowels, it occupies the mid central area and varies according to the nucleic vowel (NV) of the syllable where is occurs (Quilis, 1993; Ramírez 2006). As for its duration, it ranges on average from 32 ms to 47 ms. In addition, native Spanish speakers are unaware of the existence of the vowel, from both perceptual and production standpoints. Navarro (1957:16) points out that the production of EV is unconscious on the part of the native speakers even though it can be longer in duration than /r/.

Nevertheless, the occurrence of the EV is not generally accepted. Widdison (2004a:70) argues that in the case of the Cr clusters, the EV is an automatic and unconscious result of the articulatory restrictions in the transition from an obstruent to a vibrant. Regarding the Cl cluster, Widdison claims that although there is not an abrupt rupture as in the Cr cluster, which would

result in an epenthetic vowel, the lateral has spectral characteristics that are similar to those of a vowel. Thus, in the articulation of [l] the transitions in its onset and release phases present the most cues for its identification as a consonant but between the target and release phases it is has spectral energy very similar to that of vowels: formant-like formations at low and mid frequencies as illustrated in Figure 6.1.

I argue, in contrast to Widdison, that the EV is part of the articulatory plan which ultimate goal is the perceptual clarity of the cluster. As it is shown in this study, the EV is not unconscious and nonfunctional; it occurs more often and longer in the least perceptually recoverable linguistic contexts.

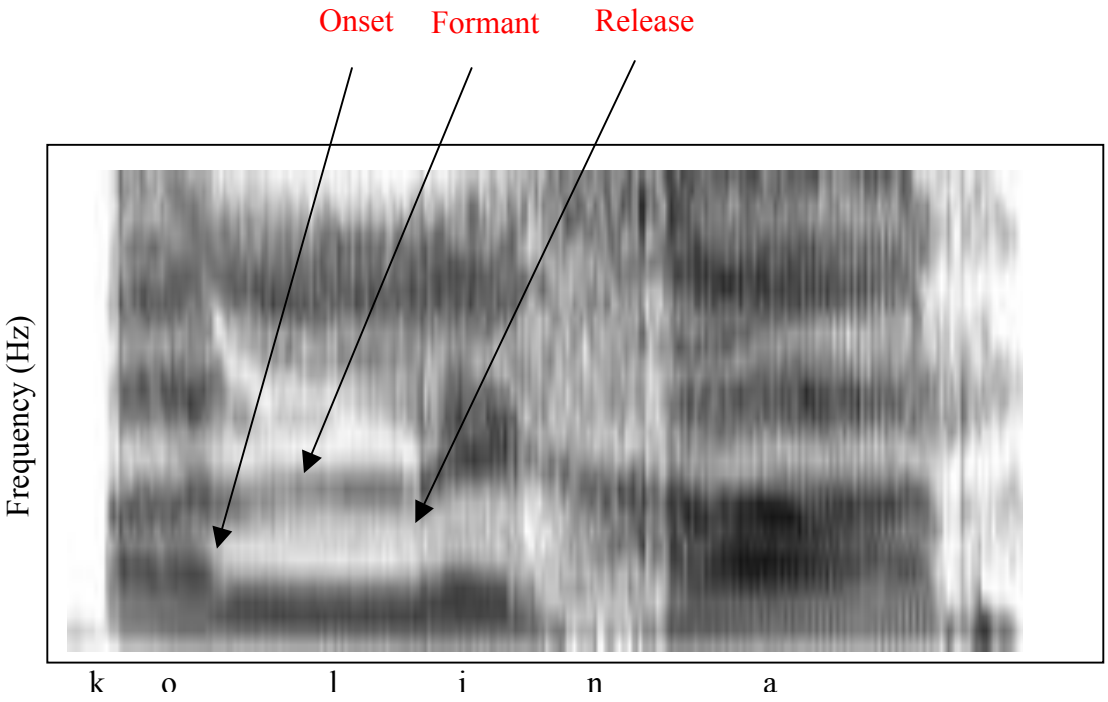

Figure 6.1. Spectrogram of *Colina* 'hill' as produced by a 28-year-old male from Colombia.

However, cases of epenthetic vowels are registered independently from the liquid in Cl clusters as illustrated in Figure 6.2.

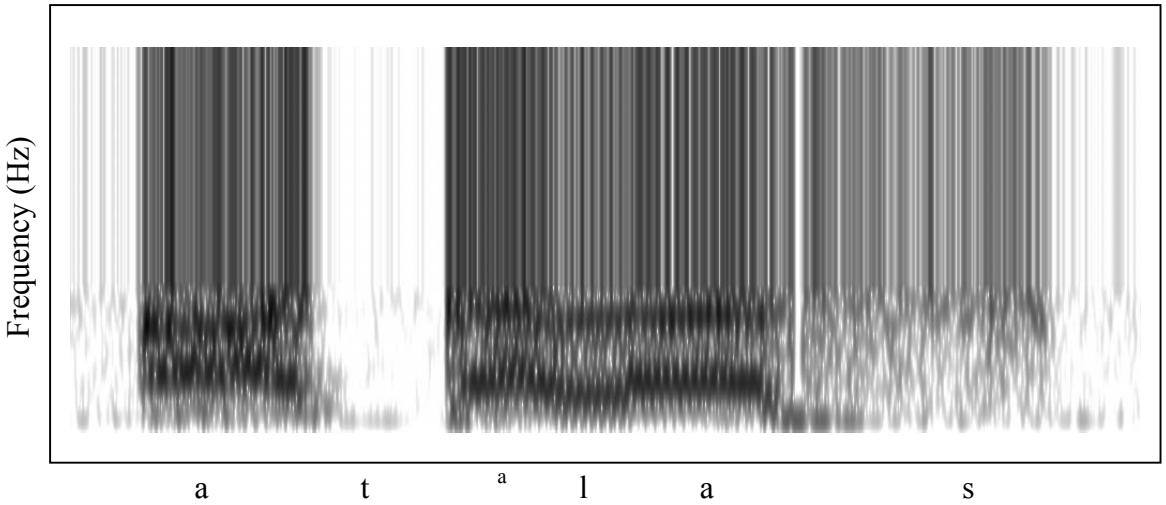

Figure 6.2. Spectrogram of the word *atlas* 'atlas'

In addition, Widdison points out that the liquids, like the vowels, are inherently long segments. The author claims that the similarities between liquids and vowels give the former a vowel-like characteristic that creates marked acoustic differences between the CC and CL contexts. Thus the former offers an "*entorno vocálico empobrecido*" (a poor vocalic context), while the latter gives an "*entorno vocálico enriquesido*" (70) (an enriched vocalic context), which leads the author to argue that diachronic changes from Latin to Spanish such as *eglesia* > *egelesia* and Cristo > *Kiristo* are the product of auditory hypercorrection in which listeners identify as a vowel what in fact was a vowel-like characteristic of the liquids.

In order to identify the minimal duration for a vowel to be perceived as such in a CL context, Widdison conducted a recognition test. The test was intended to identify at what point native Spanish speakers distinguished between minimal pairs such as *átalas tie*-imp.-fem-pl. 'tie them' and *atlas* 'atlas'. The stimuli consisted of tokens in which the vowel was progressively

cut every 4-5 glottal pulses. The results for the study showed that listeners tend to recognize very short vowel sounds (around 17 ms) in the CC context (the cutting recognition rate was 50%), whereas in the Cl context the vowel was not reported below 27 ms. In a similar study for Cr clusters, Widisson (2004b) found that no vowel was reported under 45ms. The author concludes that the Cl context is propitious for the erroneous perception of a vowel due to the vowel-like characteristics of /l/. Nonetheless, there has not been research on the cues that the listener obtains for the recoverability of consonant clusters.

The next section discusses some psycholinguistic assumptions we adopt in the analysis of perception.

6.2. Levels of speech processing

We assume along with Werker and Logan (1985) that speech can be processed at three different levels:

- Acoustic level: processing of fine, non-linguistic distinctions; that is, listeners evaluate physical identity such as fundamental frequency, amplitude, etc.

- Phonetic level: At this level there is processing of linguistically relevant information only, both contrastive and non-contrastive, as well as normalization of non-linguistic differences.

- Phonemic level: at this stage there is processing of contrastive information only; normalization of non-linguistic and non-distinctive information.

Werker and Logan (1985) showed that in an experimental setting the different levels of processing can be accessed depending on the stimulus, specifically the duration of the inter-stimulus intervals (ISI) in the task. Werker and Tees (1984) have shown linguistic processing at

the different levels. For instance, in a category change procedure, native English speakers could not distinguish contrasts that are not phonemic in English; they failed to discriminate between Hindi dental and retroflex stops, and between Thompson Salish velar and uvular stops in a category change procedure. Nevertheless, in an AX procedure, English native speakers were able to discriminate these contrasts. Werker and Tees suggest that these results show evidence of phonemic processing in the former experiment and phonetic processing in the latter one.

Brannen (2002) adopted Werker and Logan´s (1985) proposed distinction in a study of the cross-linguistic differential substitution of the interdental fricative [θ] by Japanese (JP), European French (EF), and Quebec French (QF) learners of English as a Second Language. In their second language (L2) oral production, JP and EF usually replaced [θ] by [s] whereas QF replaced it with [t]. Brannen used the AXB protocol with both short ISI (250 ms) and long ISI (1500 ms) to tap into phonetic and phonemic perceptual processing respectively. The expected difference when using the two types of intervals is that with a long interval between the two stimuli, by the time the person listens to the second stimulus, the non-distinctive phonetic features of the first stimulus have faded, and what remains are only the distinctive phonemic features. With the short ISI, on the other hand, the two stimuli are close enough that both the phonetic and the phonemic features are processed.

Werker and Logan (1985) found phonetic processing at 250 ms ISI, whereas phonemic processing was registered at 1500ms ISI, at least for the first two blocks of their between-subjects study. However, Brannen found in the results of her study that the AXB protocol only elicited phonetic processing despite using the same difference in ISI values proposed by Werker and Logan, which seems to indicate that the AXB protocol is not conducive to accessing language processing at the phonemic level (Brannen 2002).

Based on Werker and Logan (1985) and Brennan (2002), and since the goal of this study is to analyze linguistic processing at the phonetic level, the short ISI of 250 ms. is adopted.

The objective of this experiment is to test whether native English speakers are aware of the epenthetic vowel in the context of a consonant cluster (CC), and also to identify what perceptual cues L1 and L2 speakers use in order to recover the consonant cluster information.

As for the differences in L1, in native Spanish speakers the production of the epenthetic vowel is unconscious (Navarro, 1963) and its perception requires it to be at least 27ms in duration (Widisson, 2004). In contrast, English does not present epenthetic vowels between the members of consonant clusters. Thus, we anticipate that English speakers will be more sensitive to the presence of the epenthetic vowel than Spanish speakers. That is, we anticipate that English speakers will hear more differences between clusters and CVC syllables. We also foresee that they will mistake CvC for CVC, and vice versa.

Furthermore, we anticipate an effect of proficiency level; as learners obtain more exposure to the L2, their interlanguage becomes more similar to the target language. Therefore, the working hypotheses are the following.

Hypothesis 1: L1 has an effect on the discrimination of the CC and CvC sequences: English speakers reach higher accuracy levels of discrimination than Spanish speakers.

Hypothesis 2: The perceptual cues used at higher levels of L2 proficiency resemble the cues used by native speakers.

6.3. Test Design

6.3.1. Subjects

The subjects in these experiments came from two groups: native Spanish speakers and native English speakers learning Spanish. In turn, the native English speakers were divided into three proficiency levels – beginner, intermediate, and advanced - as indicated by the Spanish course in which they were registered: first, second, or third-year language level. The group of native Spanish speakers served as the control group.

The participants were recruited into the experiment by an open invitation to participate made in their Spanish classes and by posting notices on campus at the University of Houston. Upon the first contact with the researcher, the participants were given a questionnaire that served to identify their linguistic background, namely their first language, their use and knowledge of other languages, and the age at which they started learning the L2 (See Appendix 4).

One of the main criteria used to screen the English speaking participants was that they had to have started learning Spanish after the age of seven, the purported critical period for phonological acquisition (Scovel, 1988); thus, they were not considered native bilinguals. This criterion allows us to be sure that we were examining second-language learners and not bilingual individuals.

The English speakers' level of proficiency in Spanish was determined by the level of the last Spanish language class completed. Thus, beginner learners had not taken any Spanish classes (23% of the students) or had taken a maximum of one year of Spanish in high school at least three years prior (77%). The intermediate learners were students enrolled into second year Spanish. The pre-requisites to be in that class were to have successfully completed first year

Spanish (94% of the subjects) or to have been put into that level by a placement test[21] (6%). The advanced learners were students who were enrolled in third-year Spanish classes. They would have either completed second-year Spanish courses (98%) or would have been placed into this level by a placement examination (2%). The native Spanish speakers were adults who had been in an English-speaking environment for less than18 months. None of the participants of any group reported having a speech or hearing impediment. Subject information is given in Table 6.1.

Table 6.1

Number of subjects and mean age per group

Group	N	Mean age
Native-Spanish speakers	63	22.8 (s.d. 7.1)
Beginners	67	19.1 (s.d. 6.3)
Intermediate	77	21.1 (s.d. 9.6)
Advanced	36	23.8 (s.d. 10.2)

6.3.2. The discrimination task

For the discrimination test, an AXB forced choice procedure was employed (Best and Strange, 1992; Brannen, 2002). In this procedure, each item is a triad in which two stimuli are similar and one is different. The subjects listen to the three stimuli and then decide whether A is similar to X or B is similar to X. The AXB procedure was chosen over the more common ABX and AX procedures because it is less cognitively demanding (Beddor and Gottfried, 1995; Brannen 2002). Macmillan, Kaplan, and Creelman (1977) showed that an ABX procedure tends to underestimate performance when discrimination is difficult (e.g., intra-category discrimination

[21] The placement test was composed of a written part that examined their knowledge of grammar and vocabulary, and an oral interview carried put by an experienced instructor. The instructor would make an assessment of the overall linguistic proficiency and then place the student in the appropriate Spanish class.

in categorical perception studies) and overestimate performance when discrimination is relatively easy (e.g., cross category discrimination).

In the ABX procedure, the subject has to retain item A in memory in order to compare it with X; whereas in the AXB the subject does not have to retain A in working memory[22] for the same time. Once the subject listens to X, he/she can make a decision about the similarity of the stimuli, holding the two stimuli in working memory to compare A to X and X to B.

Regarding the AX procedure, Beddor and Gottfried (1995) point out that it is more prone to bias than AXB because in the former, the subject may tend to choose either the same item most of the time, or all different items. Brannen (2002) adds that if, in the AX procedure, a subject responds that the items are different, there is no basis for identifying the possible differences; it could be due to different volume, pitch, etc. The adoption of the AXB procedure avoids this deficiency and presents low response bias.

The test items consisted of 66 test triads and 34 distractors for a total of 100 triads. The Interval between stimuli was, as mentioned above, the short ISI of 250 ms. Each of the participants heard all 100 triads.

The participants were tested using the phonetic analysis software Praat (Boersma and Weenink, 2010) and used Sanyo HS™ earphones. In some cases the participants took the AXB protocol tests in the same session as the identification and the goodness-rating tests (presented in chapters 5 and 7 respectively). To avoid task bias, the presentation of the tasks was alternated in such a way that approximately half of the participants took the identification test first, then the

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[22] We adopt Baddeley and Hitch´s (1974:17) definition of working memory as the "ability to store and retrieve information while performing other mental computations."

goodness-ranking test and lastly the discrimination test, while the other half took the discrimination test first, then the goodness-ranking test and then the identification test. Between the first and the second task students had a five-minute break, and between the second and the third tasks they filled in the language use questionnaire (see Appendix 5). The goodness-ranking test was always done between the identification and discrimination tests to provide time for the subjects to start the new task without any possible carry-over effects.

In the AXB protocol, participants were presented with a total of 100 triads. The triads were divided into four blocks of 25 each. After each block, they could take a short break if they wanted to do so. The order of the triads was randomly assigned by the software so that each person heard them in a different order from all the other participants. Each task started with a training session in which they could get familiarized with the task. The training consisted of only three triads and its focus was on the understanding of the task, the testing of the volume level, and familiarization with the equipment and the mechanics of the task, such as the area of the screen they were to touch in order to make the selection.

The participants were told that they would hear words in Spanish. The goal of this instruction was to have them predisposed to use their L2 Spanish grammar rather than their L1 grammar; this stems from the assumption of modularity of grammars (Brannen 2002).

6.3.3. Materials

The stimuli employed were the same used for the identification test (See chapter 5 for details on recording procedure). They consisted of non-words which were phonologically and morphologically possible in Spanish.

The stimuli were designed as minimal pairs in which one word contained a consonant cluster and the other word contained a vowel breaking the cluster, i.e. *adrunar* and *adurunar*. All the clusters contained EVs; their durations ranged from 31 ms to 40ms. Only clusters with a clear EV, on the spectrographic analysis, were used. Distractor items were included at a ratio of 2:1.

The factors considered are the same as those from the identification test:

a) place of articulation: dorsal, coronal, and labial

b) Manner of articulation: occlusive, and spirantized (fricativized).

c) Voicing: voiced, voiceless.

d) Stress: pre-tonic, tonic, and post-tonic syllables.

Liquid type: lateral (/l/), and tap (/r/).

6.4. Results

Every participant's response for each triplet was coded '1' if correct or '0' if incorrect; and the participant's accuracy was calculated for each contrast. Statistical analysis was done using generalized estimating equations (GEE), which is the equivalent of the mixed logit models used in the analysis of the production data (Chapter 3).

The remainder of this section is organized as follows: section 6..4.1 presents the results by comparing and contrasting all the groups. It presents the results of the clusters formed by Obstruent plus liquid /l/ or /r/; section 6.4.2 shows the analysis of the cluster Obstruent + /r/ and 6.4.3 shows the results for the Obstruent + /l/ cluster. The analysis of the liquids independently stems from the consideration that although /l/ and /r/ are both liquids, they behave in a different manner and have subspecifications that are particular to each language.

The results of the GEE approach are presented as main effects and interaction effects.

6.4.1. Main effects models with odd ratios

6.4.1.1. Main effect for Group

The mean values and standard deviations from the two way interaction, EVs and NVs in Cr and Cl clusters, are presented in Table 6.2.

Table 6.2.

Mean values and S.D. for EV and CVs by Cluster by Group

		Cr				**Cl**				
Group		**EV**		**NV**		**EV**			**NV**	
	n	M	S.D.	M	S.D.	M		S.D.	M	S.D.
Beginners	1206	.82	.385	.85	.012	1005	.76	.014	.81	.013
Intermediate	1386	.86	.352	.87	.011	1155	.78	.013	.85	.012
Advanced	648	.84	.369	.84	.017	540	.77	.018	.80	.018
Spanish Native Speakers	612	.89	.319	.89	.014	510	.83	.017	.88	.014

Table 6.3 presents the odds ratios by group. The results present a clear trend: although not statistically significant ($p = .37$), intermediate learners are 11% more likely to accurately discriminate clusters from CVC syllables than beginner learners (the baseline for comparison =1). Advanced learners are 34% more likely to accurately discriminate than beginners (difference marginally significant $p = .07$), and Native Spanish speakers are are 64% more likely to discriminate clusters and CVC syllables than beginners (difference highly significant with $p < .000$).

Since all the clusters contain EVs, the results suggest that beginner learners are more likely to confuse the epenthetic vowels with the nucleic vowels than intermediate and advanced learners, and Spanish native speakers. These results also indicate that time of exposure - through

classroom instruction - affects perception. In this case, learners who have had more exposure to the language discriminate the stimuli more accurately than learners with less exposure.

Table 6.3.

Results of perception of consonant clusters by group

	$C^{v}rV$			
Predictor	e^{β} (odds ratio)	SE β	Z	*P*
Beginner learners	1			
Intermediate learners	1.11	.14	0.90	0.37
Advanced learners	1.34	.22	1.78	0.07
Spanish Native speakers	1.64	.28	2.87	0.00

6.4.1.2. Epenthetic vs. Nucleic Vowel

The odds ratio results for type of vowel (Table 6.4) shows that listeners are .8 as likely to discriminate clusters correctly as compared to CVC syllables (the baseline for comparison = 1). That is to say that listeners are more likely to confuse the EV in a cluster with a full vowel than confusing a full vowel with an EV.

Table 6.4.

Results of odds ratios for Epenthetic Vowel and Full Vowel

	e^{β} (odds ratio)	SE β	Z	*P*
Full vowel (CVC)	1			
Epenthetic vowel	.80	.05	-7.24	0.000

6.4.1.3. Type of Liquid (//l/, /r/)

Table 6.5.

Results of discrimination of clusters and non-clusters by type of liquid

	Epenthetic vowel				Full vowel (CVC)			
Predictor	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	z	*P*
Cl	1				1			
Cr	.36	.049	-7.44	0.000	.92	.143	-0.49	0.622

The analysis of the type of vowel (EV or NV) also reveals differences in perception according to the type of liquid (Table 6.5). The results show that the listeners present no statistical difference in discriminating between syllables with full vowels and either liquid (l or r). That is to say that listeners were likely to correctly discriminate CVr and CVl syllables at the same rate.

In contrast, the discrimination rate of consonant clusters (with EV) varies according to the type of liquid: clusters formed with /r/ are likely to be discriminated correctly at only 36% of the rate for clusters formed with /l/. These results show that there are great odds of confusing Cr clusters with CVr syllables than Cl clusters with CVl syllables, which shows that listeners are more sensitive to EV in clusters with /r/ than in those with /l/.

Similar to the results found for production of consonant clusters (Chapter 3), the type of liquid is a significant predictor, as measured by accurate perceptual discrimination. Therefore, the data were analyzed by type of liquid. The following section presents the results of Cr and Cl clusters.

6.4.2. Obstruent + /r/ (Cr)

6.4.2.1. Interactions

The analysis of interactions shows that the only interactions that are significant are Voicing x Stress and Voicing x Nucleic vowel. The results for each interaction are as follows.

6.4.2.1.1. Voicing x Stress

The interaction voicing x stress is a significant predictor of the accurate perceptual discrimination of clusters with EV (Wald's χ^2 = 418.37, and Prob. χ^2 = .000). The results in Table 6.6 show the significant category within the voicing x stress interaction: voiceless x stress. The results show that, compared to the voiceless x pretonic context (the baseline =1), clusters in the voiceless x tonic syllables (i.e., clusters in which the first consonant is voiceless and occurs in a stressed syllable) are 107% more likely to be discriminated correctly. The voiceless x post-tonic context is 122% more likely to be discriminated correctly than the same cluster in a pre-tonic position.

From a perceptual point of view, the stressed (tonic) position is strongest and the post-tonic position is the weakest one. But the fact that the clusters in the context of voiceless x post-tonic are discriminated at a higher rate than in tonic and pre-tonic position supports the hypothesis that the EV enhances the recoverability of the cluster.

Table 6.6.

Interaction voicing X stress

	e^{β} (odds ratio)	SE β	Z	*P*
VoicelessXpre-tonic	1			
VoicelessXtonic	2.07	.36	4.16	0.000
VoicelessXpost-tonic	2.22	.37	4.77	0.000

Test		Wald's χ^2	χ^2	Prob. χ^2
		418.37	27.23	0.000

Further analysis of voicing shows that clusters with voiceless consonants are a consistent predictor of perceptual discrimination. The results for voicing (Table 6.7) show that clusters formed by voiceless consonants are more likely to be accurately discriminated than clusters formed by voiced consonants. The clusters with voiceless consonants are 63% more likely to be discriminated correctly than those with voiced ones (the baseline for comparison).

Table 6.7.

Odds ratios for Voicing

	e^{β} (odds ratio)	SE β	Z	*P*
Voiced	1			
Voiceless	1.63	1.65	4.11	0.000

The results of perceptual discrimination by stress and voicing support the findings obtained in the analysis of speech production (Chapter 3). In production, clusters with EVs were 4.4 times more likely to occur in the post-tonic position, and since that is the position that is more likely to be perceived correctly, it leads us to conclude that it is the higher rate of

occurrence of EVs - in the post-tonic position - that enhances its higher perceptibility, which is otherwise the weakest position perceptually. These results support our hypothesis that EVs occur more frequently in the weakest context to enhance perceptibility.

6.4.2.1.2. Place of articulation

The results for place of articulation (Table 6.8) show that there is a difference in discrimination of non-clusters (CVC) and clusters (CrV). In non-clusters there is no significant difference in perception, whether they occur after a labial, a coronal, or a dorsal consonant. However, there is a 39% better chance of correct discrimination after a bilabial than after a dorsal, marginally significant at p = .095. Discrimination after a coronal is only slightly lower than after a dorsal (8%) and this does not approach significance.

In contrast, the accuracy of discrimination between consonant clusters is significantly different when the first consonant is a coronal. There is only a .36 chance of correct discrimination compared to the rate for a dorsal (the baseline), while the bilabials are .3 times more likely to be discriminated correctly when compared to the rate for dorsals. The low rates of correct discrimination in the context of a coronal are consistent with those found in production: more EVs were found in the context of dorsals and bilabials than coronals. Thus, the results suggest that EVs strengthen the perceptual salience of the cluster in which they occur.

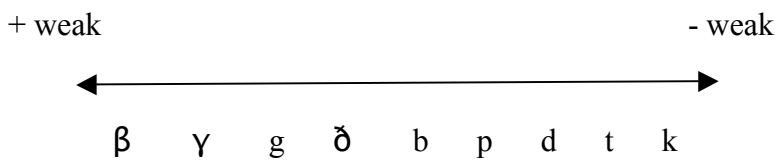
Table 6.8.

Results of perceptual discrimination by place of articulation in Cr clusters and non-clusters

	$C^{v}rV$				CVrV			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Dorsal	1				1			
Coronal	.36	.049	-7.44	0.000	.92	.143	-0.49	0.622
Labial	1.37	.267	1.61	0.106	1.39	.274	1.67	0.095

The overall results support the hypothesis that more EVs occur in the context of weaker consonants as discussed in Chapter 3 and repeated in (6.1) for the sake of convenience. Among the weakest contexts we find that of the labial [β] and dorsals [g] and [γ]. Among the least weak consonants, and therefore those that would be less favourable to the occurrence of EV, we find the coronals [t] and [d]. However, further study is required on the distinction of manner of articulation (spirantization vs. occlusion) and perceptibility in consonant clusters. Such an analysis could shed light on the distinction between consonants that share place of articulation but differ on manner of articulation.

(6.1)



6.4.3. Obstruent + /l/ (Cl)

6.4.3.1. Main effects models with odds ratios

6.4.3.1.1. Stress

Table 6.9 presents the results for stress in the Cl clusters. It shows that although both clusters and non-clusters have higher odds of correct perception in post-tonic position, it is the former that presents the highest odds (5.83) when compared to pre-tonic –the baseline for comparison.

Table 6.9.

Results of discrimination by stress on clusters and non-clusters

	$C^{v}lV$				CVlV			
	e^{β} (odds ratio)	SE β	Z	*P*	e^{β} (odds ratio)	SE β	Z	*P*
Pre-tonic	1				1			
Tonic	2.08	.47	3.21	0.001	1.58	.27	2.60	0.009
Post-tonic	5.83	2.04	5.03	0.000	3.84	.1.37	3.76	0.000

Similar to the perception of the Cr clusters, the Cl clusters present higher odds of being correctly discriminated in the post-tonic position followed by the stressed position and last the pre-tonic position. However, the perception of Cl does not support the results found through speech production. In the production study the stressed and the post-tonic positions were less likely to present EVs (52% and 53% respectively compared to the pre-tonic position). Thus these results do not support the hypothesis that more EVs would enhance the perceptual salience of the clusters.

These results could be explained by the fact that the length of the lateral /l/ requires less enhancement and therefore shows less occurrence of EV. In addition, the results could indicate a cue trade-off: listeners seem not to rely on stress as a cue to recover the cluster perceptually.

6.4.3.1.2. Place of articulation

The results for place of articulation (Table 6.10) show that Cl clusters and non-clusters, in the context of coronals and labials, are less likely to be correctly perceived as compared to dorsals. In particular, coronals are less likely to be perceived correctly (.6 and .15 of the values for dorsals, for clusters and non-clusters respectively), whereas clusters and non-clusters with

labials are more likely to be perceived correctly (.74 and .89 of the values for dorsals respectively).

The extremely low rate of accurate perception in the context of coronals (.06) as compared to dorsals (the baseline), shows that listeners heard the ClV clusters as the CVLV non-clusters. A possible factor in the low rate of accurate perception of clusters with a coronal is item frequency. In Spanish, there is only one cluster formed by coronal plus liquid (tl). The cluster /dl/ is not productive in Spanish, and /tl/ is restricted to very few words (*atlas* 'atlas' and *atlántico* 'Atlantic' and its derivatives, and a few other words from indigenous languages with very restricted dialectal use). The data seem to indicate that the low frequency of these clusters affects the listeners at the time of discrimination. It is accepted that the listener, when faced with an unfamiliar or illegal sequence in his/her language, presents a longer reaction time or simply categorizes the token as a mistake or inaudible (Ramirez 2002, Seo 2004). However, the magnitude of the effect of item frequency is not covered in this study. It is a question for future research.

Table 6.10

Results of perceptual discrimination by place of articulation in Cl clusters and non-clusters

	C^{v}lV				CVlV			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Dorsal	1				1			
Coronal	.06	.022	-8.26	0.000	.15	.039	-7.31	.000
Labial	.74	.207	-1.607	0.28	.89	.21	-0.47	.63

6.4.3.1.3. Voicing

The results for voicing (Table 6.11) show that the clusters and non-clusters formed by a voiceless consonant are more likely to be perceived correctly. In the case of clusters, those

formed with a voiceless consonant are three times more likely to be discriminated correctly than clusters formed with a voiced consonant.

Table 6.11.

Results of perceptual discrimination by voicing in Cl clusters and non-clusters

	$C^{v}lV$				CVlV			
	e^{β} (odds ratio)	SE β	Z	*P*	E^{β} (odds ratio)	SE β	Z	*P*
Voiced	1				1			
Voiceless	3.0	.79	4.19	0.000	1.99	.50	2.72	.006

6.5. Discussion and Conclusions

The results of this experiment are relevant in two ways. Firstly, they shed light on which cues the listeners use for discrimination purposes. Secondly, they provide information as to the effect of L1 and L2 proficiency in the use of those cues. A summary of the results is presented in Table 6.12. They are presented in a ranking based on their salience for perceptual discrimination.

Table 6.12.

Significant main affects across consonant clusters

Factor	**Cluster**	
	Cr	Cl
Place of articulation	labial > dorsal > coronal	dorsal > labial > coronal
Voicing	voiceless > voiced	voiced / voiceless
Stress	post-tonic > tonic > pre-tonic	post-tonic > tonic > pre-tonic
Type of vowel	epenthetic / full	full / epenthetic
Group	L1 / L2	L1 > L2

/ No significant difference was found, > significant difference

Depending upon place of articulation, listeners correctly identify clusters as such (that is, as not being CVC sequences) with dorsal consonants over both coronal and labial ones at a significantly higher rate. These results partially conform to the universal salience ranking for

stops (Jun, 1995) in which the clusters with dorsals are identified at a significantly higher rate than those with coronals or labials (as discussed in Chapter 2). However, for the Cr clusters there does not seem to be any difference in the rate of discrimination between dorsal and labials. It would be expected that dorsals would be identified at a higher rate. It is not clear what cues cause the labials to be as salient as the dorsals. It is not likely attributable to the interaction with the epenthetic vowel since the results for clusters and non-clusters present the same pattern. These results show that listeners confound clusters with CVC sequences, at a significant rate, when the first consonant is a coronal or a bilabial, which seems to argue for the more salient perceptual characteristics of the dorsal consonants.

Regarding voicing, the results are consistent for Cr and Cl clusters. They present higher odds of being accurately discriminated, as clusters in the context of a voiceless, rather than a voiced consonant. That is to say that Cr and Cl clusters are heard as CVC syllables more often when the first consonant of the cluster is a voiced consonant.

In terms of lexical stress, consonant clusters in post-tonic syllables are identified as such at a higher rate than when they occur in tonic and pre-tonic syllables. Surprisingly, the results show that listeners confound clusters with CVC syllables in pre-tonic and tonic positions, which are commonly accepted as the more perceptually salient contexts, whereas both Cr and Cl clusters are correctly identified at a higher rate in post-tonic position, which is considered the perceptually weakest position. The results support the hypothesis that EVs enhance perceptibility of the clusters in a perceptibly weaker context: the post-tonic position, which is usually the word-final position.

The results for Group show that non-native speakers are more likely to hear an EV as a NV, but, through exposure to the language, they gradually improve their perceptual discrimination between clusters and CVC syllables.

The data shows that, in contrast to native speakers, non-native speakers are more sensitive to the EV. They are more likely to perceive EVs and therefore they mix up EVs and full vowels at a significantly higher rate than native speakers. These results support hypothesis 1: English speakers pay attention to the EV, even in the context of a consonant cluster, at a higher rate, as expected.

Regarding the type of liquid, the native speakers and L2 learners perceived the Cl clusters at a significantly different rate. The results suggest that L2 learners, even at advanced stages of the learning process, do not reach the accuracy level of listening of the native speakers. They do not seem to have adjusted their mode of L2 listening to reach native-like performance, and their L1 listening strategies do not seem to give them an advantage, which disproves hypothesis #2.

The odds of accurate discrimination improve with time of exposure to the language so that advanced learners discriminate more accurately than beginners. The progression in accurate discrimination suggests that they form a more native-like perceptual system. However further analysis of individual cues in each group is necessary to identify whether the language learners use, at some point in the language learning process, the same cues as the native speakers.

With respect to the identification of the most salient cues, the results are not straightforward. The listening strategies for discrimination in general seem to be a composite processes in which the different segmental and prosodic variables contribute in a cue weighting process, as suggested by the interaction of effects at the highest levels for both Cr and Cl

clusters. In addition, cues used for discrimination vary from cluster to cluster. For discrimination of Cr clusters, the variables Voicing and Stress are significant for all groups, whereas place of articulation is only significant for L2 learners. For Cl, the variables Place of articulation and Stress are significant for all the groups. The data also suggest that for the identification of Cl clusters, L2 learners rely more heavily on clues from place of articulation, stress and to a lesser degree on type of vowel. The variable voicing seems to play little or no role in the discrimination between CC and CVC sequences. Thus, based on the consistent behavior of variables across the L2 levels, the ranking in 6.2 for the discrimination of the Cl cluster is proposed. The dashed line represents exchangeable positions.

(6.2) Ranking of perceptual cues for identification of Cl clusters

Stress ⋮ Place of articulation > Type of vowel

On the other hand, the identification of Cr clusters relies more heavily on the variables stress and voicing, and to a lesser degree on place of articulation. Identification relies on type of vowel at the lowest level. Thus, I argue that the discrimination of the Cr cluster follows the ranking proposed in (6.3).

(6.3) Ranking of perceptual cues for identification of Cr clusters

Stress ⋮ Voicing > Place of articulation

Thus, the type of vowel is especially important for the clear identification of Cl clusters, and even more so in Cr ones.

Methodologically, the study shows that the inter stimulus interval used for the experiment (250 ms) tapped for the most part into perceptual processing at the phonetic level. However, when illegal low frequency clusters were used, processing at the phonological level was activated. I argue that, as a result of the change in processing level, the participants classified perfectly legal phonemic sequences as erroneous. For instance, the [tl] cluster, which is allowed but with low frequency, had only a 6% chance of being accurately perceived. Instead, it was perceived as tVl.

Chapter 7
The Role of EVs in the Perception of Foreign Accent

This chapter presents the results of a goodness rating experiment on words containing digitally manipulated epenthetic vowels. It analyses the factors that determine the accentedness in consonant clusters and the interaction of those factors. The chapter is organized as follows, section 1 presents a discussion on the methodology, section 2 presents the experimental results, and section 3 presents the results and discussion.

7.1. Overview

Previous research has shown that listeners can detect a foreign accent after listening to samples of speech as short as 30 ms (Flege & Hammond, 1982; Flege, 1984). These results indicate that listeners do not filter out subcategorical or subphonemic differences. On the contrary, listeners are sensitive to subtle differences in phonetic structure that contribute to foreign accent (Magen 1998). It also has been shown that even untrained listeners are reliable at judging foreign accent (Brennan and Brennan, 1981; Anderson-Hsieh and Koehler, 1988; Cunningham-Andersson and Engstrand, 1989; Munro, 1995; Flege and Fletcher, 1992; Derwing, Munro and Wiebe, 1998), although experienced listeners (with training in linguistics and/or familiarity with other languages) are sometimes more reliable (Thompson, 1991). Although listeners generally perceive a foreign accent holistically, it is important to understand the contributions of various phonetic factors to the perception of global foreign accent not just for

the understanding of the second language acquisition process, but also for the teaching of a second language (Munro & Derwing, 1995; Magen, 1998).

The aim of the study presented here is to contribute to the understanding of the nature of accentedness from the point of view of the L1 and L2 listeners, and to assess the perceptual significance of the epenthetic vowel in the overall accentedness at the lexical level.

7.1.1. Acoustic salience

As discussed in Chapter 3, Jun (1995) argues that the robustness of the acoustic cues is derived from long transitional cues: the tongue gesture is long and slow, such as in dorsals and labials, whereas the coronals have short transitional cues: the tongue gesture is short and rapid. Therefore, the author proposes a perceptibility scale as in (7.1), after Jun (1995).

(7.1) dorsals > labials > coronals

In addition, Hume, Johnson, Seo, and Tserdanelis (1999) point out that dorsals are more salient because they contain additional acoustic cues for place of articulation from the convergence of F2 and F3 of neighbouring vowels. Thus, the salience of a segment can be enhanced or diminish by the context.

It has been shown that low perceptual salience is triggered when two segments in the cluster are phonetically similar to each other (Kawasaki, 1982; Ohala 1992, 1993; Seo, 2004). Furthermore, low perceptual salience can also be triggered by the occurrence of a segment in contexts with weaker phonetic cues (Wright, 1996; Côte 2000, and 2003; Seo, 2004).

Kochetov and Co (2007), in a study of the cross-linguistic perception of place of articulation of released syllable-final stops, found support for Jun's perceptibility scale. They

found higher salience of dorsals over labials and coronals, and marginal salience of labials over coronals. Kochetov and Co (2007), along with Byrd (1992), attribute the labial vs. coronal asymmetry to the "relatively slow movement of the lips resulting in more robust VC [vowel-consonant] transitions, compared to the more rapid movement of the tongue resulting in weaker VC transitions" (7). The dorsals' salience is enhanced by the relative duration of stop releases: stops articulated at the back of the oral cavity tend to have longer and more acoustically robust releases (VOT) (Cho & Ladefoged 1999). The authors concluded that there is strong support for "language-independent perceptual salience differences between places of articulation" (8).

In a study of stop place perception, Hume, Johnson, Seo, and Tserdanelis (1999) also found support for Jun's (1995) perceptibility scale. Nevertheless, they found that dorsals' salience is especially affected by the neighbouring vowel. For instance, they found that in the context of /i/ the dorsals are less salient than both the labial and coronal, but more salient in the context of /u/ and /a/. Therefore they conclude that vowel context affects the salience of different consonants, except for coronals, which always come out as the lowest ranked in the salience hierarchy (2077).

In terms of the relation between the perception and production of consonant clusters, Colantoni and Steele (2006) proposed that there is negative language transfer. In the case of consonant clusters in the Spanish-English interlanguage, the authors point out that a native-like production of a consonant cluster would entail the mastering of three phonetic properties: the mastering of the flap, the mastering of the EV and the voicing of the stops[23]. In a study of the acquisition of native Spanish-like pronunciation by ten English native speakers, Colantoni and

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[23] In Spanish prevoicing can be a *sufficient* voicing cue in contrast to English where the short vs. long lag is needed to mark such difference (See Williams 1976, Lisker and Abramson 1964, Quilis, 1993)

Steele (2006) found that the mastery of native-like pronunciation after puberty is exceptional, only one student out of ten achieved native-like performance (indistinguishable from a Spanish-native-speaker control). They suggest that that “a complete explanation of learners’ failure to master target phonetic properties must look both at perception (i.e., input and cues) and production (i.e., articulatory constraints)” (71). The results of this study will shed some light on the perceptual aspect of consonant clusters formed with [r].

7.1.2. Predictions

Assuming that the perceptibility correlates with the strength of the context and the articulatory similarity of the segments, it is expected that listeners will identify as more accented those words with changes in EVs in context of dorsals rather than in context of labials and coronals, along the perceptibility scale (i.e., dorsal > labial > coronal). It is also expected that there will be language-particular differences; since English does not present intrusive vowels between consonant clusters, and the schwa vowel is more frequent in English, it is expected that English speakers will identify accentedness at a higher rate than native Spanish speakers.

7.2. Methodology

This study follows current trends in L2 research in assessing phonetic similarity using perceptual assimilation tasks. In these tasks, monolingual listeners are presented with speech stimuli, and asked to indicate to which L1 phonetic category each L2 token is most similar, and to rate its "goodness" as an exemplar of that category (Ingram and Park, 1997; Flege, Bohn and Jang, 1997; Strange, 1999). Three different results may be obtained: a) an L2 sound is consistently categorized as a good instance of one L1 vowel (that is, consistently assimilated to one L1 vowel) and attains good ratings, b) an L2 sound is consistently assimilated to one L1

vowel but with poor goodness ratings, or c) an L2 sound is identified as instances of multiple L1 vowels with poor goodness ratings. These three alternatives correspond to the classification of L2 sounds as 'identical,' 'similar', and 'new', respectively, in Flege's (1987) equivalence classification hypothesis.

7.2.1. Participants

This study examined two groups: Spanish native speakers and English speakers learning Spanish (L2). In turn, the participants in the latter group were divided according to their level of proficiency in Spanish: beginners, intermediate and advanced. The native Spanish speakers (NSS) were monolingual speakers with no experience in an English-speaking environment. They were first-year-university students in Colombia who voluntarily agreed to participate. The L2 speakers were native speakers of English who started learning Spanish after the age of eight years, the purported critical period for phonological acquisition (Scovel, 1988); therefore they are not considered to be bilinguals. They were students at a university in the United States.

The distribution of participants by group and average age at which they started learning the language is presented in Table 7.1 below.

Table 7.1.

Distribution of participants by group, age and age of initial instruction in Spanish

Group	Number of participants	Females	Males	Age		Initial age of instruction in Spanish	
				Mean	SD	Mean	SD
Beginners	69	40	29	20.1	10.3	17.1	6.0
Intermediate	80	51	29	22.8	11.7	16.8	3.2
Advanced	37	22	15	22.9	8.8	16.6	4.4
Native-Spanish speakers	69	38	31	20.9	6.3		

7.2.2. Stimuli

The stimuli used in this study were real words containing tautosyllabic consonant clusters /p, t, k, b, d, g, + r/ and they were selected from the *Curpus del Español* (Davies 2002), a 100-million-word corpus of Spanish available online. To control frequency, the words were drawn from among the first 150 entries containing the target consonant cluster.

The tokens were recorded by a twenty-four year old female who is a native Spanish speaker from Colombia. The recording took place in a soundproof room. The target tokens were inserted in a carrier sentence i.e. *diga* ___ *nuevamente* 'say ___ again'. Based on the productions findings reported in Chapter 1, the tokens were distributed based on the segmental and phonotactic factors. The segmental factors considered were (i) place of articulation of the first consonant (C_1), (ii) voicing of the C_1. (iii) the quality of the nucleic vowel (a, e, i, o, u). The phonotactic factor considered was lexical stress; that is to say, the position of the cluster with respect to the stressed syllable in the word (pre-tonic, tonic or post-tonic). The total distribution of tokens was 90 (3 place X 2 voicing X 5 vowel X 3 Stress). The clusters formed by Consonant + /l/ were excluded from this study due to the small number of cases of EVs, (10 out of 15 possible conditions), which made it impossible to do a reliable statistical analysis.

Preparation of the stimuli

The spectral analysis of the tokens revealed that the epenthetic vowel was not clearly distinguishable in all the cases. Thus, only cases in which the epenthetic vowel was clearly differentiated from the surrounding segments were chosen (30 tokens). From those cases that clearly showed an epenthetic vowel we obtained the first duration variant: tokens with a full epenthetic vowel. For the second variant, half of the vowel was trimmed, cutting only in the

middle of the vowel in order to maintain the transitions intact, resulting in the tokens with half of an epenthetic vowel (Mid). For the third variant, the whole epenthetic vowel was deleted, resulting in the tokens with no epenthetic vowel (zero EV). The number of tokens was 90 (30 x 3) plus 90 distractors for a total of 180 tokens. However, the absence of a clear EV in some cluster combinations did not allow analysis of the effect of nucleic vowels. In cases such as pre-tonic coronals, for [dr] the EV was found with all five vowels whereas for [tr] it was found only in the context of one vowel (/o/). Thus, the linguistic variables that are taken into account are: voicing of the first consonant of the cluster (voiced /voiceless), lexical stress (pre-tonic/tonic/post-tonic positions), and duration of the epenthetic vowel (full epenthetic vowel, half epenthetic vowel, and no epenthetic vowel).

Although the epenthetic vowel was digitally manipulated, the stimuli overall are considered naturally produced as opposed to synthesized stimuli. It is considered that such manipulations "do not affect the naturalness of the speech stream" (Jesney, 2005). This study is in line with current methodology (see alteration for speech rate: Munro and Derwing 1998; the correction of perceived segmental errors, Magen 1998; alterations in factors such as VOT, González-Bueno, 1997; or even the removal of all segmental cues, Jilka 2000).

7.2.3. The Task: Accentedness Rating

For this experiment, a goodness judgment task using Praat (Boersma & Weenink, 2009) was designed, which presented the participants with a scale from 1 to 7 and then randomly played one of the words. The participants were asked to grade the goodness of the token, where 1 is bad (affected/accented speech) and 7 is good (natural/unaccented speech). Once the participants made a selection, a there was a two second interval before the the next word was played. The average time to complete the task was about 30 minutes.

Before the actual task a short training session was conducted using 3 tokens produced by a different speaker in order to familiarize the subjects with the equipment and the task.

Since the stimuli involved real words, the participants were explicitly instructed to assess accentedness so that they would focus more on the general assessment of the phonetics rather than the comprehensibility of the tokens.

7.2.4. Results and discussion

The results were analyzed using Repeated Measures ANOVA (RM ANOVA) with Group as the between-subjects factor. The results of the RM ANOVA (Table 7.2) show that the there is a significant three-way interaction Cluster*Epenthetic*Group ($F(30, 1522) = 3.1$, $p = .000$). Main effects are also highly significant: the factor Cluster is highly significant ($F(5, 1270) = 37.6$, $p = .000$), as well as Vowel ($F(2, 1522) = 21.582$, $p = .000$), and Group ($F(3, 761) = 23.2$, $p = .000$).

Table 7.2.

Full ANOVA summary: Main effects and interactions

Main Effect	Sum of Squares	Mean Square	df	Error	F	Sig.	Power
Cluster	727.1	145.4	5	3.8	37.6	.000	1
Group	1077.7	359.2	3	15.4	23.2	.000	1
Epenthetic duration	53.3	26.6	2	1.2	21.5	.000	1
Cluster*Group	752.5	50.1	15	3.8	12.9	.000	1
Epenthetic*Group	13.3	2.2	6	1.2	1.7	.096	.681
Cluster*Epenthetic	12.42	1.2	10	.39	3.12	.001	.987
Cluster*Epenthetic*Group	114.3	3.8	30	1.2	3.1	.000	1

Computed using alpha = .05

The mean values and standard deviations from the 3-way interaction are presented in Table 7.3.

Table 7.3.

*Mean values and S.D. from the three-way interaction Cluster*Epenthetic vowel* Group*

		Beginners		Intermediate		Advanced		Native Span. Speakers	
Cluster	Epenthetic's duration	Mean	s.d.	Mean	s.d.	Mean	s.d.	Mean	s.d.
cr	Complete	4.360	.107	4.560	.099	5.207	.145	6.242	.107
	Mid	4.377	.116	4.592	.108	4.811	.158	5.901	.116
	Zero	4.502	.121	4.342	.112	4.860	.165	5.676	.173
gr	Complete	5.097	.104	5.047	.097	5.652	.142	5.663	.104
	Mid	4.910	.108	4.925	.100	5.550	.147	5.797	.108
	Zero	4.659	.111	4.785	.103	5.279	.152	5.539	.111
tr	Complete	4.874	.126	5.000	.117	5.378	.172	5.592	.126
	Mid	5.097	.125	4.908	.116	5.234	.171	5.710	.125
	Zero	4.966	.125	4.942	.116	5.144	.170	5.575	.125
dr	Complete	4.635	.121	4.550	.112	5.204	.165	4.265	.121
	Mid	4.714	.122	4.577	.114	5.095	.167	4.112	.122
	Zero	4.493	.123	4.498	.114	4.951	.168	4.476	.123
pr	Complete	5.220	.109	4.967	.101	5.559	.149	5.329	.109
	Mid	4.949	.101	4.983	.094	5.468	.138	5.807	.101
	Zero	5.058	.111	4.946	.103	5.387	.151	5.546	.111
br	Complete	4.766	.108	4.594	.100	5.153	.148	5.307	.108
	Mid	4.626	.114	4.696	.105	4.752	.155	4.986	.114
	Zero	4.517	.114	4.298	.106	4.928	.156	5.039	.114

The results presented in Table 7.2 argue for the composite nature of perception. They indicate that there is interplay of cues. That is to say that cue recoverability is a product of cue weighting and factor interaction.

As follow up analysis, two-way Anovas (simple interaction tests) were performed on the interactions Cluster by Group and Cluster by Epenthetic duration. Table 7.4 presents the results of the two-way Anova for the interaction Cluster by Group for each level of duration of EV (Complete, Mid, and Zero EV). The results from Table 7.4 show that the interaction

Cluster*Group is highly significant in all levels of EV duration (Complete EV, F(15, 3805) = 9.3, *p* = .000; Mid EV, F(15, 3805) = 8.3, *p* = .000; and Zero EV, F(15, 3805) = 3.3, *p* = .000.)

Table 7.4.

Inte*raction of Cluster*Group by level of Epenthetic duration*

	Sum of Squares	Mean Square	Df	F	Sig.	Power
Complete EV						
Cluster	225.6	45.1	5	17.1	.000	1
Group	377.9	125.9	3	47.8	.000	1
Cluster*Group	368.5	24.5	15	9.3	.000	1
Mid EV						
Cluster	332	66.4	5	24.3	.000	1
Group	401.8	133.9	3	49.1	.000	1
Cluster*Group	339.6	22.6	15	8.3	.000	1
Zero EV						
Cluster	231.4	46.2	5	16.2	.000	1
Group	382.4	127.6	3	44.6	.000	1
Cluster*Group	145	9.6	15	3.3	.000	1

Computed using alpha = .05

The averaged results by cluster, as illustrated on Figure 7.1, show that the groups overall perceived clusters differently. Spanish speakers rank the cluster *dr* as the least natural-sounding among all the other clusters (*M* = 4.84, *SE* = .107) (Recall that the higher the score the least accented the cluster was rated), while beginner learners perceive the *cr* cluster as the least native-like (*M* = 4.413, *SE* = .210) while the cluster *br* (*M* = 4.39, *SE* = .19) and *cr* (*M* = 4.79, *SE* = .165) were the least native-like for intermediate and advanced learners respectively. The cluster *cr* presents the most dissimilar ranking; while beginning learners ranked it as the more accented (least natural) of all the clusters, native Spanish speakers ranked it as the most natural sounding.

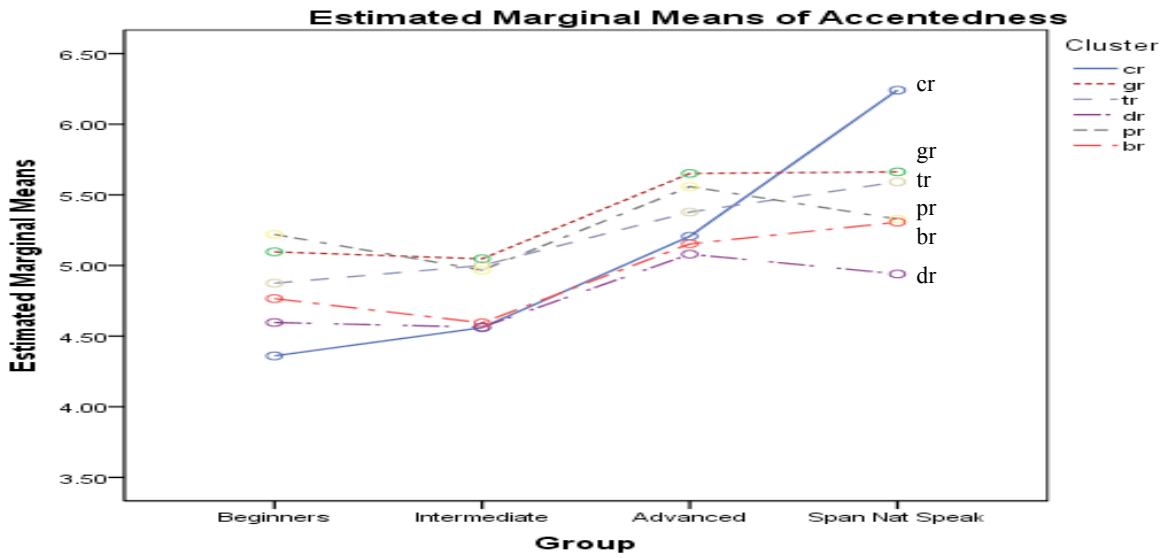


Figure 7.1 Perceived Accentedness of Cluster by Group.

Follow up two-way Anova analyses were performed as simple interaction tests. Table 7.5 shows the results of the two-way Anovas for the interaction Epenthetic duration x Cluster for each level of group. The results show that the interaction EV duration x Cluster is highly significant for the group of native Spanish speakers ($F(10, 7610) = 2.7$, $p = .002$); but not for English speakers who are learners of Spanish (Beginners, $F(10, 7610) = 1.38$, $p = .181$; Intermediate learners, $F(10, 7610) = 1.38$, $p = .746$; and Advanced learners, $F(10, 7610) = .23$, $p = .993$.)

Table 7.5.

*Interaction of Epenthetic duration*Cluster* by Group level*

	Sum of Squares	Mean Square	Df	F	Sig.	Power
Beginners						
Epenthetic duration	9.8	4.9	2	1.74	.175	.368
Cluster	205.8	41.1	5	14.5	.000	1
Epenthetic duration *Cluster	39.2	3.9	10	1.38	.181	.716
Intermediate						
Epenthetic duration	16.4	8.2	2	3.34	.035	.634
Cluster	243.3	48.6	5	14.5	.000	1
Epenthetic duration *Cluster	16.6	1.68	10	1.38	.746	.366
Advanced						
Epenthetic duration	24.6	12.3	2	4.7	.009	.790
Cluster	90.6	18.1	5	6.9	.000	.999
Epenthetic duration *Cluster	6	.6	10	.23	.993	.133
Spanish Native speakers						
Epenthetic duration	6.4	3.2	2	1.06	.3.44	.239
Cluster	1087.3	217.4	5	71.7	.000	1
Epenthetic duration *Cluster	83.6	8.3	10	2.7	.002	.973

Computed using alpha = .05

The averaged results of clusters, as illustrated in Figure 7.2, show the tendency to rank the clusters with complete EVs as more natural and those with half (Mid) or no EV as less natural. This distinction is clearer in the clusters with dorsals ([cr] and [gr]) and labials ([br], [pr]) and less clear with coronals [tr] and [dr].

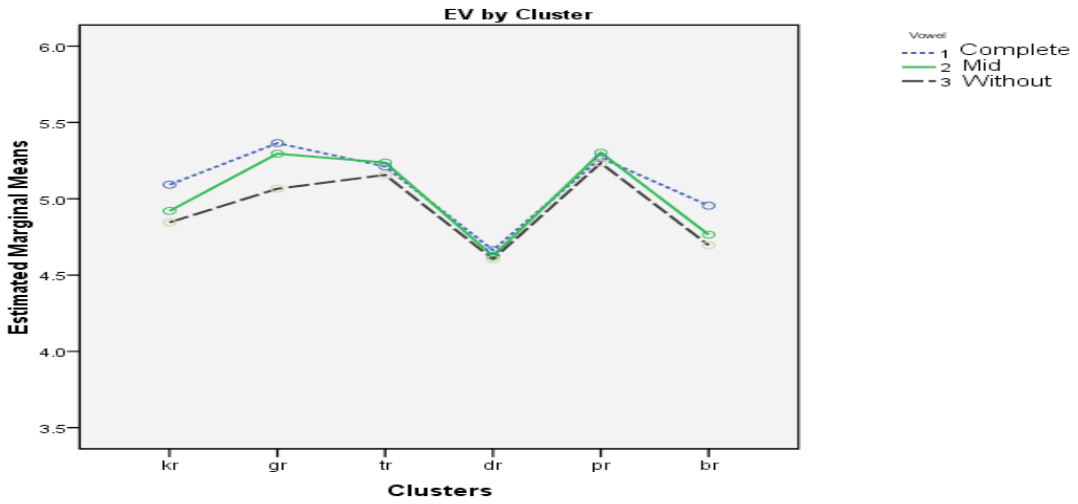


Figure 7.2 Perceived accentedness of levels of Epenthetic duration by Cluster

As follow up on the two-way Anova, paired T-tests were performed on the EV duration in each cluster (Table 7.6). The analysis of each cluster per group shows differences among groups: the L2 learners show significant differences in accentedness for the clusters [gr], [cr], and [br] between cases with Complete and zero EV, with the exception of [cr] (kr) for beginners and [br] for advanced learners. Furthermore, small differences for zero and Mid EV are perceived for [cr] and [br] by advanced learners. Advanced learners seem to perceive the difference between Mid and zero EV in the [cr] and [br] clusters, and similarly beginners also do for the [pr] cluster. The native Spanish speakers also seem to identify differences in accentedness when there are changes in EV duration in the [cr], [pr] and [br] clusters. Nonetheless, in contrast to L2 learners, native speakers identify differences in accentedness at a significant rate in the clusters [pr] and [cr] (for all the pairs) and [dr] (for the Complete-Mid and Complete-zero EV pairs). None of the groups perceived changes in accentedness of the cluster [tr] for any of the possible pairs.

Table 7.6.

Paired samples t-test analysis of EV duration in each cluster

Cluster	Pair	Beginners	Intermediate	Advanced	Spanish speakers
cr	complete – mid	$t(68) = -.848$, $p = .399$	$t(79) = 2.167$, $p = .033$*	$t(36) = -.318$, $p = .752$	$t(68) = 2.173$, $p = .033$*
	complete – zero	$t(68) = -1.085$, $p = .282$	$t(79) = 2.154$, $p = .034$*	$t(36) = 2.445$, $p = .019$*	$t(68) = 5.575$, $p = .000$*
	mid – zero	$t(68) = -.160$, $p = .873$	$t(79) = -.312$, $p = .756$	$t(36) = 2.575$, $p = .014$*	$t(68) = 3.942$, $p = .000$*
gr	complete – mid	$t(68) = 1.930$, $p = .058$	$t(79) = 1.621$, $p = .109$	$t(36) = 1.792$, $p = .081$	$t(68) = 3.392$, $p = .001$*
	complete – zero	$t(68) = 3.432$, $p = .001$*	$t(79) = 3.130$, $p = .002$*	$t(36) = 2.717$, $p = .010$*	$t(68) = 1.231$, $p = .223$
	mid – zero	$t(68) = 1.486$, $p = .142$	$t(79) = 1.379$, $p = .172$	$t(36) = .684$, $p = .498$	$t(68) = -1730$, $p = .088$
tr	complete – mid	$t(68) = 1.153$, $p = .253$	$t(79) = -.291$, $p = .772$	$t(36) = .630$, $p = .533$	$t(68) = 1.231$, $p = .223$
	complete – zero	$t(68) = -.746$, $p = .458$	$t(79) = .548$, $p = .585$	$t(36) = 1.797$, $p = .081$	$t(68) = .141$, $p = .888$
	mid – zero	$t(68) = -1.977$, $p = .052$	$t(79) = .797$, $p = .428$	$t(36) = 1.043$, $p = .304$	$t(68) = -.999$, $p = .321$
dr	complete – mid	$t(68) = 2.029$, $p = .046$	$t(79) = 1.024$, $p = .309$	$t(36) = 1.047$, $p = .302$	$t(68) = -2.634$, $p = .010$*
	complete – zero	$t(68) = 1.464$, $p = .148$	$t(79) = .647$, $p = .520$	$t(36) = 1.900$, $p = .066$	$t(68) = -2.123$, $p = .037$*
	mid – zero	$t(68) = -.798$, $p = .428$	$t(79) = -.330$, $p = .742$	$t(36) = .906$, $p = .371$	$t(68) = 1.918$, $p = .034$*
pr	complete – mid	$t(68) = -1.077$, $p = .285$	$t(79) = .402$, $p = .689$	$t(36) = .595$, $p = .556$	$t(68) = 2.803$, $p = .007$*
	complete – zero	$t(68) = 1.640$, $p = .106$	$t(79) = .240$, $p = .811$	$t(36) = 1.305$, $p = .20$	$t(68) = -2.512$, $p = .014$*
	mid – zero	$t(68) = 2.536$, $p = .014$*	$t(79) = -.220$, $p = .826$	$t(36) = .816$, $p = .420$	$t(68) = -5.656$, $p = .000$*
br	complete – mid	$t(68) = 1.200$, $p = .234$	$t(79) = 4.267$, $p = .000$*	$t(36) = -.992$, $p = .328$	$t(68) = -.618$, $p = .538$
	complete – zero	$t(68) = 2.035$, $p = .046$*	$t(79) = 2.890$, $p = .005$*	$t(36) = 1.289$, $p = .206$	$t(68) = 2.534$, $p = .014$*
	mid – zero	$t(68) = 1.318$, $p = .192$	$t(79) = -1.143$, $p = .257$	$t(36) = 2.245$, $p = .031$*	$t(68) = 3.384$, $p = .001$*

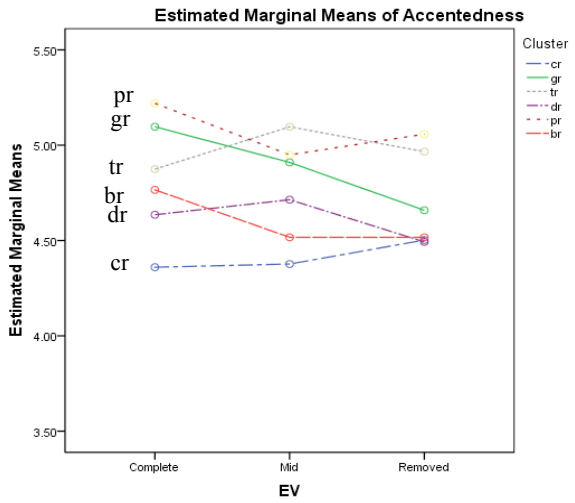
* Statistically significant. Computed using alpha = .05

The significant differences in the perception of accentedness in cases of presence vs. absence of EV (complete vs. Zero EV) seem to depend on the L1. Spanish native speakers identify differences in accentedness at a significant rate when most of the clusters have a complete EV versus when they do not have an EV. The exceptions where this difference is not perceived are the clusters *gr* and *tr*. As for the L2 learners, there is variability in the perception of accentedness: while all the L2 groups perceive significant differences between compete and zero EV in the *gr* clusters, only intermediate and advanced learners perceive it for the *cr* cluster; and for the *br* cluster, only beginners and intermediate learners perceive it. This variation seems to indicate the changing nature of accent perception. In the case of the *cr* cluster, beginners do not perceive the difference between complete and zero EV, but intermediate and advanced learners do. This change in perception seems to evolve towards native-like perception. However, the changes do not always seem to be maturational in the sense that they do not always progress towards a native-like perception. For example, in the case of cluster *br*, beginners and intermediate learners perceive the differences, but not advanced learners. Given that native Spanish speakers perceive the difference, we would expect that advance learners would resemble native-like perception more than any other group. The results indicate that the perception of the complete-zero EV distinction is lost as the learning process and exposure to the language increases. However, in the clusters *cr* and *br* advanced learners (but no beginners or intermediate learners) perceive differences between mid and zero EV, which are very small differences in EV duration.

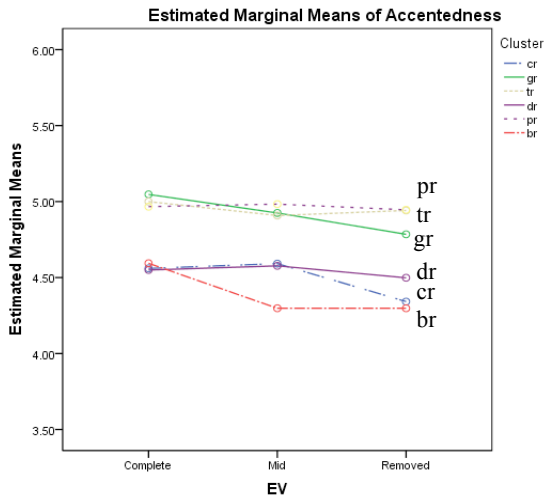
In cases where the epenthetic is cut in half, compared to full and no epenthetic (clusters [cr], [gr], [dr], [pr] and [br]), the results suggest that listeners perceive the presence/absence of the epenthetic and even smaller changes as cutting the epenthetic in half; considering that the

average duration of the epenthetic is 26.98 ms, the reduction to one half (approx. 13.5 ms) would fall below the accepted perceptibility threshold of about 25 ms.

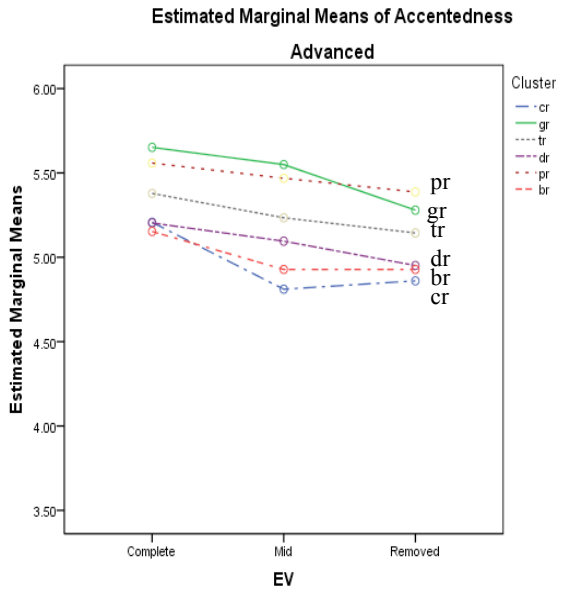
a) EV duration by Cluster for Beginners



b) EV duration by Cluster for Intermediate learners



c) EV duration by Cluster for Advanced learners



d) EV duration by Cluster for Spanish native speakers

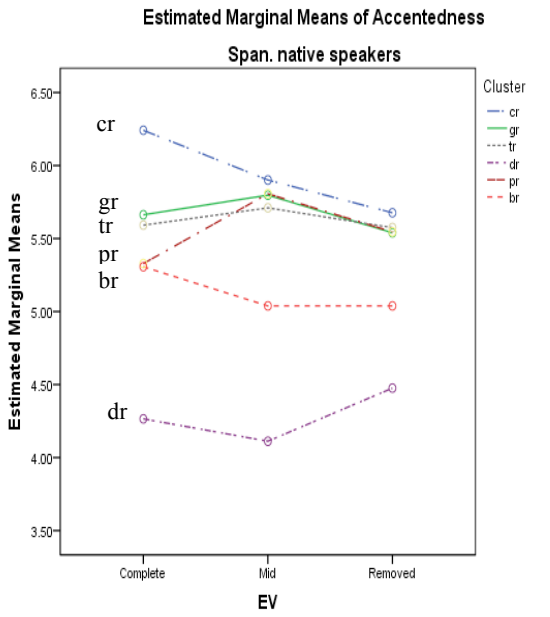


Figure 7. 3 EV duration*Cluster in all Groups.

Figure 7.3 illustrates the rating of native-like accentedness assigned to the different durations of EV by each group. In the group of beginners (3a), there is more variability: the clusters *gr* and *br* are perceived as more native-like when they have complete EVs, less native-like with mid EVs, and the least native-like are those without EVs. But, for the clusters *tr* and *dr*, those cases with half EV are rated as more native-like; and for the cluster cr, the tokens without EV were rated as more native-like. For the intermediate learners, most of the lines are fairly horizontal (for *pr*, *tr* and *dr*), which suggest that there is no clear differentiation of accentedness. Nonetheless, the clusters *cr* and *br* without EV are perceived as less native-like at a significant rate.

For the advanced learners the results are as expected: the clusters with complete EVs are rated as more native-like, slightly less native-like with mid EVs, and the least native-like without EVs. A variation of the pattern are the clusters *br* and *cr* where cases with mid EV are rated as the least native-like, but still those with complete EV are considered more native-like.

In the case of Spanish native speakers (3d), the ratings of the consonant clusters do not present a consistent pattern. For the clusters *gr*, *tr*, and *pr,* the tokens with mid EV are rated as more native-like, while for *br* and *cr* the tokens with complete EV are rated more native-like. The case of the cluster *dr* is particular in that NSS rate it, overall, as the least native-like, as compared to any ratings by any of the L2 groups. NSS rate the tokens without EV as the more native-like of the tokens.

I argue that the overall low rating of *dr* clusters is due to the role of attention. That is to say that native speakers, when prompted to listen carefully to words containing the *dr* cluster, note that there is an EV, or that it sounds affected if it is cut or elided, and without being

conscious that they themselves produce EV in their speech, they rate it as accented. However, some context allows clearer recoverability.

In terms of linguistic factors, on the whole, NSS seem to perceive changes in accentedness in the context of dorsals, bilabials, and the voiced coronal. Within the dorsals, all variants of the EV were significantly perceived in the context of the voiceless segment /k/, while in the context of the voiced segment /g/ only the complete-mid pair was perceived, which suggests that the voiceless is the strongest context for perceptibility. Similarly, for bilabials, salience is stronger in the context of the voiceless /p/ compared to the context of the voiced segment /b/ (where only the complete-zero and the mid-zero pairs were perceived as significantly different). As for the coronals, the voiced segment /d/ seems to be the salient context since in the environment of the /t/ no changes are perceived. The lack of perceptibility in context of /t/ suggests that this is a weak context. I argue that the cluster [tr] forms a weak environment due the shared place of articulation between the elements, which requires very short and fast transitions. In the case of /d/, its more common realization is the spirantized (approximant) allophone [ð] (found after vowels and consonants except /l/ and nasals. Thus, between the two coronal segments, the dento-alveolar /t/ has a more common place of articulation with the also alveolar /r/ than with the [ð]. This difference in place of articulation would account in part for the perceptibility of EVs in context of /d/.

Regarding voicing, the results indicate that within each place of articulation, the EV has the most salience after the voiceless element. Hence, perceptually, the voiced EV is more recoverable after a voiceless segment than after a voiced one. For instance, the EV is more recoverable after the voiceless sound [k] (spelled with c) in the cluster [k^{v}r] than after the voiced [g] in [g^{v}r]. Besides this, the voiced nature of [r] makes the EV less recoverable in the latter case

since it is embedded between two voiced segments. The exception of the recoverability in context of voiceless over voiced segments is the case of coronals, where, as discussed above, the voiceless /t/ coincides in place of articulation with /r/ creating short and quick transitions that in turn mask the perceptibility of the EV.

The results suggest an ordering in salience as presented in (7.2)

(7.2) Salience hierarchy for NSS

labials > dorsals > coronals
p > b > k > g > d

As for the perception of accentedness in the contexts of dorsals, the data show that L2 learners perceive more differences in the context of the voiceless /k/, whereas in the context of /g/, all the groups of L2 learners consistently distinguish only the complete-without EV difference. This indicates that in the context of the voiced consonant, only extreme differences in EV duration (absence vs. presence) are perceived, while in the context of voiceless consonants more subtle differences (absence vs. half EV and half EV vs. full EV) are perceived.

As for the labials, the context of the voiced /b/ seems to be by far more salient than the voiceless /p/. In the latter, only beginners were able to distinguish the mid-complete pair. This marked difference between labials could be attributed to the intrinsic articulatory energy of the segments: voiceless have higher articulatory energy than voiced segments. As discussed in Chapter 2, the EVs are longer after voiced segments, which are recovered in perception by L2 learners.

Interestingly, L2 listeners do not perceive any changes in accentedness in the context of a coronal either voiced or voiceless. This lack of perception of the EV in the context of coronals

may be an effect of the L1. In English the rhotic /r/ is realized as the retroflex [ɹ] (and in some cases it is affricated such as after /t/). It is also accepted that the pronunciation of the Spanish tap [ɾ] is one of the most difficult tasks to achieve for Spanish L2 learners (Navarro Tomás, 1946; Lado, 1948, Colantoni and Steele, 2006), even though English speakers produce the tap [ɾ] in cases such as [leiɾi] 'lady' and sweater [sweɾeɹ]. Hence, a possible explanation for the impossibility in recovering perceptual cues for the EV after coronals is related to the L1's transfer, which makes it difficult to produce the Spanish flap [ɾ] and the consonant clusters in a native-like manner.

In terms of the relation between perception and production of the consonant clusters in the Spanish English interlanguage, the results of the current study suggest that L2 learners do not perceive any differences in the EV in the context of a coronal, which maybe the source of the learners' failure to attain native-like performance in the pronunciation of consonant clusters. However, further research is required on the L2 production in different linguistic contexts.

The data suggest an ordering in perceptual salience (recoverability) as presented in (7.3).

(7.3) Salience hierarchy for L2 learners

dorsals > labials > coronals
g > k > b > p > d, t

The order in (3) supports Jun's (1995) and Hume et al.'s (1999) findings that for English listeners the dorsals are more salient than the labials and coronals, which indicates that even at advanced levels of L2 proficiency listeners transfer their L1 perceptual strategies. That is to say, in L2 perception they tend to recover information from the same cues they use in L1 listening.

7.3. Conclusions

The analysis of the relationship of variables considered here (place of articulation, voicing, stress) shows that the perception of accentedness in consonant clusters does not depend exclusively on one factor. These results argue for the composite nature of perception: cue recoverability is a product of cue weighting and factor interaction.

The results of this study show that the salience hierarchy of voiceless stops is language specific. While English speakers learning Spanish follows the dorsal > labial > coronal hierarchy, Spanish native speakers obtain cues in the hierarchical order labial > dorsal > coronal. Thus, our prediction is partially supported by the data. Nonetheless, it is necessary to note that as shown in Chapter 3, the salience of a segment is affected by the proximity of a vowel and possibly by other factors. The interaction of factors influencing the salience of stops is outside the scope of this study, but for future research it will be important to determine the salience of the segments in different conatexts linguistics in English as well as Spanish.

The results for length of EV by groups show that NSS perceive accentedness when there is a manipulation of the EV, whereas the overall results for L2 learners is not significant, although it is perceived in some clusters. These results do not support our predictions regarding the role of L1 in the perception of the EV. The results indicate that L2 learners do not yet perceive changes in accentedness when there are differences in the duration of the EV. In contrast, NSS do perceive changes in accentedness. Although NSS are not aware of the occurrence of the EV, they are perceptually sensitive to changes, even small changes, in its duration. These results support findings in current perception research. Furthermore, they present evidence that listeners can detect subcategorical differences in durations as shorts as 13ms,

which fall below the perceptual threshold. The results indicate that the failure to produce EVs in Spanish clusters is a source of accentedness among L2 learners' speech production

In sum, the results of this study show that some environments that are richer than others in terms of acoustic/perceptual cues, and that the relationship between acoustics and perception of the EV is not straightforward since acoustically rich environments do not correspond with the most perceptually recoverable environments. The dorsal and labial place of articulation are among the most perceptually recoverable environments, whereas the context of a coronal segment make for a perceptually poor environment (less recoverable).

Chapter 8. Conclusions

This chapter summarizes the findings of the production and perceptual studies and evaluates them in light of the results of other studies. In addition, it discusses the new findings of this research and their implications for the fields of linguistics and second language acquisition. Finally, it discusses directions for future research in this area.

8.1 Production

The occurrence of EVs in Cr and Cl clusters.

In a production experiment, eight native speakers of Spanish produced 100 tokens in contexts defined by voicing, place of articulation of the consonant, type of liquid in the cluster and its position with respect to the stressed syllable. These were analyzed acoustically in an attempt to discover the relative frequencies of EVs within the different contexts. This study showed that the Epenthetic Vowel in consonant clusters is an unconscious strategy from the speaker to make the clusters more perceptible.

Despite the high degree of variability that is part of speech, the findings shed light on some of the factors that affect the production of the EV in consonant clusters. Among the main contributions of this study is the analysis of the occurrence of EV in Cl clusters. In contrast to previous studies that dismissed the study of Cl clusters due to the low occurrence of EVs (Colantoni and Steele, 2005, 2006; Schmeiser, 2006), this study found that Cl clusters had EVs 40% of the time while Cr clusters presented EVs 75% of the time. While the percentage of EVs in Cl clusters is significantly lower than in Cr ones, the percentage is not negligible and thus we consider that it provides valuable information on the occurrence of EVs.

According to the hypothesis proposed, it would be expected that we would find more EVs in the context of the more sonorous liquid, and therefore the weakest context (the context formed with the lateral /l/). However, more EVs appear consistently in the context of /r/ than in the context of /l/. I argue that these results are due to the well-known fact that /l/ is of much longer duration as compared to /r/. Thus there is a trade-off between duration and sonority. Since the /l/ is long it does not frequently require the occurrence of EVs in order to be perceptually recovered, despite the fact that it otherwise forms a weak perceptual context. In contrast to previous studies, I found considerable numbers of EVs in Cl clusters, although significantly lower than in Cr clusters. The difference in comparison to other studies could be due to differences in speech rate or the dialectal differences.

Regarding the main predictors of the occurrence of EVs, the type of liquid (r or l) is a strong predictor. Therefore, the analysis of the clusters was divided into Cl and Cr. Table 8.1 presents the ranking of the predictors by type of cluster.

Table 8.1.

Significant main affects across consonant clusters in speech production.

Factor	**Cluster**	
	Cr	**Cl**
Place of articulation	dorsal > labial > coronal	labial > coronal > dorsal
Voicing	voiceless > voiced	voiceless > voiced
Stress	post-tonic > pre-tonic > tonic	pre-tonic > tonic /post-tonic
Nucleic vowel	u > a/e > i/o/	o/a> i/o > u

/ = no significant difference.

The results show that although /r/ and /l/ share some features (i.e., the liquid manner and alveolar point of articulation), they affect cluster production in different ways. I argue that the main difference is due to the inherently longer duration of /l/ (M = 90.7ms, SD = 24.1ms) as compared to /r/ (M = 18.8ms, SD = 15.2ms).

8.1.1. Place of articulation

In terms of place of articulation, the results for the Cr cluster show that there is a higher chance of occurrence of an EV after dorsal consonants than in any other place of articulation. Since Spanish consonants do not show significant differences in sonority/strength due to place of articulation (see Lavoie, 2001), I argue that, for place of articulation, articulatory principles take precedence over perceptual recoverability. In other words, the occurrence of the EV is not determined mainly by the perceptual strength of the context, but by the clear differentiation of articulatory gestures. Thus, in the articulation of clusters such as dorsal + r (which is coronal), the displacement of the tongue (the active articulator) is longer than it is for clusters formed by coronal + coronal. The longer displacement of the tongue avoids the overlapping of the articulatory gestures that occurs with shorter displacements such as the coronal + coronal cluster. The smaller overlap allows a more clear identification of the EVs. In the case of clusters formed by labial + r, there are two active articulators in the articulation of the cluster, the lips and the tongue, which allows for completion of the articulatory gesture with little or no overlap.

In the case of the Cl clusters, there is a higher chance of an EV occurring in a labial + l than in other positions. In contrast to Cr clusters, the position with the least chance of having an EV is that with dorsal consonants.

The results for the Cr cluster partially support the scale of consonant strength for Spanish consonants proposed in chapter 1, and presented in Figure 8.1. There are higher odds of an EV occurring with consonants that form a weak context such as the dorsals [g] and [ɣ], and lower odds with coronals [t] and [d] ad labials [p] and [b] since they form a strong context.

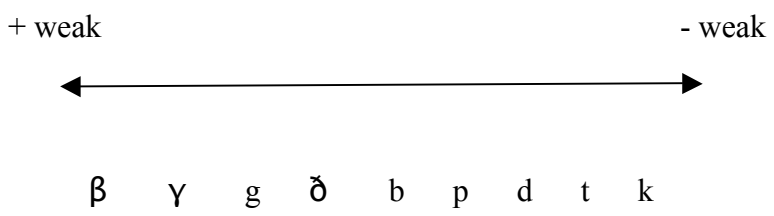


Figure 8.1. Scale of strength for Spanish consonants

Cl clusters, in comparison, do not necessarily follow a consonant-strength hierarchy. The Cl clusters require less enhancement from the EVs since the /l/ has longer duration, as compared to /r/. The occurrence of the EV in a Cl cluster seems to be determined by the differentiation required in the articulatory gestures. However, further research is required in this area.

In terms of the effect of speech rate on the occurrence of EVs, the results show that speech rate accounts for 14% of the variability in an EV's length. In the case of Cl clusters, EVs lengthen in inverse proportion to the rate of speech. Thus, at higher speech rates, when the segments tend to be shorter, we find that longer EVs occur. We can infer that the motivating factor for the compensatory lengthening of EVs is perceptual recoverability. However, in the case of the Cr clusters, there is no correlation between word duration and the duration of the EV, or between the duration of the EV and the nucleic vowel. Further research is needed to identify the role of speech rate in Cr clusters.

8.1.2. Voicing

The results reveal that the voicing of the first segment of the cluster has an effect on the rate of occurrence of the EV. The EV has a higher rate of occurrence in the context of a voiced consonant than in the context of a voiceless one. The results confirm our assumption regarding voicing: more EVs are likely to occur in the context of a voiced consonant since it creates a perceptually weaker context as opposed to the voiceless consonants. These results can also be interpreted as support for the assumption that more EVs occur in the context of voiced segments to compensate for the inherently short duration of the segments. However, if that were that the case, we would expect to find a different trend in Cl clusters where the /l/ has a longer duration than the /r/. This, however, is not the case. The results of this study confirm the findings of Blecua (2001), Colantoni and Steele (2005), Ramírez (2006), and Schemeiser (2006).

8.1.3. Stress

The results show that lexical stress is a significant factor in the EV's rate of occurrence. In the case of Cr clusters, there is a higher chance of an EV occurring in the post-tonic position than in the pre-tonic and tonic positions. These results support our hypothesis that EVs are more likely to occur in the weakest context. Post-tonic position is generally the word-final position, in which there is usually articulatory laxness. This causes overlapping of gestures and undershooting in the enunciation of segments. These conditions create a weak context and therefore benefit from the occurrence of an EV to enhance the perceptibility of the cluster.

Regarding the Cl clusters, the results show that there are higher odds of an EV occurring in the pre-tonic position (word-initial position) than in the tonic and post-tonic positions. These results follow the pattern of speech production in which the first part of the word is articulated

more clearly than any other parts of the word. Part of the more careful pronunciation is the low level of overlapping among segments, which allows the occurrence of an EV without being overlapped by its contiguous segments.

These results contrast with results from previous studies that report the stressed position as the most likely to present EV (Colantoni and Steele 2005, Schmeiser, 2006). When we consider that lengthening of the vowels and consonants is a result of stress, it is not unexpected to find that longer EVs occur in stressed positions. Thus, any other possible motivation for the occurrence of an EV in stressed syllables is confounded by the prosodic effects of lexical stress.

8.1.4. Nucleic vowel

I argue that that the formant transitions between the consonant and the following vowel are part of the same articulatory plan. Thus, the /r/ and /l/ in consonant clusters are affected by the following vowel.

The hypothesis put forward in terms of vowels was that more EVs were expected with the low vowel /a/ (the perceptually weakest vowel), and fewer EVs with the high vowels /i/ and /u/ (the perceptually strongest vowels). With respect to Cr clusters, the results show that EVs are more likely to occur with the high back vowel /u/ whereas it is less likely that they will occur in the context of the high front vowel /i/. These results do not support the hypothesis, but rather seem to differ in terms of the frontness or backness of the articulation of the vowel. However, further research is required on the role of the nucleic vowel on the occurrence and length of the EV.

In contrast, with the Cl clusters, the results support the hypothesis proposed: more EVs are more likely to occur in the context of the mid-low vowel /a/ whereas and fewer EVs tend to

occur with the high-back vowel /u/. We infer from these results that the EV enhances the perceptibility of the weaker contexts.

8.1.5. Speech rate

In terms of the effect of speech rate on the occurrence of EVs, the results show that speech rate accounts for 14% of the variability in an EV's length. In the case of Cl clusters, EVs lengthen in inverse proportion to the rate of speech. Thus, at higher speech rates, when the segments tend to be shorter, we find that longer EVs occur. We can infer that the motivating factor for the compensatory lengthening of EVs may be perceptual recoverability. However, in the case of the Cr clusters, there is no correlation between word duration and the duration of the EV, or between the duration of the EV and the nucleic vowel. Further research is needed to identify the role of speech rate in Cr clusters.

8.2. Perception

In this study speech perception is operationalized as identification and discrimination. To this end, an identification and a discrimination experiment were conducted. For the identification task, 179 speakers judged a total of 100 tokens. They had to match the token they heard to one of four possible written items. For the discrimination experiment, 243 speakers used an AXB forced choice procedure in which they had to listen to the three stimuli and then decide whether A is similar to X or B is similar to X. They heard a total of 100 triads.

The results of the perceptual experiments shed light on the cues that are used for perceptual recoverability as indicated in the identification and discrimination experiments. The results from the two experiments show differences in the hierarchy of cues based on the type of cluster. The results for the clusters Cr and Cl are presented in Table (8.2).

Table 8.2.

Significant main affects across perceptual tasks

Factor	Cluster	Identification	Discrimination
Place of articulation	Cr	labial > dorsal > coronal	labial > dorsal > coronal
	Cl	labial > dorsal > coronal	dorsal > labial > coronal
Voicing	Cr	voiceless > voiced	voiceless > voiced
	Cl	voiced / voiceless	voiced / voiceless
Stress	Cr	tonic /pre-tonic >post-tonic	post-tonic > tonic > pre-tonic
	Cl	tonic > post-tonic/pre-tonic	post-tonic > tonic > pre-tonic
Type of vowel	Cr	epenthetic > full vowel	epenthetic / full vowel
	Cl	epenthetic / full vowel	epenthetic / full vowel
Group	Cr	L1 > beginner/intermediate	L1 / L2
	Cl	L1 > beginner/intermediate	L1 > L2

> Significant difference in hierarchical order.

/ No significant difference was found.

Despite the fact that identification and discrimination are two different perceptual tasks that use different cues (Mayo, 2000), they show considerable similarities, as discussed in the next sections.

8.2.1. Place of articulation

In the case of Cr clusters, the results of the identification and discrimination tests show that the clusters formed with a labial are more likely to be perceived correctly than those with dorsals and coronals. For Cl clusters, the results are mixed: the identification results show that clusters with labials have better odds of being correctly identified, but in discrimination, the chances are better after dorsal segments. These results partially support Jun's (1995) universal salience ranking, according to which coronals are the least perceptually salient. That is to say, they provide the weakest context for perceptibility. However, the data indicate that the labials and not the dorsals form the perceptually strongest context, with the exception of the discrimination of Cl clusters that conform to the hierarchy.

Overall, the results from the identification and discrimination tests show that the perception of phonetic features is highly related to the salience of the context in which it appears. The results support Hume, Johnson, Seo, and Tserdanelis's (1999) claim that the salience ranking proposed by Jun (1995) for voiceless stops is language-specific. While the salience scale for English follows Jun's raking, as presented in (8.2), in Spanish the labials seem to be more salient as presented in (8.3).

(8.2) Salience ranking for English native speakers (After Jun, 1995)

more salient ⟷ less salient

dorsals > labials > coronals

g > k > b > p > d / t

(8.3) Salience ranking for Spanish native speakers

more salient ⟷ less salient

labials > dorsals > coronals

p > b > k > g > d / t

Nonetheless, it is important to note that this rank is not static. It is conditioned by the different contexts in which cues interact, presumably to make communication efficient.

8.2.2. Voicing

In terms of voicing, the results from the identification and discrimination experiments show that there are better chances of correctly perceiving the Cr clusters clusters after voiceless segments as compared to voiced ones. These results contrast with the results from speech production according to which there are higher odds of an EV occurring after a voiced segment, as opposed to a voiceless one. These results suggest that the occurrence of EVs, although they enhance the perceptibility of the clusters, is not enough to surpass the voiceless consonants, which have an intrinsic higher articulatory energy (Quilis 1993: 67.) In addition, the fact that voiceless consonants are inherently longer than the voiced consonants makes them easier to perceive.

In the case of the Cl cluster, the results show that there is no significant difference in perceptibility due to the voicing of the first segment in the cluster. In terms of production, more EVs occur in the context of a voiced segment as compared to a voiceless one. The data suggest that the enhancement of the voiced context brings it up to par with the voiceless context. In other words, the occurrence of an EV allows clusters with voiced consonants to be as perceptually recoverable as those with voiceless ones.

8.2.3. Stress

In terms of lexical stress, the results indicate that the perceptual identification of both sets of clusters –Cr and Cl- follows the pattern of speech perception: the most perceptually salient parts of the word are those in word-initial position (pre-tonic) and stressed position (tonic). The Cr clusters present the greatest odds of being correctly identified in pre-tonic or tonic positions over the post-tonic position, whereas for the Cl clusters in the tonic position has the greatest likelihood of being correctly identified. The results support the generally accepted notion that edge effects make word final position the articulatory weakest position and the least attended to by listeners. Thus, we find less accurate identification of clusters in post-tonic position, which in Spanish is usually word final position.

In terms of production, EVs are more likely to occur in post-tonic position for Cr clusters and in pre-tonic position for Cl clusters. Perceptually, the post-tonic position is the least salient context. These results suggest that the word-edge effects outrank prosodic effects such as lexical stress. More than perceptual effects, word-edge effects are attentional (Kohler, 1992:207); listeners focus their attention on word-initial information and pay less attention to word-final information, which makes the latter less salient despite its articulatory energy and/or other factors affecting its perceptual prominence.

Thus, I argue that the EVs are more frequent in word-final position (the perceptually weakest context) to enhance the perceptual recoverability of the cluster so that it is minimally distinguishable, whereas in the pre-tonic position (generally word-initial position in this study) EVs are not required since it is a context that is much more focused upon by the listener.

However, it could be argued that low levels of accuracy are indeed high levels of confusion between clusters (C^vCV) and non-clusters (CVC). Thus, the data would suggest that listeners confound more clusters and non-clusters in the post-tonic position because they perceive more EVs in clusters and confound them with full vowels (non-clusters). Thus, listeners, especially L2 learners, seem to perceive EVs at such rate that there is only 53% accurate identification of Cr clusters in post-tonic position. In the case of the Cl clusters, EVs do not to create the same degree of erroneous identification in post-tonic position: clusters and non-clusters are identified or discriminated correctly more (.2 and 5.8 times respectively) than in pre-tonic positon.

The case of word stress is a good example of cue interaction; the analysis of the interaction voicing x stress shows that listeners are more likely to correctly identify clusters and non-clusters in post-tonic position than in tonic or pre-tonic position (4.8 and 2.2 times in identification and discriminations task respectively). The results of the interaction argue for the role of EVs to enhance perceptual recoverability in the weakest context.

8.2.1.4. Type of vowel

Regarding the effect of the presence of the EV or the full vowel, the results in Table 8.2 show that it is only statistically different for the Cr cluster in the identification task. Overall, the results show that there is no perceptual confusion over CvC and CVC cases. It suggests that the EV maintains a range in duration so that it is long enough to differentiate the two similar segments in the cluster, but short enough so that it does not become so salient as to create ambiguity in cases such as *prosa* and *porosa* for instance. Nevertheless, in the diachronic development of the language there have been cases in which the epenthetic vowel becomes so perceptually salient that it is incorporated as part of the word or its lexical derivates. For

instance, *chacra* 'farm' is derivationally related to *chacarero* 'farmer' and not *chacrero* as it would be expected.

8.2.1.5. Group

The results for group show that Spanish L2 learners are significantly better than native speakers of Spanish at perceiving the EVs. That is they confound EVs with full vowels at a significant rate, while native speakers confound them at a significantly lower rate. However, the results show that, in the identification of the Cr and Cl clusters, the advanced L2 group performs almost the same as the native speakers. These results show that L2 learners develop a more native-like perception due to exposure to the L2. In this case the exposure was given mainly through class instruction.

The almost native-like accuracy of the advanced L2 group demonstrates high levels of perceptual adaptability, especially considering that the occurrence of EVs in consonant clusters is not taught in Spanish classes. Since native Spanish speakers are not aware of the EV, it is not incorporated in the program of Spanish L2 language or into specific pronunciation classes.

In contrast with the results for perception, L2 speech production at native-like levels is very rare when learned after the age of 10 (Munro, Flege, and MacKay, 1996). Colantoni and Steele (2005) found that only one out of 10 subjects produced Spanish consonant clusters at comparable levels to a native speaker after 3 years of instruction.

In terms of perceptual discrimination of the Cr cluster, there was no significant difference between the performances of L1 and L2 groups, but there was, however, a significant difference for the Cl clusters: L2 listeners perceived (discriminated) EVs at a statistically significant rate, whereas L1 learners confused the EVs and the nucleic vowels.

Most of the studies have focused on the Cr clusters in part because of the clear differences between the English and Spanish realizations of /r/, but Cl clusters seem to pose a perceptual difficulty that had not been documented previously. As far as I am aware, this is the first study that examines the production and perception of Cl clusters. However, further study comparing production and perception of L1 and L2 speakers is required.

8.3. Perception of Accentedness

The analysis of the relationship of the variables place of articulation, voicing, and stress shows that the perception of accentedness in consonant clusters does not depend exclusively on one factor. These results argue for the composite nature of perception: cue recoverability is a product of cue weighting and factor interaction.

The results of this study show that the salience hierarchy of voiceless stops is language specific. While English speakers learning Spanish follow the hierarchy from L1 (English): dorsal > labial > coronal, Spanish native speakers obtain cues in the hierarchical order labial > dorsal > coronal. Thus, our prediction is partially supported by the data. Nonetheless, it is necessary to note that the salience of a segment is affected by the proximity of a vowel and possibly by other factors. The interaction of factors influencing the salience of stops is outside the scope of this study, but for future research it will be important to determine the salience of the segments in different linguistics in English as well as Spanish.

8.4. Hypothesis testing

This study confirms the complexity of the analysis of speech production and perception. It is generally accepted that very small changes, such as changes in the rate of speech of a speaker, the sociolinguistic register, or the loudness, among others, will have an effect on the

speech stream (see Mayo 2000). Therefore, cues do not only vary between speakers, "but also between two productions of the same utterance by the same speaker" (9). While it is true that there is variability in production, the signal still has to be accurately perceived. Among the processes used to maintain a perceptually accurate signal, the signal undergoes alterations to adjust to changes in the phonetic environment. Among the adjustments in the signal, there is cue weighting, cue trading, and cue redundancy.

The results show that advanced L2 learners develop a more native-like perception than beginners and intermediate learners. However, they rarely achieve native-like rates of accuracy. Thus, hypothesis 1 is borne out.

Furthermore, L2 perception, like L2 production, is a developmental process that requires an extended period of time in order to achieve native-like accuracy. The results from Group show that the length of time of exposure to the language is a factor in the degree of perceptual accuracy. Learners with more exposure to the language performed better (more native-like) than learners with little exposure. Therefore, hypothesis 2 is also borne out.

It is expected that, given more time, learners could reach native-like accuracy. However, it could also be the case that L2 learners will never achieve native-like rates of accurate perception. Thus, hypothesis 3 is inconclusive. There is not enough evidence to prove or disprove it.

The findings of this study are relevant to second language instruction, specifically for advanced learners of Spanish as L2. Explicit instruction about the occurrence of EVs in consonant clusters could raise consciousness about this feature and might help learners to acquire a more native-like pronunciation. However, this would require both the development of

appropriate teaching tools and a research design for testing the outcomes against students who do not receive this kind of intervention.

In addition, this study contributes to a new approach in the statistical analysis in the study of epenthetic vowels. As far as I am aware, this is the first study in this research area that considers item and individual variation as part of the statistical analysis. With the use of mixed logit models for statistical analysis, binary outcomes such as presence/absence of EVs can be analysed with both speaker and participant variation being taken into account at the same time. In particular, this approach avoids the use of analysis of variance on categorical data, either with or without transformations (e.g. Jaeger 2008).

8.5. Further directions in research

This study is relevant in two ways: first, it sheds light on what cues are used by the listener for discrimination and identification purposes; and second, it provides some information as to the effect of L1 and L2 proficiency in the use of those cues.

Nonetheless, future research should examine the effects of prosodic contexts such as different rates of speech and their effect on epenthetic and nucleic vowels since the possible differences in their behavior may indicate different motivations for the EV. Furthermore, it is necessary to examine the effect of segmental features in the perceptual salience of the EV, specifically the role of the nucleic vowel under controlled and uncontrolled rates of speech.

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Appendices

Appendix 1.

List of quasi-minimal pairs used in the production study. List of quasi-minimal pairs used in the production study.

blindado	contabilidad
platal	palatal
capricornio	pirineo
bravo	barado
blanco	balando
planta	palanca
Prusia	arampurú
diablus	discóbulus
vértebrum	táburus
dirigido	escudriñar
trujillo	turumillo
podría	pediría
atlántico	talante
estrado	tarado
vidrio	adórdiris
cuadrus	cándurus
atlas	átala
Sumatra	alcántara
glisemia	anguilizado
aclimatado	kilimanjaro
cruzado	curumaní
gurú	grupo
mezclado	poco calado
Anglés	Anguilis
pácurus	sucrus
flirteo	filigrana
frustrado	enfuruñarse
Flandes	falange
péteflun	gónfulun
sofrito	zafirito

Appendix 2. Identification Test:

List of test words and distracters. The underlined word in each set of test words indicates the correct target word (the word played in the recording). The set of distracter words do not have an underlined word.

Identification Test

Choose the word that matches the word you hear.

(The target answer is underlined, and the highlighted numerals indicate the distracter items)

1. a) gabres b) cábres c) gáberes d) láberes	61. a) aburunar b) adurunar c) adrunar d) abrunar
2. a) casple b) rasple c) cáspele d) gáspele	62. a) báfarra b) báfara c) bárafa d) bálafa
3. a) merétecrus b) mértecurus c) néntecrus d) mértecrus	63. a) neguluco b) nefuluco c) negluco d) nefluco
4. a) jantal b) janatal c) janantal d) jarantal	64. a) porreta b) poneta c) poreta d) poleta
5. a) gandras b) cándaras c) candras d) gandaras	65. a) nefruco b) nefluco c) nefuluco d) nefuruco
6. a) agueles b) ajueles c) águeles	66. a) odrisa b) ofrisa c) odirisa

d) ajles	d) ofirisa
7. a) alcáncala b) algáncala c) alcákela d) algánkela	67. a) calosta b) calorta c) calomsa d) calonsa
8. a) bunaelente b) boenalente c) bonaelente d) dunaelente	68. a) nabarar b) nabrar c) naparal d) naprar
9. a) barado b) brado c) berado d) bredo	69. a) cásbele b) casple c) cáspele d) casble
10 a) pereguís b) bereguís c) preguís d) priguís	70. a) eblugo b) epluco c) epuluco d) ebulugo
11. a) tirito b) diritito c) drito d) dirito	71. a) aclifa b) aglifa c) aguilifa d) aquilifa
12. a) síntono b) míntono c) minitono d) níntono	72. a) adrisa b) ablisa c) adlisa d) atlisa
13. a) gabres b) gábiris c) gáberes d) caberes	73. a) corosamí b) curusaní c) curusamí d) gurusaní
14. a) blicol b) plicol	74. a) dresa b) dlesa

c) bilicol d) blincol	c) tlesa d) blesa
15. a) anfusitiodo b) anfrustiodo c) anfustiodo d) anfestiodo	75. a) senflos b) sínfolos c) sífolos d) siflos
16. a) curupado b) acurupado c) crupado d) cripado	76. a) sujallo b) sunallo c) sojallo d) sujello
17. a) adliso b) antliso c) atliso d) atiliso	77. a) blafal b) plafal c) balafal d) palafal
18. a) dempre b) témpere c) tempre d) tempere	78. a) líguiri b) línquili c) lingri d) lincri
19. a) aldrunar b) adrunar c) andrunar d) adurunar	79. a) anguiente b) arguiente c) anguimente d) alguimente
20 a) garalo b) gralo c) garralo d) garanlo	80. a) lengru b) lencru c) léncuru d) lénguru
21. a) cúperos b) cúpros c) cúporos d) cúberos	81. a) gulupol b) glubol c) glupol d) gulupol
22. a) trodir	82.

b) drodir
c) torodir
d) tredir

23.
a) balasal
b) palasal
c) plasal
d) blasal

24.
a) gulupol
b) glupol
c) culupel
d) gulupel

25.
a) mablas
b) nábalas
c) nablas
d) mábalas

26.
a) nabrar
b) nalbrar
c) nabarar
d) malbrar

27.
a) securuno
b) secruno
c) segruno
d) seguruno

28.
a) aclifa
b) aglifa
c) aquilifa
d) aclisa

29.
a) cáncolo
b) cángolo
c) canglo
d) canclo

30.
a) brito
b) dirito
c) drito
d) birito

83.
a) secrumo
b) securuno
c) securumo
d) secruno

84.
a) renfeno
b) lenfeno
c) lenfino
d) resfeno

85.
a) ratris
b) ládiris
c) rátiris
d) ladris

86.
a) sestle
b) séstele
c) sértele
d) sertle

87.
a) pereguís
b) perleguís
c) prequís
d) preguís

88.
a) fibló
b) fiboló
c) fimboló
d) fimbló

89.
a) feretul
b) foretul
c) froentul
d) frentul

a) torodir
b) doronir
c) tonodir
d) trodir

31.
a) grotel
b) corotel
c) crotel
d) gorotel

32.
a) eltodía
b) eltodría
c) eltondía
d) eltosdía

33.
a) feraneo
b) ferineo
c) firineo
d) frineo

34.
a) gralo
b) garalo
c) garralo
d) caralo

35.
a) golunja
b) guluja
c) glunja
d) gluja

36.
a) gandras
b) cándaras
c) canedras
d) candras

37.
a) alifibación
b) afilibación
c) alibación
d) aflibación

90
a) sunfri
b) súnfiri
c) funfri
d) fúnfiri

91.
a) corotel
b) crotel
c) grotel
d) gorotel

92.
a) glufa
b) gluja
c) guluja
d) gulufa

93.
a) lámpico
b) clámpico
c) lámbico
d) lámpeco

94.
a) cangló
b) cángolo
c) cándolo
d) canglo

95.
a) fundri
b) funfri
c) fúnyiri
d) fúnfiri

96.
a) arcla
b) árgala
c) argla
d) árcala

97.
a) torocante
b) dorocante
c) borocante
d) porocante

98.

38.
a) robubión
b) robulión
c) rolubión
d) rulubilión

39.
a) burufir
b) brufir
c) purufir
d) prufir

40.
a) branqueta
b) balenqueta
c) blanqueta
d) barenqueta

41.
a) sífolos
b) siflos
c) sísfolos
d) sífelos

42.
a) adrela
b) atrela
c) aderela
d) aterela

43.
a) beletín
b) belebín
c) plebín
d) pelebín

44.
a) fápolo
b) sápolo
c) jápolo
d) gápolo

45.
a) balicol
b) bilicol
c) blicol
d) plicol

a) saperente
b) saprente
c) sabrente
d) saferente

99.
a) aglas
b) aclas
c) aplas
d) aclás

100.
a) afaldo
b) afando
c) jafando
d) janfado

101.
a) banalse
b) panalse
c) fananse
d) pananse

102.
a) ágoros
b) ágorros
c) agorós
d) áboros

103.
a) irledo
b) ilsedo
c) isredo
d) esredo

104.
a) burufir
b) brufir
c) brujir
d) burutir

105.
a) ebuluco
b) epluco
c) epuluco
d) ebluco

46. a) equilonado b) oquilonado c) aquilonado d) aquelonado	106. a) flasista b) flanista c) falamista d) falanista
47. a) árcala b) árgala c) arcla d) argla	107. a) séstele b) féstere c) sestle d) festle
48. a) adliso b) atiliso c) atliso d) atoliso	108. a) tulugan b) turugan c) dlugal d) tlugan
49. a) ferentul b) fretul c) jerentul d) gretul	109. a) jibró b) fibló c) fiboló d) fipoló
50. a) grimal b) girinal c) grinal d) girimal	110. a) ratris b) rastris c) rátiris d) rástiris
51. a) ijros b) icros c) igros d) igrós	111. a) naflas b) nábalas c) nablas d) nápalas
52. a) sífolos b) siblos c) síflos d) síflus	112. a) cúpuros b) cudros c) cupros d) cúduros
53. a) cónjurus b) cáncorus c) gáncurus d) cáncurus	113. a) quirinal b) girinal c) grinal d) crinal

54. a) bodirida b) podilida c) bodlida d) podirida	114. a) acolobar b) agolobar c) atlobar d) aclobar
55. a) abeste b) ameste c) aneste d) ajeste	115. a) flanista b) falanista c) glanista d) calanista
56. a) tulugan b) dulugan c) tlugan d) tlucan	116. a) ofrisa b) ocrisa c) oquirisa d) ofirisa
57. a) llínguiri b) lingri c) llingri d) línguiri	117. a) balnado b) barnado c) parnado d) balmado
58. a) saperente b) saprente c) sabrente d) sapirente	118. a) aberela b) aterela c) adrela d) atrela
59. a) palafal b) plabal c) palabal d) plafal	119. a) agrobar b) aclobar c) acolobar d) ajolobar
60 a) léncuru b) lencro c) lencru d) léngoro	120. a) táglima b) táglina c) dáglina d) tánglina

Appendix 3.

Test Triads for the AXB discrimination protocol. For each triad I present the combinations in which they were presented to the listeners.

Test triads.

Test triads	
1) aclifa,aclifa,aquilifa	34) gabres,gáberes,gáberes
2) aclifa,aquilifa,aquilifa	35) garalo,gralo,gralo
3) acolobar,aclobar,aclobar	36) gralo,garalo,garalo
4) acolobar,acolobar,aclobar	37) gulupol,gulupol,glupol
5) adrunar,adurunar,adurunar	38) glupol,glupol,gulupol
6) adurunar,adrunar,adrunar	39) grinal,grinal,guirinal
7) árcala,arcla,arcla	40) guirinal,guirinal,grinal
8) árcala,árcala,arcla	41) guluja,gluja,gluja
9) aterela,atrela,atrela	42) guluja,guluja,gluja
10) aterela,aterela,atrela	43) lencru,léncuru,léncuru
11) atiliso,atiliso,atliso	44) léncuru,lencru,lencru
12) atliso,atliso,atiliso	45) lingri,línguiri,línguiri
13) brufir,burufir,burufir	46) línguiri,lingri,lingri
14) brufir,brufir,burufir	47) nábalas,nábalas,nablas
15) candras,cándaras,cándaras	48) nablas,nablas,nábalas
16) candras,candras,cándaras	49) nabarar,nabarar,nabrar
17) casple,cáspele,cáspele	50) nabrar,nabrar,nabarar
18) casple,casple,cáspele	51) palafal,plafal,plafal
19) corotel,corotel,crotel	52) plafal,palafal,palafal
20) crotel,crotel,corotel	53) preguís,pereguís,pereguís
21) bilicol,blicol,blicol	54) preguís,preguís,pereguís
22) blicol,bilicol,bilicol	55) rátiris,ratris,ratris
23) cángolo,canglo,canglo	56) ratris,rátiris,rátiris
24) cángolo,cángolo,canglo	57) saperente,saprente,saprente
25) cupros,cupros,cúpuros	58) saprente,saperente,saperente
26) cúpuros,cúpuros,cupros	59) secruno,securuno,securuno
27) drito,dirito,dirito	60) secruno,secruno,securuno
28) drito,drito,dirito	61)sestle,séstele,séstele
29) epuluco,epuluco,epluco	62)sestle,sestle,séstele
30) epluco,epluco,epuluco	63) tlugán,tlugán,tulugán
31) fibló,fiboló,fiboló	64)tulugán,tulugán,tlugán
32) fiboló,fibló,fibló	65)trodir,trodir,torodir
33) gáberes,gabres,gabres	66) torodir,torodir,trodir

Table 2. Distracters.

Distracters	
1) ágoros alter ,ágoros1,ágoros	18) curusaní,curusaní ,curusanífi
2) ágoros1,ágoros ,ágoros alter	19) dorocant ,dorocant,dorocante
3) agueles1,agueles1,agueles	20) fápolo alter ,fápolo1,fápolo
4) alcáncala ,alcáncala ,alcáncala1	21) feretul1,feretul,fretul
5) alifibaciónfl,alifibación1,alifibaciónaefl	22) feretul,fretul1,fretul
6) alter ,dlesa1,dlesa	23) firineo alter ,firineo1,firineo
7) ameste ,ameste1,ameste alter	24) firineo alter ,firineo1,firineo
8) anfustiodo ,anfustiodo ,anfustiodo1	25) flanista,falanista,falanista1
9) anguiente,anguiente ,anguiente1	26) flanista,flanista1,falanista
10) aquilonado ,aquilonado1,aquilona	27) funfri,fúnfiri1,fúnfiri
11) balnado ,balnado1,balnado alterfi	28) igros alter ,igros1,igros
12) barenqueta,barenqeta1,barenqueta	29) igros1,igros ,igros alter
13) bonaelente,bonaelente ,bonaelente1	30) isredo alter ,isredo1,isredo
14) brado1,brado ,brado alter	31) jafando1,jafando,jafando alter
15) calosta,calosta,calosta1	32) janatal ,janatal ,janatal1
16) cáncurus,cáncurus1,cáncurus alter	33) mértecrusfl,mértecrus ,mértecrus1
17) crupado1,crupado1,crupado	34) míntono1,míntono ,míntono